Linear Algebra and Its Applications

Chapter 2

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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2.1 Matrix operations

A matrix has a *shape*: an $m \times n$ matrix has m rows and n columns. Addition requires identical shapes. In AB, the number of columns of A must equal the number of rows of B; the answer then has the outside shape. If A is $m \times n$ and B is $n \times p$, then

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}, \qquad AB = [Ab_1 \ \cdots \ Ab_p].$$

The first formula takes a row against a column; the second says that one column of the answer can be computed without the others. The transpose interchanges rows and columns. In particular $(AB)^T = B^T A^T$: its reversal of order is essential. Matrix multiplication generally does not commute.

Practice problem 2.1.P1: A vector and its transpose

With $A = \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix}$ and $x = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$, first compute $Ax = (5-9, -10+12)^T = (-4, 2)^T$. Thus

$$(Ax)^T = [-4 \ \ 2], \qquad x^T A^T = [5 \ \ 3] \begin{bmatrix} 1 & -2 \\ -3 & 4 \end{bmatrix} = [-4 \ \ 2].$$

The outer product multiplies each entry in the column by the whole row:

$$xx^T = \begin{bmatrix} 25 & 15 \\ 15 & 9 \end{bmatrix}, \qquad x^T x = 5^2 + 3^2 = 34.$$

Finally A^T is 2×2 and x^T is 1×2. The inner sizes $2, 1$ do not agree, so $A^T x^T$ is undefined. An outer product and an inner product are different operations, even when they use the same entries.

Practice problem 2.1.P2: Choose the order of a computation

What we must show. Compute $A^2 x$ efficiently for A of shape 4×4. Associativity permits $A(Ax)$. A matrix–vector product uses four multiplications for each of four entries, or 16 total. Computing Ax then $A(Ax)$ uses $16 + 16 = 32$. In contrast, A^2 has 16 entries, each requiring four multiplications: 64; the final multiplication by x uses 16 more. The second route uses 80. Both routes give the same vector, but the first avoids computing matrix entries that are not needed.

Exercise 2.1.1: Check shapes before calculating

For the matrices supplied in the exercise, A, B are 2×3 and C, D are 2×2. Scalar multiplication and subtraction act entrywise:

$$-2A = \begin{bmatrix} -4 & 0 & 2 \\ -8 & 10 & -4 \end{bmatrix}, \qquad B - 2A = \begin{bmatrix} 7-4 & -5-0 & 1+2 \\ 1-8 & -4+10 & -3-4 \end{bmatrix} = \begin{bmatrix} 3 & -5 & 3 \\ -7 & 6 & -7 \end{bmatrix}.$$

The product AC is undefined: the inner sizes are 3 and 2. For CD, the inner sizes match, and taking rows against columns gives

$$CD = \begin{bmatrix} 1 \cdot 3 + 2(-1) & 1 \cdot 5 + 2 \cdot 4 \\ -2 \cdot 3 + 1(-1) & -2 \cdot 5 + 1 \cdot 4 \end{bmatrix} = \begin{bmatrix} 1 & 13 \\ -7 & -6 \end{bmatrix}.$$

Exercise 2.1.2: Two defined expressions and two shape failures

Entrywise addition gives

$$A+3B=\begin{bmatrix}2+21 & 0-15 & -1+3\\ 4+3 & -5-12 & 2-9\end{bmatrix}=\begin{bmatrix}23 & -15 & 2\\ 7 & -17 & -7\end{bmatrix}.$$

The vector E is 2×1, so $2C-3E$ is undefined: subtraction cannot combine a 2×2 matrix with a 2×1 matrix. For the defined product,

$$DB=\begin{bmatrix}3\cdot 7+5\cdot 1 & 3(-5)+5(-4) & 3\cdot 1+5(-3)\\ -7+4 & 5-16 & -1-12\end{bmatrix}=\begin{bmatrix}26 & -35 & -12\\ -3 & -11 & -13\end{bmatrix}.$$

Finally EC would have inner sizes 1 and 2, so it is undefined.

Exercise 2.1.3: A scalar identity matrix in a sum and a product

For $A=\begin{bmatrix}2 & -5\\ 3 & -2\end{bmatrix}$,

$$3I_2-A=\begin{bmatrix}3-2 & 0-(-5)\\ 0-3 & 3-(-2)\end{bmatrix}=\begin{bmatrix}1 & 5\\ -3 & 5\end{bmatrix},\qquad (3I_2)A=3A=\begin{bmatrix}6 & -15\\ 9 & -6\end{bmatrix}.$$

Subtracting a scalar multiple of I changes diagonal entries only; multiplying by that matrix scales *every* entry.

Exercise 2.1.4: The same distinction in three dimensions

Subtract 5 from each diagonal entry of the supplied A:

$$A-5I_3=\begin{bmatrix}0 & -1 & 3\\ -4 & -2 & -6\\ -3 & 1 & -3\end{bmatrix}.$$

Multiplication instead scales each row, giving

$$(5I_3)A=5A=\begin{bmatrix}25 & -5 & 15\\ -20 & 15 & -30\\ -15 & 5 & 10\end{bmatrix}.$$

Exercise 2.1.5: Compute a product by columns and by entries

(a) Use the two columns of B.

For $A=\begin{bmatrix}-1 & 3\\ 2 & 4\\ 5 & -3\end{bmatrix}$,

$$Ab_1=4\begin{bmatrix}-1\\ 2\\ 5\end{bmatrix}-2\begin{bmatrix}3\\ 4\\ -3\end{bmatrix}=\begin{bmatrix}-10\\ 0\\ 26\end{bmatrix},\qquad Ab_2=-2\begin{bmatrix}-1\\ 2\\ 5\end{bmatrix}+3\begin{bmatrix}3\\ 4\\ -3\end{bmatrix}=\begin{bmatrix}11\\ 8\\ -19\end{bmatrix}.$$

Place them in order: $AB = \begin{bmatrix} -10 & 11 \\ 0 & 8 \\ 26 & -19 \end{bmatrix}$.

(b) Check by the row–column rule.

The six dot products are $-1(4)+3(-2) = -10$, $-1(-2)+3(3) = 11$, $2(4)+4(-2) = 0$, $2(-2)+4(3) = 8$, $5(4)-3(-2) = 26$, and $5(-2)-3(3) = -19$. They give exactly the same matrix, as the two definitions require.

Exercise 2.1.6: A second product, with zeros included

(a) Compute the columns separately.

The columns of A are $(4,-3,0)^T$ and $(-3,5,1)^T$. Hence

$$Ab_1 = \begin{bmatrix} 4 \\ -3 \\ 0 \end{bmatrix} + 3\begin{bmatrix} -3 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} -5 \\ 12 \\ 3 \end{bmatrix}, \qquad Ab_2 = 4\begin{bmatrix} 4 \\ -3 \\ 0 \end{bmatrix} - 2\begin{bmatrix} -3 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 22 \\ -22 \\ -2 \end{bmatrix}.$$

(b) Verify the entries.

The row–column products are $4-9 = -5$, $16+6 = 22$, $-3+15 = 12$, $-12-10 = -22$, $0+3 = 3$, and $0-2 = -2$. Thus $AB = \begin{bmatrix} -5 & 22 \\ 12 & -22 \\ 3 & -2 \end{bmatrix}$.

Exercise 2.1.7: Recover an unknown shape

A 5×3 matrix can multiply B only if B has 3 rows. The product has as many columns as B; since AB has 7 columns, B has shape $\boxed{3 \times 7}$.

Exercise 2.1.8: Rows of a product

The number of rows of BC is the number of rows of B. Therefore B has $\boxed{5}$ rows. The given product does not determine how many columns B has; that number only has to match the rows of C.

Exercise 2.1.9: A parameter making two products equal

Multiplying in the two orders gives

$$AB = \begin{bmatrix} -7 & 3k+18 \\ -4 & k-9 \end{bmatrix}, \qquad BA = \begin{bmatrix} -7 & 12 \\ -k-6 & k-9 \end{bmatrix}.$$

Equality requires equality in *every* position. The upper-right position gives $3k+18 = 12$, or $k = -2$; the lower-left gives $-4 = -k-6$, also $k = -2$. The diagonal entries already match. Thus $\boxed{k = -2}$ is both necessary and sufficient.

Exercise 2.1.10: Why cancellation can fail

Direct row–column multiplication gives

$$AB = \begin{bmatrix} -21 & -21 \\ 7 & 7 \end{bmatrix} = AC,$$

although, for example, $B_{11} = -1 \neq -3 = C_{11}$. Indeed $B - C = \begin{bmatrix} 2 & 6 \\ 1 & 3 \end{bmatrix}$ and each column is a multiple of $(2, 1)^T$, which A sends to zero. Thus $A(B - C) = 0$ need not imply $B - C = 0$. Cancelling a matrix requires additional hypotheses.

Exercise 2.1.11: Diagonal multiplication and a commuting example

$$AD = \begin{bmatrix} 5 & 6 & 6 \\ 10 & 12 & 10 \\ 15 & 15 & 12 \end{bmatrix}, \qquad DA = \begin{bmatrix} 5 & 10 & 15 \\ 6 & 12 & 15 \\ 6 & 10 & 12 \end{bmatrix}.$$

The jth column of AD is the jth column of A multiplied by d_{jj}; only the jth entry of the jth column of D is nonzero. Similarly, left multiplication scales row i by d_{ii}. For the last request, choose $B = 2I_3$. This is neither zero nor the identity, and $AB = 2A = BA$ by the identity and scalar-multiplication rules.

Exercise 2.1.12: Build a nonzero matrix annihilated by another

For $A = \begin{bmatrix} 3 & -6 \\ -2 & 4 \end{bmatrix}$, the equation $Ax = 0$ reduces to $x_1 = 2x_2$. Choose two different nonzero solutions, $(2, 1)^T$ and $(4, 2)^T$, and make them columns:

$$B = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}, \qquad AB = [A(2,1)^T \;\; A(4,2)^T] = [0 \;\; 0].$$

Having nonzero columns does not prevent their images from being zero.

Exercise 2.1.13: Pack several matrix–vector products together

Set $R = [r_1 \;\cdots\; r_p]$, an $n \times p$ matrix. The column definition of multiplication gives

$$[Qr_1 \;\cdots\; Qr_p] = QR.$$

The factors are the supplied matrix Q and the matrix of input columns, not auxiliary identity matrices. The sizes are $(m \times n)(n \times p)$. If one of these supplied factors happens itself to be an identity matrix, the literal requirement that neither factor be an identity can still be met: write the product as $(\lambda Q)(\lambda^{-1} R)$. Choose a nonzero scalar λ avoiding any value for which either factor is an identity. Each factor excludes at most one scalar, so at least one of $2, 3, 4$ works.

Exercise 2.1.14: Read the units in a cost product

Column j of Q gives dollars of output of the two products in quarter j. Multiplication by U converts that output vector into three dollar costs: materials, labor, and overhead. Thus UQ is 3×4; its entry (i, j) is the total dollar cost in category i during quarter j. The sum over products is exactly the row–column rule: cost per output dollar times output dollars, added over the two products.

Exercise 2.1.15: Five true–false statements about products and transposes

(a) **False.** Two column vectors a_1, b_1 of shape 2×1 cannot be multiplied as $a_1 b_1$. The correct column formula is $AB = [Ab_1 \;\; Ab_2]$.

(b) **False.** Columns of AB are combinations of columns of A, with weights from columns of B. The roles in the statement are reversed. For example, take $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$. The actual first column of AB is $(1, 2)^T$, whereas the proposed combination using the first column of A as weights gives $b_1 = (1, 1)^T$.

(c) **True.** The left distributive law says $A(B + C) = AB + AC$.

(d) **True.** Entry (i, j) of both $A^T + B^T$ and $(A + B)^T$ is $a_{ji} + b_{ji}$.

(e) **False.** Transposing a product reverses order: $(AB)^T = B^T A^T$. For example, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ give different $B^T A^T$ and $A^T B^T$.

Exercise 2.1.16: Five more structural statements

(a) **True.** Entry j of the first row of AB is $\sum_k a_{1k} b_{kj}$, exactly entry j of $(\text{row}_1 A)B$.

(b) **False.** The proposed expression has one column, whereas AB has three. Its correct form is $[Ab_1 \;\; Ab_2 \;\; Ab_3]$.

(c) **True.** $(A^2)^T = (AA)^T = A^T A^T = (A^T)^2$.

(d) **False.** Two reversals give $(ABC)^T = C^T B^T A^T$. The stated middle order is incorrect; taking $C = I$ reduces the claim to the false same-order product rule in the preceding exercise.

(e) **True.** Transpose distributes over sums, entry by entry.

Exercise 2.1.17: Find two unknown input columns

Write $B = [b_1 \; b_2]$. Since $AB = [Ab_1 \; Ab_2]$, solve the two systems with right sides $(-3, 1)^T$ and $(-11, 17)^T$. For a right side $(c, d)^T$, the equations are $x_1 - 3x_2 = c$ and $-3x_1 + 5x_2 = d$. Replacing the second equation by itself plus three times the first gives $-4x_2 = d + 3c$. For $(c, d) = (-3, 1)$, this gives $x_2 = 2$, $x_1 = 3$; for $(-11, 17)$, it gives $x_2 = 4$, $x_1 = 1$. Thus

$$b_1 = (3, 2)^T, \qquad b_2 = (1, 4)^T, \qquad B = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}.$$

Substitution into both original systems checks the two columns.

Exercise 2.1.18: A zero input column

The third column is Ab_3. If $b_3 = 0$, then $Ab_3 = A0 = 0$. Thus the third column of AB is zero, regardless of the other columns.

Exercise 2.1.19: A sum relation survives multiplication

If $b_3 = b_1 + b_2$, then $Ab_3 = A(b_1 + b_2) = Ab_1 + Ab_2$ by distributivity. Therefore column 3 of AB is the sum of columns 1 and 2.

Exercise 2.1.20: Equal columns have equal images

If $b_1 = b_2$, multiplying the equality by A gives $Ab_1 = Ab_2$. These are precisely the first two columns of AB; they are equal. No conclusion is forced about any later column.

Exercise 2.1.21: A nonzero vector sent to zero

Let b be the last column of B. By hypothesis $b \neq 0$, but the last column of AB is $Ab = 0$. Writing this as a linear combination of the columns of A, the coefficients are the entries of b, not all zero. That is a nontrivial dependence relation. Thus the columns of A are linearly dependent.

Exercise 2.1.22: A dependence relation cannot be destroyed by a linear map

If $b_1, \ldots, b_p$ are dependent, there are scalars c_j, not all zero, with $\sum_j c_j b_j = 0$. Multiply by A and distribute:

$$\sum_j c_j(Ab_j) = A\left(\sum_j c_j b_j\right) = A0 = 0.$$

The *same* nonzero coefficient vector witnesses dependence of the columns Ab_j of AB. Some images may even be zero; that only makes dependence easier, not harder.

Exercise 2.1.23: A left inverse forbids free variables

Suppose $Ax = 0$ and $CA = I_n$. Multiplying by C yields $x = I_n x = (CA)x = C(Ax) = C0 = 0$. Thus the homogeneous system has no free variables: all n columns of A must be pivot columns. Since there is at most one pivot in each row, A must have at least n rows. Therefore it cannot have more columns than rows.

Exercise 2.1.24: Construct a right inverse one column at a time

Since the columns of A span $\mathbb{R}^3$, each equation $Ad_j = e_j$ is consistent for $j = 1, 2, 3$. Choose one solution of each, for example by row reduction, and set $D = [d_1\ d_2\ d_3]$. Each d_j has n entries, so D is

$n \times 3$. Then $AD = [Ad_1\ Ad_2\ Ad_3] = [e_1\ e_2\ e_3] = I_3$. If solutions are nonunique, any choices work; uniqueness was not assumed.

Exercise 2.1.25: A left and right inverse force a square matrix

From $CA = I_n$, the argument of Exercise 23 gives $n \le m$. From $AD = I_m$, every $b \in \mathbb{R}^m$ has solution $x = Db$, so A has a pivot in every row and $m \le n$. Thus $m = n$. Associativity then gives

$$C = CI_m = C(AD) = (CA)D = I_nD = D.$$

One must first respect the sizes; the two inequalities explain why both sorts of inverse can occur together only in the square case.

Exercise 2.1.26: A right inverse guarantees all outputs

For arbitrary $b \in \mathbb{R}^m$, choose $x = Db$. Then $Ax = A(Db) = (AD)b = I_mb = b$. This proves consistency for every right side. Hence A has a pivot in each of its m rows. There cannot be more pivots than columns, so $m \le n$: no more rows than columns.

Exercise 2.1.27: Inner and outer products in coordinates

For $u = (-3, 2, -5)^T$ and $v = (a, b, c)^T$, multiplication gives

$$u^Tv = v^Tu = -3a + 2b - 5c,$$

$$uv^T = \begin{bmatrix} -3a & -3b & -3c \\ 2a & 2b & 2c \\ -5a & -5b & -5c \end{bmatrix}, \qquad vu^T = \begin{bmatrix} -3a & 2a & -5a \\ -3b & 2b & -5b \\ -3c & 2c & -5c \end{bmatrix}.$$

The scalar product pairs corresponding coordinates and adds; the outer product keeps all pairwise products, which explains its different shape.

Exercise 2.1.28: Relationships between the two orders

For real vectors, $u^Tv = \sum_i u_iv_i = \sum_i v_iu_i = v^Tu$. The outer products need not be equal, but $(uv^T)^T = (v^T)^Tu^T = vu^T$. Their (i, j) and (j, i) entries are both u_iv_j. The real-scalar assumption matters: complex inner products usually use a conjugate transpose instead.

Exercise 2.1.29: Prove both distributive laws entrywise

What we must show. Establish Theorem 2(b) and 2(c), with all products of matching sizes. For left distributivity, take an arbitrary entry and expand a finite sum:

$$[A(B + C)]_{ij} = \sum_k a_{ik}(b_{kj} + c_{kj}) = \sum_k a_{ik}b_{kj} + \sum_k a_{ik}c_{kj} = [AB + AC]_{ij}.$$

For right distributivity the same scalar rule gives

$$[(B + C)A]_{ij} = \sum_k (b_{ik} + c_{ik})a_{kj} = [BA + CA]_{ij}.$$

The index k runs over the common inner dimension in each formula. Because every entry agrees, the matrices agree.

Exercise 2.1.30: Move a scalar through a product

For an arbitrary position (i, j),

$$[(rA)B]_{ij} = \sum_k (ra_{ik})b_{kj} = r\sum_k a_{ik}b_{kj} = \sum_k a_{ik}(rb_{kj}) = [A(rB)]_{ij}.$$

The middle quantity is also $[r(AB)]_{ij}$. We used only distributivity of finite sums and associativity of real multiplication. Thus $r(AB) = (rA)B = A(rB)$.

Exercise 2.1.31: The identity on the left

Write $A = [a_1 \ \cdots \ a_n]$, where every $a_j \in \mathbb{R}^m$. By the column definition, $I_mA = [I_ma_1 \ \cdots \ I_ma_n] = [a_1 \ \cdots \ a_n] = A$, using the supplied fact that $I_mx = x$ for each vector in $\mathbb{R}^m$.

Exercise 2.1.32: The identity on the right

The jth column of I_n is e_j, whose only nonzero entry is a 1 in position j. Thus $Ae_j = 0a_1 + \cdots + 1a_j + \cdots + 0a_n = a_j$. Consequently $AI_n = [Ae_1 \ \cdots \ Ae_n] = A$.

Exercise 2.1.33: Prove the reversal rule for transposes

Let A be $m \times n$ and B be $n \times p$. Both $(AB)^T$ and B^TA^T have size $p \times m$. At position (i, j),

$$[(AB)^T]_{ij} = (AB)_{ji} = \sum_{k=1}^{n} a_{jk}b_{ki} = \sum_{k=1}^{n} (B^T)_{ik}(A^T)_{kj} = [B^TA^T]_{ij}.$$

The middle reordering uses commutativity of scalar entries, not commutativity of matrix multiplication. Equality of all entries proves the rule.

Exercise 2.1.34: Transpose a product with a vector

Treat x as a one-column matrix and reverse the order of all three factors:

$$(ABx)^T = x^T(AB)^T = \boxed{x^TB^TA^T}.$$

If A is $m \times n$, B is $n \times p$, and x is $p \times 1$, the final shape is $1 \times m$, as a transposed output vector should have.

Exercise 2.1.35: Create standard matrices in a matrix program

One concrete choice of software is Python with NumPy. These commands create the four requested arrays without entering every entry:

```
import numpy as np
np.zeros((4,5))
np.ones((5,3))
np.eye(5)
np.diag([3,4,2,5])
```

A tuple such as $(4, 5)$ specifies rows first, columns second. The diagonal constructor places the listed numbers on the diagonal and zeros elsewhere. These commands were executed to verify their shapes.

Exercise 2.1.36: Generate reproducible random arrays

The following uses a fixed seed so another reader can repeat the experiment:

```
rng = np.random.default_rng(21036)
R = rng.random((5,6))
Z = rng.integers(-9,10,size=(4,4))
```

Entries of R lie in $[0, 1)$; the endpoint convention differs slightly from the idealized $0 < x < 1$ in the hint and does not affect the task. Entries of Z are integers in $\{-9, -8, \ldots, 9\}$: the upper limit 10 is excluded. The dedicated integer generator avoids having to round scaled real numbers. A seed controls reproducibility, not a mathematical claim that all resulting matrices have a special property.

Exercise 2.1.37: Test commutativity with four random pairs

```
rng = np.random.default_rng(21037)
for j in range(4):
    A = rng.integers(-9,10,(4,4))
    B = rng.integers(-9,10,(4,4))
    print(np.max(np.abs(A@B-B@A)))
```

The executed outputs are $163, 136, 158, 156$. Here $\max|M|$ means the largest absolute entry, not a determinant. Each positive result proves that that particular commutator $AB - BA$ is nonzero, so $AB \neq BA$ in all four trials. One counterexample already disproves a universal commutativity claim. The trials do not prove that matrices *never* commute: I commutes with every square matrix.

Exercise 2.1.38: A valid and an invalid difference-of-squares identity

Use integer arrays to avoid roundoff in testing these polynomial expressions:

```
rng = np.random.default_rng(21038)
for j in range(3):
    A = rng.integers(-9,10,(5,5)); I = np.eye(5,dtype=int)
    print(np.max(np.abs((A+I)@(A-I)-(A@A-I))))
for j in range(3):
    A = rng.integers(-9,10,(4,4))
    B = rng.integers(-9,10,(4,4))
    print(np.max(np.abs((A+B)@(A-B)-(A@A-B@B))))
```

The first three outputs are $0, 0, 0$; the next three are $182, 188, 189$. Algebra explains the difference:

$$(A + I)(A - I) = A^2 - A + A - I = A^2 - I,$$

whereas

$$(A + B)(A - B) = A^2 - B^2 + BA - AB.$$

Thus the latter proposed identity holds precisely when $AB = BA$; the experiments found three pairs for which this extra condition fails.

Exercise 2.1.39: Test three transpose rules

For three random pairs, execute

```
rng = np.random.default_rng(21039)
for j in range(3):
    A = rng.integers(-9,10,(4,4)); B = rng.integers(-9,10,(4,4))
    errors = [(A+B).T-A.T-B.T, (A@B).T-B.T@A.T,
              (A@B).T-A.T@B.T]
    print([np.max(np.abs(E)) for E in errors])
```

The outputs are $(0, 0, 173)$, $(0, 0, 141)$, and $(0, 0, 223)$. Thus the sum rule and reversed-product rule pass all three exact-integer checks, while the same-order product rule fails all three. The proofs in the preceding exercises establish the first two rules for every matching pair; random tests by themselves would not do that.

Exercise 2.1.40: Powers of a shift matrix

The given S satisfies $S(x_1, x_2, x_3, x_4, x_5)^T = (x_2, x_3, x_4, x_5, 0)^T$. Each multiplication shifts entries one more place and inserts another zero. Consequently

$$S^2 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad S^3 = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

$$S^4 = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \qquad S^5 = S^6 = 0_{5\times 5}.$$

Equivalently $(S^k)_{ij} = 1$ exactly when $j = i + k$ is a valid column index. Once $k \geq 5$ there is no such position; the matrix is nilpotent.

Exercise 2.1.41: Powers that settle toward an averaging matrix

For the supplied symmetric matrix A, exact rational powers followed by decimal conversion give, for example,

$$A^5 \approx \begin{bmatrix} .333912 & .334877 & .331211 \\ .334877 & .335134 & .329990 \\ .331211 & .329990 & .338799 \end{bmatrix}, \quad A^{10} \approx \begin{bmatrix} .333341 & .333344 & .333315 \\ .333344 & .333350 & .333306 \\ .333315 & .333306 & .333379 \end{bmatrix}.$$

By A^{20} and A^{30} every entry rounds to 0.333333 at six decimal places. The observed behavior is convergence toward the matrix with every entry $1/3$: repeated multiplication increasingly averages the coordinates. For a reproducible numerical calculation, enter the fractions rather than rounded initial entries and use `np.linalg.matrix_power(A,k)`. As an optional explanation using later eigenvalue theory, A has eigenvalues $1, (1+\sqrt{13})/12, (1-\sqrt{13})/12$. The last two have magnitude less than 1 and their powers tend to zero; the surviving eigenspace is the line spanned by $(1,1,1)^T$. Its orthogonal projection is exactly the averaging matrix. This explains the trend rather than extrapolating from a few powers.

2.2 The inverse of a matrix

An inverse is a *two-sided* undoing operation: for a square matrix, $AB = BA = I$ means $B = A^{-1}$. Multiplication order matters. In particular, $(AB)^{-1} = B^{-1}A^{-1}$ when both inverses exist. For a 2×2 matrix,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad \text{provided } ad - bc \neq 0.$$

For larger matrices, apply the *same row operations* to both halves of $[A \mid I]$. If the left half becomes I, the right half is A^{-1}. If a pivot is missing on the left, the inverse does not exist. To solve $Ax = b$, one usually row reduces $[A \mid b]$ rather than calculating an entire inverse; both routes are explained where requested below.

Practice problem 2.2.P1: Three determinant tests

Compute the scalar $ad - bc$ in each case:

(a) For $\begin{bmatrix} 3 & -9 \\ 2 & 6 \end{bmatrix}$ it is $3 \cdot 6 - (-9)2 = 36 \neq 0$: invertible.

(b) For $\begin{bmatrix} 4 & -9 \\ 0 & 5 \end{bmatrix}$ it is $4 \cdot 5 - (-9)0 = 20 \neq 0$: invertible.

(c) For $\begin{bmatrix} 6 & -9 \\ -4 & 6 \end{bmatrix}$ it is $36 - 36 = 0$: not invertible. Indeed row 2 is $-2/3$ times row 1.

Practice problem 2.2.P2: Detect singularity while seeking an inverse

Begin with the augmented matrix and eliminate its first column:

$$\left[\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ -1 & 5 & 6 & 0 & 1 & 0 \\ 5 & -4 & 5 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3 - 5R_1]{R_2 \leftarrow R_2 + R_1} \left[\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 0 & 3 & 5 & 1 & 1 & 0 \\ 0 & 6 & 10 & -5 & 0 & 1 \end{array}\right].$$

Now $R_3 \leftarrow R_3 - 2R_2$ makes row 3 equal to $[0\ 0\ 0 \mid -7\ -2\ 1]$. The coefficient half has only two pivots, so it cannot reduce to I_3. Therefore the given matrix has no inverse. A nonzero row in the *right* half does not repair a missing pivot in A.

Exercise 2.2.1: An inverse with determinant two

Here $ad - bc = 8 \cdot 4 - 6 \cdot 5 = 2$. Swap the diagonal entries, negate the other two, and divide by 2:

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 4 & -6 \\ -5 & 8 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ -5/2 & 4 \end{bmatrix}.$$

For example, the first row of AA^{-1} is $(16-15, -24+24) = (1, 0)$, and the second is $(10-10, -15+16) = (0, 1)$. The reverse product is also I_2, either by direct multiplication or the verified 2×2 inverse formula.

Exercise 2.2.2: An inverse with determinant negative one

For $A = \begin{bmatrix} 3 & 2 \\ 8 & 5 \end{bmatrix}$, $ad - bc = 15 - 16 = -1$. Hence

$$A^{-1} = -\begin{bmatrix} 5 & -2 \\ -8 & 3 \end{bmatrix} = \begin{bmatrix} -5 & 2 \\ 8 & -3 \end{bmatrix}.$$

The negative denominator changes *every* sign, not just the diagonal. Multiplication gives diagonal entries $-15 + 16 = 1$ and $16 - 15 = 1$, and off-diagonal entries $6 - 6 = 0$ and $-40 + 40 = 0$.

Exercise 2.2.3: Keep track of a negative determinant

For $A = \begin{bmatrix} 7 & 3 \\ -6 & -3 \end{bmatrix}$, $ad - bc = -21 + 18 = -3$. Therefore

$$A^{-1} = -\frac{1}{3}\begin{bmatrix} -3 & -3 \\ 6 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -2 & -7/3 \end{bmatrix}.$$

As a check, AA^{-1} has rows $(7 - 6, 7 - 7) = (1, 0)$ and $(-6 + 6, -6 + 7) = (0, 1)$.

Exercise 2.2.4: An inverse with fractional entries

The determinant is $2(-6) - (-4)4 = 4 \neq 0$, so

$$A^{-1} = \frac{1}{4}\begin{bmatrix} -6 & 4 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} -3/2 & 1 \\ -1 & 1/2 \end{bmatrix}.$$

Fractions cause no problem: inverse entries need not be integers even when all entries of the original matrix are integers.

Exercise 2.2.5: Use the first inverse to solve a system

Multiply the right side by the inverse from Exercise 1:

$$x = A^{-1}b = \begin{bmatrix} 2 & -3 \\ -5/2 & 4 \end{bmatrix}\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 + 3 \\ -5 - 4 \end{bmatrix} = \begin{bmatrix} 7 \\ -9 \end{bmatrix}.$$

Check in the original equations: $8(7) + 6(-9) = 2$ and $5(7) + 4(-9) = -1$. Since A is invertible, this solution is unique.

Exercise 2.2.6: Solve using the inverse from Exercise 3

$$x = \begin{bmatrix} 1 & 1 \\ -2 & -7/3 \end{bmatrix}\begin{bmatrix} -9 \\ 4 \end{bmatrix} = \begin{bmatrix} -5 \\ 18 - 28/3 \end{bmatrix} = \begin{bmatrix} -5 \\ 26/3 \end{bmatrix}.$$

Substitution gives $7(-5) + 3(26/3) = -9$ and $-6(-5) - 3(26/3) = 4$. Both equations, including their signs, are satisfied.

Exercise 2.2.7: Four right sides, one set of row operations

(a) Compute the inverse and multiply each right side.

The determinant of $A = \begin{bmatrix} 1 & 2 \\ 5 & 12 \end{bmatrix}$ is $12 - 10 = 2$, so $A^{-1} = \begin{bmatrix} 6 & -1 \\ -5/2 & 1/2 \end{bmatrix}$. For any $b = (c, d)^T$, its solution is $(6c - d, (-5c + d)/2)^T$. Applying this formula to the four supplied columns gives, in order,

$$(-9, 4)^T, \quad (11, -5)^T, \quad (6, -2)^T, \quad (13, -5)^T.$$

(b) Solve the four systems together.

Use $R_2 \leftarrow R_2 - 5R_1$, then $R_2 \leftarrow R_2/2$, then $R_1 \leftarrow R_1 - 2R_2$:

$$\left[\begin{array}{cc|cccc} 1 & 2 & -1 & 1 & 2 & 3 \\ 5 & 12 & 3 & -5 & 6 & 5 \end{array}\right] \longrightarrow \left[\begin{array}{cc|cccc} 1 & 2 & -1 & 1 & 2 & 3 \\ 0 & 2 & 8 & -10 & -4 & -10 \end{array}\right]$$

$$\longrightarrow \left[\begin{array}{cc|cccc} 1 & 0 & -9 & 11 & 6 & 13 \\ 0 & 1 & 4 & -5 & -2 & -5 \end{array}\right].$$

Every column on the right has undergone the same operations as its own separate augmented system, so the final block lists exactly the four solutions.

Exercise 2.2.8: Undo a similarity expression

From $A = PBP^{-1}$, multiply on the left by P^{-1} and on the right by P:

$$P^{-1}AP = P^{-1}(PBP^{-1})P = (P^{-1}P)B(P^{-1}P) = B.$$

Thus $\boxed{B = P^{-1}AP}$. Putting both new factors on the same side would not cancel the original factors.

Exercise 2.2.9: True–false tests for inverses

(a) **True.** The definition of inverse requires $AB = BA = I$. For square matrices one equality eventually implies the other, but both equalities are still true when B is the inverse.

(b) **False.** The correct order is $(AB)^{-1} = B^{-1}A^{-1}$. For instance take $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $B = \operatorname{diag}(1, 2)$. Then $A^{-1}B^{-1}$ is $\begin{bmatrix} 1 & -1/2 \\ 0 & 1/2 \end{bmatrix}$, whereas the actual inverse is $\begin{bmatrix} 1 & -1 \\ 0 & 1/2 \end{bmatrix}$.

(c) **False.** The determinant is $ad - bc$, not $ab - cd$. The singular matrix $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has $ab - cd = 2 - 8 = -6 \neq 0$, disproving the assertion.

(d) **True.** For each b, $x = A^{-1}b$ satisfies $Ax = AA^{-1}b = b$.

(e) **True.** Reverse the underlying row operation: a swap undoes itself, nonzero scaling is undone by its reciprocal, and adding r times another row is undone by adding $-r$ times that row. The corresponding elementary matrix is the inverse.

Exercise 2.2.10: Five further inverse statements

(a) **False.** Operations taking A to I collectively multiply by A^{-1}. Applied to A^{-1} they give A^{-2}, usually not I. For $A = 2I$, dividing rows by 2 takes $A^{-1} = I/2$ to $I/4$.

(b) **True.** $A^{-1}A = AA^{-1} = I$ says precisely that A is the inverse of A^{-1}.

(c) **False.** The product is invertible, but inverses occur in reverse order; the counterexample in Exercise 9(b) applies.

(d) **True.** Solvability for each e_j lets us place solutions in columns of D with $AD = I$. Alternatively, every $b = \sum_j b_j e_j$ is an output of A by linearity, so A has a pivot in every row. Since A is square, its reduction is I, proving invertibility.

(e) **True.** This is the row-reduction characterization: applying the same operations to I produces the inverse.

Exercise 2.2.11: Solve a matrix equation and prove uniqueness

If $AX = B$, multiply on the left by A^{-1} to obtain $X = A^{-1}B$. This proves every solution must have that form. Conversely, $A(A^{-1}B) = (AA^{-1})B = B$, so that form really is a solution. The two arguments together prove existence and uniqueness. Shapes agree: $A^{-1}B$ is $n \times p$, just as X must be.

Exercise 2.2.12: A right inverse of an invertible matrix is its inverse

Multiply $AD = I$ on the left by A^{-1}:

$$D = (A^{-1}A)D = A^{-1}(AD) = A^{-1}I = A^{-1}.$$

The existing inverse of A justifies the cancellation; it is not a general rule for arbitrary rectangular or singular matrices.

Exercise 2.2.13: Cancellation and the necessity of invertibility

If $AB = AC$ and A is invertible, multiply both sides on the left by A^{-1} to get $B = C$. Without invertibility this need not hold. For example, $A = \operatorname{diag}(1, 0)$, $B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, and $C = 0$ satisfy $AB = 0 = AC$ but $B \neq C$. Thus the hypothesis performs real work.

Exercise 2.2.14: Cancel on the correct side

Starting from $(B - C)D = 0$, multiply on the *right* by D^{-1}: $B - C = (B - C)(DD^{-1}) = 0D^{-1} = 0$. Therefore $B = C$. Left multiplication would not put D next to its inverse.

Exercise 2.2.15: What the augmented reduction computes

Let the row operations be represented by elementary matrices $E_1, \ldots, E_k$, applied in that order. Their combined multiplier is $E = E_k \cdots E_1$. Reducing $[A \mid B]$ to $[I \mid X]$ means $EA = I$ and $EB = X$.

Since A is invertible, $E = EAA^{-1} = A^{-1}$, whence $X = A^{-1}B$. The right block has been transformed along with A, avoiding a separate full inverse calculation.

Exercise 2.2.16: Recover an invertible factor from its product

Put $C = AB$, which is invertible by hypothesis. Right multiplication by B^{-1} gives $A = CB^{-1}$. This is a product of two matrices whose inverses are known: C^{-1} and B. Indeed BC^{-1} is an inverse of A, because

$$(CB^{-1})(BC^{-1}) = I, \qquad (BC^{-1})(CB^{-1}) = I.$$

Thus A is invertible without assuming the desired result in advance.

Exercise 2.2.17: The inverse of three factors

Choose $D = C^{-1}B^{-1}A^{-1}$. Associativity allows the adjacent cancellations

$$(ABC)D = AB(CC^{-1})B^{-1}A^{-1} = AA^{-1} = I,$$

$$D(ABC) = C^{-1}B^{-1}(A^{-1}A)BC = C^{-1}C = I.$$

Both sides have been checked, so D is an inverse of ABC.

Exercise 2.2.18: Solve without pretending that factors commute

From $AB = BC$, right-multiply by B^{-1}: $\boxed{A = BCB^{-1}}$. Substitution checks $(BCB^{-1})B = BC$. In general BCB^{-1} is not C; cancelling the two B factors across C would be invalid.

Exercise 2.2.19: An equation with two outside inverses

Multiply $C^{-1}(A+X)B^{-1} = I$ first on the left by C, then on the right by B. This gives $A+X = CB$, hence $\boxed{X = CB - A}$. Substitution gives $C^{-1}(CB)B^{-1} = I$, proving that a solution always exists under the stated hypotheses. The rearrangement also proves uniqueness.

Exercise 2.2.20: An unknown matrix appearing inside an inverse

(a) First prove that the required inverse exists.

From $(A - AX)^{-1} = X^{-1}B$, left multiplication by X gives $B = X(A - AX)^{-1}$. Both factors on the right are invertible by hypothesis, so B is invertible.

(b) Remove the inverse, then collect the unknown on one side.

The same equality gives $B(A - AX) = X$, or $BA - BAX = X$. Therefore

$$(I + BA)X = BA.$$

This also shows $I + BA = BAX^{-1}$, a product of invertible matrices, so the next inverse is justified. Left multiplication yields

$$\boxed{X = (I + BA)^{-1}BA}.$$

The original assumptions already require an admissible solution X; we used them to prove that every inverse in this expression exists.

Exercise 2.2.21: Invertibility implies independent columns

Suppose a combination of the columns is zero, equivalently $Ax = 0$. Multiplying by A^{-1} gives $x = 0$. Thus all coefficients in the combination are zero, which is exactly linear independence.

Exercise 2.2.22: Invertibility implies spanning

For every $b \in \mathbb{R}^n$, the vector $x = A^{-1}b$ satisfies $Ax = b$. But Ax is the linear combination of A's columns with coefficients from x. Thus every b is such a combination: the columns span $\mathbb{R}^n$.

Exercise 2.2.23: No homogeneous freedom means every column is a pivot

A nonpivot column corresponds to a free variable and would produce a nontrivial solution of $Ax = 0$. The hypothesis rules out that possibility, so all n columns are pivot columns. In an $n \times n$ matrix this means all rows have pivots too. Its reduced echelon form has a leading 1 in each column and zeros elsewhere in those columns, hence is I_n. The inverse algorithm then proves that A is invertible.

Exercise 2.2.24: Every right side means every row has a pivot

If a coefficient row lacked a pivot, some right side would produce a contradictory augmented row $[0 \ \cdots \ 0 \mid 1]$. Consistency for every right side therefore forces n pivot rows. With only n columns there are also n pivot columns, so the reduced echelon form is I_n. Again the inverse algorithm proves invertibility.

Exercise 2.2.25: A zero determinant produces a nonzero null vector

What we must show. Prove singularity directly when $ad - bc = 0$. If $(a, b) \neq (0, 0)$, use $x = (-b, a)^T \neq 0$. Multiplication gives $Ax = (-ab + ba, -bc + da)^T = 0$. If $a = b = 0$ but $(c, d) \neq (0, 0)$, use $x = (-d, c)^T \neq 0$; both rows again give zero. If all four entries vanish, any nonzero x, such as $(1, 0)^T$, works. Thus both $x = 0$ and a nonzero x solve $Ax = 0$. An inverse would imply $x = A^{-1}0 = 0$, contradicting the nonzero solution. Hence A is not invertible.

Exercise 2.2.26: Verify the two-by-two formula

Let $\Delta = ad - bc \neq 0$ and $B = \Delta^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Then

$$AB = \frac{1}{\Delta} \begin{bmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{bmatrix} = I_2,$$

$$BA = \frac{1}{\Delta} \begin{bmatrix} da - bc & db - bd \\ -ca + ac & -cb + ad \end{bmatrix} = I_2.$$

Division by Δ is legal by hypothesis. Both products equal the identity, so $B = A^{-1}$ by definition.

Exercise 2.2.27: Elementary matrices select and change rows

(a) Row i of I_3 has a 1 in position i and zeros elsewhere. Consequently $(\text{row}_i\, I_3)A$ is 1 times row i of A plus zero times the other rows: it equals $\text{row}_i\, A$.

(b) Set $E = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. The three rows of EA are, by part (a), rows $2, 1, 3$ of A. Thus it swaps rows 1 and 2, exactly as the same operation on I_3 does.

(c) Set $E = \text{diag}(1, 1, 5)$. Its rows select row 1 of A, row 2 of A, and five times row 3 of A. Thus EA performs the desired scaling, and E is obtained by scaling row 3 of I_3 by 5.

Exercise 2.2.28: An elementary row replacement and a printed typo

The intended elementary matrix is

$$E = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Its rows show that EA has rows $\text{row}_1\, A$, $-3\,\text{row}_1\, A + \text{row}_2\, A$, and $\text{row}_3\, A$, as required. To construct E from I, the replacement must be $\text{row}_2(I) \leftarrow \text{row}_2(I) - 3\,\text{row}_1(I)$. The supplied edition prints $\text{row}_1(A)$ in that last construction. Read literally that is incorrect: it would make E depend on A and need not be the elementary matrix for this operation. The computation above states and verifies the corrected version explicitly.

Exercise 2.2.29: Find an inverse by the augmented algorithm

Start with $[A \mid I]$ and perform the indicated operations:

$$\left[\begin{array}{cc|cc} 1 & -3 & 1 & 0 \\ 4 & -9 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - 4R_1} \left[\begin{array}{cc|cc} 1 & -3 & 1 & 0 \\ 0 & 3 & -4 & 1 \end{array}\right].$$

Divide row 2 by 3, then add three times the new row 2 to row 1. The result is $\left[\begin{array}{cc|cc} 1 & 0 & -3 & 1 \\ 0 & 1 & -4/3 & 1/3 \end{array}\right]$.

Thus $A^{-1} = \begin{bmatrix} -3 & 1 \\ -4/3 & 1/3 \end{bmatrix}$.

Exercise 2.2.30: A second augmented inverse calculation

For $[A \mid I]$, first use $R_1 \leftarrow R_1/3$, then $R_2 \leftarrow R_2 - 4R_1$:

$$\left[\begin{array}{cc|cc} 3 & 6 & 1 & 0 \\ 4 & 7 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{cc|cc} 1 & 2 & 1/3 & 0 \\ 0 & -1 & -4/3 & 1 \end{array}\right].$$

Negate row 2 and replace row 1 by $R_1 - 2R_2$. This gives

$$[I \mid A^{-1}] = \left[\begin{array}{cc|cc} 1 & 0 & -7/3 & 2 \\ 0 & 1 & 4/3 & -1 \end{array}\right].$$

Both left-hand pivots exist, so the inverse is the displayed right block.

Exercise 2.2.31: A full three-by-three inverse

Eliminate below the first pivot of $[A \mid I]$ with $R_2 \leftarrow R_2 + 3R_1$, $R_3 \leftarrow R_3 - 2R_1$, then $R_3 \leftarrow R_3 + 3R_2$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ -3 & 1 & 4 & 0 & 1 & 0 \\ 2 & -3 & 4 & 0 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{array}\right].$$

Next divide row 3 by 2 and add twice this new row to each of rows 1 and 2. The coefficient block is now I_3, and the right block is

$$\boxed{A^{-1} = \begin{bmatrix} 8 & 3 & 1 \\ 10 & 4 & 1 \\ 7/2 & 3/2 & 1/2 \end{bmatrix}}.$$

For instance the first row $[1, 0, -2]$ of A times this block is $[8, 3, 1] - 2[7/2, 3/2, 1/2] = [1, 0, 0]$; the other two rows similarly check the identity. Exact multiplication verifies all nine entries.

Exercise 2.2.32: A failed inverse algorithm is useful information

Apply $R_2 \leftarrow R_2 + 4R_1$ and $R_3 \leftarrow R_3 + 2R_1$ to $[A \mid I]$:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ -4 & -7 & 3 & 0 & 1 & 0 \\ -2 & -6 & 4 & 0 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 4 & 1 & 0 \\ 0 & -2 & 2 & 2 & 0 & 1 \end{array}\right].$$

Now $R_3 \leftarrow R_3 + 2R_2$ gives a zero coefficient row, with right block row $[10, 2, 1]$. The coefficient block has only two pivots and cannot become I_3. Therefore the inverse does not exist. Stopping here is a conclusion about singularity, not an unfinished calculation.

Exercise 2.2.33: Invert a cumulative-sum matrix

For the 3×3 lower-triangular matrix of ones, apply $R_3 \leftarrow R_3 - R_2$ first and then $R_2 \leftarrow R_2 - R_1$ to $[A \mid I]$. The order ensures that the subtracted rows are still the original rows. The left block becomes I and the right becomes

$$B_3 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}.$$

For the 4×4 case, work upward with $R_4 \leftarrow R_4 - R_3$, $R_3 \leftarrow R_3 - R_2$, $R_2 \leftarrow R_2 - R_1$:

$$B_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

In general B has 1 on the diagonal, -1 immediately below it, and zero elsewhere. For $j < n$, column j of AB is $Ae_j - Ae_{j+1}$. The two cumulative columns agree below row j, and differ by 1 exactly in row j, so this is e_j. The last column is $Ae_n = e_n$. Thus $AB = I_n$; the same upward row-reduction construction shows B is indeed the inverse for every n.

Exercise 2.2.34: A scaled cumulative-sum matrix

Let L be the lower-triangular ones matrix and $D = \operatorname{diag}(1, 2, \ldots, n)$. The given A is DL, because row i consists of i's up through its diagonal entry. Thus its inverse is $B = L^{-1}D^{-1}$. Explicitly,

$$b_{ii} = 1/i, \qquad b_{i,i-1} = -1/(i-1) \ (i \geq 2),$$

with every other entry zero. For example, $B_3 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1/2 & 0 \\ 0 & -1/2 & 1/3 \end{bmatrix}$; B_4 appends a last row $(0, 0, -1/3, 1/4)$ to the corresponding four-column pattern. For $j < n$, column j of AB is $(Ae_j - Ae_{j+1})/j = e_j$, since the two columns differ only in row j, where their difference is j. The last column is $Ae_n/n = e_n$. Hence $AB = I_n$, verifying the formula.

Exercise 2.2.35: Find one inverse column without the others

The third column z of A^{-1} satisfies $Az = e_3$, since $AA^{-1} = I$. Its equations are

$$-z_1 - 7z_2 - 3z_3 = 0, \quad 2z_1 + 15z_2 + 6z_3 = 0, \quad z_1 + 3z_2 + 2z_3 = 1.$$

Add twice the first equation to the second: $z_2 = 0$. Add the first to the third: $-4z_2 - z_3 = 1$, so $z_3 = -1$. The first then gives $z_1 = 3$. Thus the requested column is $\boxed{(3, 0, -1)^T}$. Substitution verifies all three equations. The coefficient matrix has pivots in all three columns by this elimination, so the use of its inverse is justified.

Exercise 2.2.36: Two inverse columns in one augmented system

Solve $Az = e_2$ and $Aw = e_3$ together. No first inverse column is needed. Start with $[A \mid e_2 \ e_3]$. To avoid early fractions, multiply row 2 by 25 and add 536 times row 1; multiply row 3 by 25 and add 154 times row 1. These are compositions of legal elementary operations:

$$\left[\begin{array}{ccc|cc} -25 & -9 & -27 & 0 & 0 \\ 536 & 185 & 537 & 1 & 0 \\ 154 & 52 & 143 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|cc} -25 & -9 & -27 & 0 & 0 \\ 0 & -199 & -1047 & 25 & 0 \\ 0 & -86 & -583 & 0 & 25 \end{array}\right].$$

Replacing row 3 by $199R_3 - 86R_2$ gives its coefficient row $[0, 0, -25975]$ and right side $[-2150, 4975]$. Back-substitution gives

$$z = \frac{1}{1039}\begin{bmatrix} 117 \\ -583 \\ 86 \end{bmatrix}, \qquad w = \frac{1}{1039}\begin{bmatrix} -162 \\ 1047 \\ -199 \end{bmatrix}.$$

For example $z_3 = 2150/25975 = 86/1039$; the preceding row determines z_2, and the first row determines z_1. The same steps give w. All three triangular pivots are nonzero; exact substitution gives $Az = e_2$ and $Aw = e_3$ and verifies both requested columns.

Exercise 2.2.37: A rectangular left inverse need not be a right inverse

For the supplied 3×2 matrix, choose

$$C = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 0 \end{bmatrix}.$$

All entries are allowed. In CA, its first row gives $[1,2] + [1,3] - [1,5] = [1,0]$; the second gives $-[1,2] + [1,3] = [0,1]$. Thus $CA = I_2$. In the other order,

$$AC = \begin{bmatrix} -1 & 3 & -1 \\ -2 & 4 & -1 \\ -4 & 6 & -1 \end{bmatrix} \neq I_3.$$

The unequal dimensions prevent a general reversal of the identity.

Exercise 2.2.38: A rectangular right inverse

The first and fourth columns of the given A are e_1 and e_2. Select them by taking

$$D = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Its entries are only $0, 1$, and $AD = [e_1\ e_2] = I_2$. There cannot be a 4×2 matrix C with $CA = I_4$. Indeed the nonzero vector $(0,1,1,0)^T$ satisfies $Ax = 0$. If $CA = I_4$, then $x = CAx = C0 = 0$, a contradiction.

Exercise 2.2.39: Convert forces into deflections

The model is $y = Df$: flexibility times force gives deflection. With $f = (40, 50, 30)^T$ pounds, the three entries are

$$y_1 = .011(40) + .003(50) + .001(30) = .62,$$

$$y_2 = .003(40) + .009(50) + .003(30) = .66, \qquad y_3 = .001(40) + .003(50) + .011(30) = .52.$$

Thus the deflections are $\boxed{(.62, .66, .52)^T}$ inches. Multiplying units, inches per pound times pounds gives inches.

Exercise 2.2.40: Stiffness and a prescribed third deflection

Row reduction, or exact rational arithmetic on the decimal entries, gives

$$D^{-1} = \begin{bmatrix} 100 & -100/3 & 0 \\ -100/3 & 400/3 & -100/3 \\ 0 & -100/3 & 100 \end{bmatrix}.$$

To check rather than trust the calculation, write $D = 10^{-3}\begin{bmatrix}11 & 3 & 1\\ 3 & 9 & 3\\ 1 & 3 & 11\end{bmatrix}$. Its first row times the displayed stiffness matrix gives $[1100-100, -1100/3+400-100/3, -100+100] = [1000, 0, 0]$; the other rows give $[0, 1000, 0]$ and $[0, 0, 1000]$. The factor 10^{-3} therefore makes the product I. For $y = .04e_3$, use only column 3 of D^{-1}:

$$f = D^{-1}y = .04(0, -100/3, 100)^T = (0, -4/3, 4)^T \text{ pounds.}$$

The negative second force points opposite to the chosen positive direction; it is needed to keep the second deflection at zero.

Exercise 2.2.41: Four measured deflections determine four forces

Let $S = 1000D$, an integer matrix, and use the measured y in centimeters. Then $Df = y$ is equivalent to

$$\begin{bmatrix}13 & 5 & 2 & 1\\ 5 & 10 & 4 & 2\\ 2 & 4 & 10 & 5\\ 1 & 2 & 5 & 13\end{bmatrix} f = \begin{bmatrix}70\\ 120\\ 160\\ 120\end{bmatrix}.$$

Exact row reduction produces

$$[I \mid f], \qquad \boxed{f = (20/21, 130/21, 80/7, 80/21)^T} \text{ newtons.}$$

For a transparent check with integers, write $f = (20, 130, 240, 80)^T/21$. Multiplying by S gives $(1470, 2520, 3360, 2520)^T/21 = (70, 120, 160, 120)^T$, which verifies all four equations. The inverse used in this computation is

$$D^{-1} = \frac{1000}{21}\begin{bmatrix}2 & -1 & 0 & 0\\ -1 & 3 & -1 & 0\\ 0 & -1 & 3 & -1\\ 0 & 0 & -1 & 2\end{bmatrix}.$$

Multiplication by the given D gives I_4, so the forces are unique. The form of this inverse makes the next requested calculation especially simple.

Exercise 2.2.42: Prescribe only the second deflection

A unit deflection e_2 requires $D^{-1}e_2$, precisely column 2 of the stiffness matrix. By linearity, a deflection $.22e_2$ requires .22 times that column. Using the inverse from Exercise 41,

$$f = .22\frac{1000}{21}(-1, 3, -1, 0)^T = \boxed{(-220/21, 220/7, -220/21, 0)^T} \text{ newtons.}$$

Multiplication by D gives $(0, .22, 0, 0)^T$ centimeters. The last force is zero, while two forces are negative: forcing a single deflection does not mean that only one applied force is needed.

2.3 Characterizations of invertible matrices

The Invertible Matrix Theorem (IMT) is a collection of equivalent tests for an $n \times n$ matrix. Invertibility, n pivots, independent columns, columns spanning $\mathbb{R}^n$, a trivial null space, and a one-to-one and onto map $x \mapsto Ax$ all describe the same situation. The word *square* is essential: a rectangular matrix can have independent columns without spanning its whole codomain.

For implications, distinguish a proof from a plausible example. A true implication needs an argument for *every* allowed matrix; a false implication needs just one counterexample with true premise and false conclusion. The zero matrix and identity matrix are useful tests, but their special properties should not be mistaken for universal properties.

Practice problem 2.3.P1: Repeated columns prevent invertibility

The three columns of the supplied matrix are $(2,2,2)^T$, $(3,3,3)^T$, and $(4,4,4)^T$. In particular column 2 is $3/2$ times column 1, so the columns are dependent. The IMT says an invertible square matrix has independent columns. Therefore this matrix is not invertible.

Practice problem 2.3.P2: Negate an existence statement correctly

Statement (g) says: for *every* $b \in \mathbb{R}^n$ there exists a solution to $Ax = b$. Its negation is: there exists *at least one* b for which $Ax = b$ is inconsistent. It does not say every system is inconsistent; indeed $Ax = 0$ always has $x = 0$ as a solution. The IMT also implies that this A is not invertible.

Practice problem 2.3.P3: Recognize a noninvertible product

The matrix AB is square. A nontrivial solution of $(AB)x = 0$ violates the trivial-null-space condition in the IMT, so AB is not invertible. We cannot conclude that *both* A and B are singular; one singular factor suffices to make a product singular. For instance $A = 0$, $B = I$ has exactly one singular factor.

Exercise 2.3.1: A two-by-two determinant test

For $A = \begin{bmatrix} 5 & 7 \\ -3 & -6 \end{bmatrix}$, $ad - bc = -30 + 21 = -9 \neq 0$. The two-by-two inverse criterion from Section 2.2 therefore gives **invertible**. No complete inverse calculation is needed.

Exercise 2.3.2: A multiple row

The second row $(6, -3)$ is $-3/2$ times the first row $(-4, 2)$. Eliminating it leaves only one pivot, fewer than two. Thus the matrix is **not invertible**. Equivalently the determinant is $12 - 12 = 0$.

Exercise 2.3.3: A triangular matrix with nonzero diagonal

The matrix is lower triangular with diagonal $3, -4, -3$. The first pivot 3 removes the entries below it; the second pivot -4 removes the next entry below it; the last pivot remains -3. None vanishes.

There is a pivot in every row and column, so the IMT gives **invertible**. The triangular structure saves a full reduction.

Exercise 2.3.4: A zero row

The second row is all zeros. Thus there can be at most two pivot rows in this 3×3 matrix. Since invertibility would require three pivots, the matrix is **not invertible**.

Exercise 2.3.5: A zero column

Column 2 is zero, so $Ae_2 = 0$ with $e_2 \neq 0$. This is a nontrivial null vector. The IMT therefore gives **not invertible**; equivalently the columns are dependent because one of them is the zero vector.

Exercise 2.3.6: A short elimination gives the last pivot

Use $R_3 \leftarrow R_3 + 3R_1$ to obtain the last row $(0, -3, -18)$, then $R_3 \leftarrow R_3 + (3/4)R_2$. The resulting echelon matrix is

$$\begin{bmatrix} 1 & -3 & -6 \\ 0 & 4 & 3 \\ 0 & 0 & -63/4 \end{bmatrix}.$$

Its three diagonal entries are nonzero pivots, so the original matrix is **invertible**. The operations preserve row equivalence.

Exercise 2.3.7: Four pivots from a few row replacements

Apply $R_2 \leftarrow R_2 + 3R_1$, $R_3 \leftarrow R_3 - 2R_1$, and then $R_4 \leftarrow R_4 - (1/4)R_2$. This gives

$$\begin{bmatrix} -1 & -3 & 0 & 1 \\ 0 & -4 & 8 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Every diagonal position is a pivot. Hence the original 4×4 matrix is **invertible**. There is no need to clear the entries above the pivots.

Exercise 2.3.8: Read pivots from an upper triangular matrix

The given matrix is already in echelon form with nonzero diagonal entries $3, 1, 2, 1$. It has four pivots, so it is **invertible** by the IMT. The entries above the diagonal do not change this conclusion.

Exercise 2.3.9: A computer-assisted exact pivot test

Using rational row reduction on the supplied integer matrix gives an echelon matrix

$$\begin{bmatrix} 4 & 0 & -3 & -7 \\ 0 & 9 & 9/2 & -3/2 \\ 0 & 0 & 71/4 & 365/12 \\ 0 & 0 & 0 & -1336/213 \end{bmatrix}.$$

For reproducibility, elimination here uses $R_i \leftarrow R_i - (a_{ik}/a_{kk})R_k$ below each successive pivot. All four pivots are nonzero; therefore the matrix is **invertible**. An exact program check is `Matrix(A).rref()`, using SymPy, which returns I_4 and pivot columns $(0, 1, 2, 3)$ (zero-based indices). These exact calculations avoid confusing a very small pivot with zero.

Exercise 2.3.10: Five pivots despite a small determinant

For the supplied 5×5 integer matrix, exact elimination, including one row interchange when a candidate pivot is zero, gives

$$\begin{bmatrix} 5 & 3 & 1 & 7 & 9 \\ 0 & 2/5 & 4/5 & -2/5 & -94/5 \\ 0 & 0 & 1 & -21 & 7 \\ 0 & 0 & 0 & 1 & 34 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}.$$

There are five nonzero pivots, hence the matrix is **invertible**. Its inverse can nevertheless be numerically sensitive; invertibility is an exact yes/no statement, while conditioning measures sensitivity. Exercise 43 quantifies that distinction. Exact SymPy row reduction returns I_5 with all five pivot columns.

Exercise 2.3.11: Five implications, with the square hypothesis retained

(a) **True.** A trivial null space means no free variables, so all n columns have pivots. For a square matrix that forces the reduced form to be I_n; this is an IMT equivalence.

(b) **True.** Spanning $\mathbb{R}^n$ forces n pivot rows. In an $n \times n$ matrix these are also n pivot columns, giving independence.

(c) **False.** Merely being $n \times n$ does not guarantee consistency for every right side. Take $A = 0$ and $b = e_1$; the equation $0x = e_1$ has no solution.

(d) **True.** A nontrivial homogeneous solution requires a free variable, so some column is not a pivot column. Thus there are fewer than n pivots.

(e) **True.** If A were invertible, then $(A^{-1})^T$ would be an inverse of A^T. The given noninvertibility of A^T excludes that possibility, so A is not invertible.

Exercise 2.3.12: Pay attention to onto versus into

(a) **True.** $AD = I$ is the right-inverse condition in the IMT, so A is invertible. Left multiplication by A^{-1} gives $D = A^{-1}$, and therefore $DA = I$.

(b) **False.** Every $n \times n$ matrix maps $\mathbb{R}^n$ *into* $\mathbb{R}^n$, including $A = 0$, whose reduced form is not I. Mapping *onto* would be a stronger premise and would imply the conclusion.

(c) **True.** For a square matrix, independent columns and columns spanning $\mathbb{R}^n$ are equivalent by the IMT.

(d) **False.** Consistency for every b implies invertibility, hence the transformation *is* one-to-one. The identity matrix is a concrete counterexample to the claimed negative conclusion.

(e) **False.** Consistency for just one right side says nothing about uniqueness. For $A = 0$, $b = 0$, every vector is a solution.

Exercise 2.3.13: When an upper triangular matrix is invertible

It is invertible *if and only if every diagonal entry is nonzero.* If all are nonzero, eliminate above the diagonal from the last row upward, then scale the rows; this reduces the matrix to I_n. For the converse, suppose diagonal entry k is zero. In an upper triangular matrix, each of the first $k-1$ columns has zero entries from row k downward. Column k also has zero entries from row k downward: its diagonal is zero, and its lower entries vanish by triangularity. Thus the first k columns live entirely in their first $k-1$ coordinates. A homogeneous system with k variables but only $k-1$ potentially nonzero equations has a free variable, so these k columns are dependent. Their dependence extends to all columns by assigning zero coefficients to the remaining columns. The IMT therefore excludes invertibility.

Exercise 2.3.14: The lower triangular case

Again the criterion is that every diagonal entry be nonzero. One way to avoid repeating the argument is to transpose: A is lower triangular exactly when A^T is upper triangular, with the same diagonal. The IMT says A is invertible exactly when A^T is invertible. Applying Exercise 13 to A^T proves both directions of the claimed criterion.

Exercise 2.3.15: Nonspanning columns exclude an inverse

No. The IMT says an invertible 4×4 matrix has columns spanning $\mathbb{R}^4$. The proposed failure of spanning is the negation of one of its equivalent conditions, so it rules out invertibility.

Exercise 2.3.16: Independence of the transposed columns

If A is invertible, then A^T is invertible, with inverse $(A^{-1})^T$. Applying the independent-column condition of the IMT to A^T shows that its columns are independent. Equivalently the rows of A are independent.

Exercise 2.3.17: Identical columns produce an explicit dependence

No. If columns i, j are equal with $i \neq j$, then $A(e_i - e_j) = a_i - a_j = 0$. The vector $e_i - e_j$ is nonzero, so the null space is nontrivial and A cannot be invertible.

Exercise 2.3.18: Identical rows also rule out invertibility

No. Equal rows of A become equal columns of A^T. Exercise 17 makes A^T singular, and the transpose equivalence in the IMT then makes A singular too.

Exercise 2.3.19: Seven independent columns give every right side exactly once

Since D is 7×7 and has independent columns, the IMT makes it invertible. For *every* $b \in \mathbb{R}^7$, the system $Dx = b$ therefore has the unique solution $x = D^{-1}b$. Both existence and uniqueness are part of the conclusion; neither needs a special choice of b.

Exercise 2.3.20: Universal consistency for a square system implies uniqueness

It is not possible to have more than one solution for any b. Universal consistency gives a pivot in every one of the five rows. Because there are five columns too, all five variables are pivot variables, leaving no free variable. The matrix is invertible, and every system has exactly one solution.

Exercise 2.3.21: Multiple solutions exclude spanning

If $Cu = v$ has two different solutions u_1, u_2, then $C(u_1 - u_2) = v - v = 0$ with $u_1 - u_2 \neq 0$. Thus C is singular by the IMT. Its columns therefore cannot span $\mathbb{R}^n$. Notice that the original right side v is still attained; the failure concerns some other right sides.

Exercise 2.3.22: A one-sided square inverse forces commutation

From $EF = I$, the IMT makes E invertible. Multiplying on the left by E^{-1} gives $F = E^{-1}$, so $FE = I$ as well. Hence $EF = FE$, which is exactly the definition of commuting matrices. This conclusion uses squareness; rectangular one-sided inverses need not reverse.

Exercise 2.3.23: A missed output implies a nontrivial null vector

If $Fx = y$ is inconsistent for some y, then F is not onto. By the IMT this square matrix is singular, so $Fx = 0$ has a nontrivial solution. In fact if $z \neq 0$ is such a solution, every tz, $t \in \mathbb{R}$, is a solution; thus the homogeneous system has infinitely many solutions.

Exercise 2.3.24: A failure to reduce to the identity

If G cannot be reduced to I_n, the IMT makes it singular. Consequently its columns are dependent and do not span $\mathbb{R}^n$. There are fewer than n pivots. These are equivalent descriptions of the same failure, not alternative cases to choose between.

Exercise 2.3.25: Verify the boxed fact about a one-sided inverse

Assume square matrices A, B satisfy $AB = I$. Then A has a right inverse, so the IMT implies that A is invertible. The equation gives $B = A^{-1}AB = A^{-1}$. Therefore B is invertible too, with inverse A,

and $BA = I$. This proves every assertion in the boxed statement preceding Example 1, rather than merely assuming a right inverse is automatically two-sided.

Exercise 2.3.26: Independent columns lead to a spanning square

Independence of the columns of square A makes A invertible. The product $A^2 = AA$ is then invertible, with inverse $A^{-1}A^{-1}$. Applying the spanning condition of the IMT to A^2 proves that its columns span $\mathbb{R}^n$. Equivalently, A maps onto $\mathbb{R}^n$ and doing so twice still reaches every vector.

Exercise 2.3.27: An invertible product forces the first factor invertible

Let $W = (AB)^{-1}$. Then $A(BW) = (AB)W = I$. Thus A has a square right inverse BW, and the IMT makes A invertible. This proof does *not* apply the inverse-of-a-product formula to unverified inverses of A, B; their existence is what must be proved.

Exercise 2.3.28: An invertible product forces the second factor invertible

Suppose $Bx = 0$. Multiplying by A gives $(AB)x = 0$. Invertibility of AB implies $x = 0$. Hence B has trivial null space, and since B is square the IMT proves it is invertible. Alternatively, $(WA)B = I$ with $W = (AB)^{-1}$ gives a left-inverse test.

Exercise 2.3.29: One-to-one endomorphisms are also onto

The one-to-one condition says $Ax = 0$ has only $x = 0$, or equivalently the columns are independent. For $n \times n$ matrices the IMT makes A invertible and hence onto $\mathbb{R}^n$. Thus the transformation is a bijection with inverse $x \mapsto A^{-1}x$. The equal dimensions are crucial.

Exercise 2.3.30: Failure of injectivity also forces failure of surjectivity

Two solutions of $Ax = b$ give a nonzero homogeneous solution by subtraction. Therefore A is singular. By the IMT the associated transformation is not onto $\mathbb{R}^n$ either: at least one right side has no preimage. It can still attain the particular b mentioned in the premise.

Exercise 2.3.31: Prove uniqueness directly from pivot counting

What we must show. Use row reduction, without Theorems 5 or 8. Consistency for all b forces a pivot in every row; otherwise an appropriate right side creates an inconsistent zero-coefficient row. The square matrix has n rows and n columns, so these n pivots use all n columns. There are no free variables. Thus, for any fixed b, back-substitution determines each variable uniquely. The hypothesis gives existence, and this pivot argument gives uniqueness, as requested.

Exercise 2.3.32: Prove existence directly from a trivial null space

What we must show. Avoid invoking the IMT. If a column were not a pivot column, its variable could be chosen freely in the homogeneous system, producing a nonzero solution. Thus all n columns

have pivots. The n pivots occupy different rows, and there are only n rows, so every row has a pivot. Hence for every augmented right side the system is consistent: no zero coefficient row can create a contradiction. This gives a solution of $Ax = b$ for every b directly.

Exercise 2.3.33: Invert a two-dimensional linear transformation

The standard matrix is $A = \begin{bmatrix} -5 & 9 \\ 4 & -7 \end{bmatrix}$, whose determinant is $35 - 36 = -1$. Thus $A^{-1} = \begin{bmatrix} 7 & 9 \\ 4 & 5 \end{bmatrix}$. For an output (y_1, y_2), its unique input is

$$\boxed{T^{-1}(y_1, y_2) = (7y_1 + 9y_2,\ 4y_1 + 5y_2)}.$$

Multiplying either pair of matrices gives I_2, which verifies both compositions $T^{-1} \circ T$ and $T \circ T^{-1}$ are the identity.

Exercise 2.3.34: A second inverse transformation

Now $A = \begin{bmatrix} 2 & -8 \\ -2 & 7 \end{bmatrix}$ has determinant $14 - 16 = -2$, so $A^{-1} = \begin{bmatrix} -7/2 & -4 \\ -1 & -1 \end{bmatrix}$. Consequently

$$\boxed{T^{-1}(y_1, y_2) = (-\tfrac{7}{2}y_1 - 4y_2,\ -y_1 - y_2)}.$$

For example applying T to this pair gives first coordinate $-7y_1 - 8y_2 + 8y_1 + 8y_2 = y_1$ and second coordinate $7y_1 + 8y_2 - 7y_1 - 7y_2 = y_2$. The reverse composition follows from the same inverse matrix calculation.

Exercise 2.3.35: Use both inverse identities directly

Let $S = T^{-1}$, so $S(T(x)) = x$ and $T(S(y)) = y$. If $T(u) = T(v)$, apply S to get $u = v$; thus T is one-to-one. For every desired output y, the input $S(y)$ satisfies $T(S(y)) = y$; thus T is onto. As a second proof, Theorem 9 identifies invertibility of T with invertibility of its square standard matrix, and the IMT then gives both injectivity and surjectivity.

Exercise 2.3.36: Two inputs sharing an output rule out onto

The equality $T(u) = T(v)$ for $u \neq v$ means T is not one-to-one. Its square standard matrix therefore fails one IMT condition, so it also fails the onto condition. Hence T cannot map $\mathbb{R}^n$ onto $\mathbb{R}^n$. One can also subtract to see $T(u - v) = 0$ with $u - v \neq 0$.

Exercise 2.3.37: A one-sided linear inverse in equal dimensions

Let A, B be the standard matrices of T, U. The hypothesis $T(U(x)) = x$ for all x implies $ABx = x$ for all x. Testing the standard basis vectors gives $AB = I$. The boxed result proved in Exercise 25 yields $BA = I$. Thus $U(T(x)) = BAx = x$ for every x. The conclusion depends on both linearity and the common finite dimension.

Exercise 2.3.38: Why the inverse function is unique

Since T is invertible, it is onto. Given arbitrary $v \in \mathbb{R}^n$, choose x with $T(x) = v$. Then

$$S(v) = S(T(x)) = x = U(T(x)) = U(v).$$

Every v was allowed, so $S = U$ as functions. We did not need to assume that S, U were linear; the surjectivity of T ensured their arguments covered the entire domain.

Exercise 2.3.39: An onto linear map in equal dimensions has an onto inverse

By the IMT applied to its standard $n \times n$ matrix, an onto T is invertible. For any x, taking $y = T(x)$ gives $T^{-1}(y) = x$; therefore T^{-1} is onto. If $T^{-1}(y_1) = T^{-1}(y_2)$, apply T to obtain $y_1 = y_2$; therefore it is one-to-one as well. So both T and its inverse are bijections of $\mathbb{R}^n$.

Exercise 2.3.40: Prove linearity of the inverse without assuming it

Let $x = S(u)$ and $y = S(v)$. The identity $T \circ S = \mathrm{id}$ gives $T(x) = u$, $T(y) = v$. By linearity of T, $T(x + y) = u + v$. Applying S and using $S \circ T = \mathrm{id}$ yields

$$S(u + v) = S(T(x + y)) = x + y = S(u) + S(v).$$

Likewise $T(cx) = cT(x) = cu$ gives $S(cu) = S(T(cx)) = cx = cS(u)$. These are additivity and homogeneity, so S is linear. Both inverse identities were used explicitly.

Exercise 2.3.41: Tiny data perturbations, large solution changes

(a) Solve both systems exactly.

For either right side (b_1, b_2), the inverse of the coefficient matrix is

$$A^{-1} = \frac{1}{4.5(1.1) - 3.1(1.6)} \begin{bmatrix} 1.1 & -3.1 \\ -1.6 & 4.5 \end{bmatrix} = \begin{bmatrix} -110 & 310 \\ 160 & -450 \end{bmatrix},$$

because $4.95 - 4.96 = -.01$. For the unrounded right side,

$$x_1 = -110(19.249) + 310(6.843) = 3.94, \qquad x_2 = 160(19.249) - 450(6.843) = .49.$$

For the rounded right side, the same products give $(2.90, 2.00)^T$. All these decimals are exact terminating decimals, not numerical approximations.

(b) Specify the meaning of percentage error.

Coordinatewise relative percentage errors are

$$100\frac{|2.90 - 3.94|}{3.94} = \frac{5200}{197}\% \approx 26.40\%, \qquad 100\frac{|2.00 - .49|}{.49} = \frac{15100}{49}\% \approx 308.16\%.$$

If a single vector error is wanted instead, the Euclidean relative error is $100\|\widetilde{x} - x\|_2/\|x\|_2 \approx 46.18\%$. Stating the norm matters: these are different legitimate summaries.

(c) Compute and interpret conditioning.

Executing `np.linalg.cond(A)` in NumPy gives the spectral (2-norm) condition number $\kappa_2(A) \approx 3362.9997$. The 1-norm and infinity-norm condition numbers both equal 4636; software conventions therefore need to be identified. The large value explains why very small changes of the measured right side can cause large changes in the recovered solution.

Exercise 2.3.42: Measure roundoff for the matrix in Exercise 9

What we must show. Compare a known vector with a solution recovered from its computed output. The following is an executed NumPy experiment using binary64 arithmetic:

```
import numpy as np
A = np.array([[4,0,-3,-7],[-6,9,9,9],
              [7,-5,10,19],[-1,2,4,-1]],dtype=float)
x = np.random.default_rng(23042).random(4)
b = A@x
x1 = np.linalg.solve(A,b)
print(np.linalg.cond(A))
print(np.max(np.abs(x1-x))/np.max(np.abs(x)))
print(np.abs((x1-x)/x))
```

The condition number is 11.3525089, and the normwise relative error $\|x_1 - x\|_\infty / \|x\|_\infty$ is approximately $4.52 \cdot 10^{-16}$. The chosen vector, rounded here for readability, is

$$x \approx (.98344656, .12402037, .75512517, .48561218)^T.$$

The worst coordinatewise relative error is about $2.13 \cdot 10^{-15}$; one coordinate was recovered exactly in this run.

Binary64 has 53 binary significant bits, about 15–16 decimal significant digits. Thus the observed normwise agreement is roughly 15 digits, with roughly 14–15 in the most sensitive coordinate. The expected worst-case loss suggested by $\log_{10} \kappa_2(A)$ is about 1.06 digits. The measured error is an observation for this vector, not a promise that every possible right side has identical accuracy. Libraries and hardware may alter the last few floating-point bits.

Exercise 2.3.43: Repeat with a more sensitive matrix

Use the same procedure but with the matrix in Exercise 10 and the seed below:

```
A = np.array([[5,3,1,7,9],[6,4,2,8,-8],
              [7,5,3,10,9],[9,6,4,-9,-5],
              [8,5,2,11,4]],dtype=float)
x = np.random.default_rng(23043).random(5)
b = A@x
x1 = np.linalg.solve(A,b)
print(np.linalg.cond(A))
print(np.max(np.abs(x1-x))/np.max(np.abs(x)))
```

The executed condition number is $\kappa_2(A) \approx 68622.4548$; the normwise relative error is approximately $6.01 \cdot 10^{-12}$. Thus this run retains about 11 decimal digits normwise, losing about 4–5 of the approximately 15–16 available digits. The theoretical sensitivity indicator $\log_{10} \kappa_2(A) \approx 4.84$ is consistent with that behavior. The exact pivot calculation in Exercise 10 proved invertibility, but it did not guarantee numerical robustness.

Exercise 2.3.44: The last inverse column of a fifth-order Hilbert matrix

To obtain column 5 of H^{-1} without the other columns, solve $Hx = e_5$, where $H_{ij} = 1/(i + j - 1)$ for $1 \le i, j \le 5$. An exact rational solve gives

$$\boxed{x = (630, -12600, 56700, -88200, 44100)^T}.$$

It can be verified by multiplying the five rows of H by this column; the results are $0, 0, 0, 0, 1$ respectively. For example the first is $630 - 6300 + 18900 - 22050 + 8820 = 0$. A numerical version is

```
H = np.array([[1/(i+j+1) for j in range(5)]
              for i in range(5)])
x = np.linalg.solve(H,np.eye(5)[:,4])
print(np.linalg.cond(H))
```

Here $\kappa_2(H) \approx 476607.25 \approx 10^{5.68}$. Starting with about 15–16 decimal digits, the condition estimate allows a loss of about six digits, so about 9–10 digits is a conservative expectation. Our executed solve has largest relative coordinate error about $1.93 \cdot 10^{-12}$ against the exact column, roughly 11–12 digits of agreement. A bound on possible loss need not be reached for this particular right side.

Exercise 2.3.45: A twelfth-order Hilbert inverse can be numerically unreliable

Use a numerical inverse, rather than an exact rational one, for this experiment:

```
H = np.array([[1/(i+j+1) for j in range(12)]
              for i in range(12)])
B = np.linalg.inv(H)
R = H@B-np.eye(12)
print(np.linalg.cond(H))
print(np.max(np.abs(R)))
print(np.linalg.norm(R,np.inf))
```

In the executed NumPy binary64 run, the outputs are approximately $1.64 \cdot 10^{16}$, 0.0892, and 0.320, respectively. Thus the largest entry of the computed product's error is nearly 0.09: HB is visibly different from the identity. The computed B_{11} is approximately 142.655, whereas the exact Hilbert inverse has $(H^{-1})_{11} = 12^2 = 144$.

The exact rational Hilbert matrix is invertible, but its condition number is comparable to the reciprocal of binary64 rounding precision. Ordinary floating-point arithmetic therefore has essentially no reliable significant digits guaranteed in an inverse computation of this size. Different numerical libraries may produce somewhat different inaccurate answers; none of that contradicts exact invertibility. Using higher precision and checking against exact rational arithmetic are ways to investigate further, not reasons to silently report the floating-point product as I.

2.4 Partitioned matrices

A block is itself a matrix. Block addition and multiplication follow the ordinary row–column rules, but every block product must have matching inner dimensions. Preserve the order of factors: AB cannot in general be changed to BA. An identity block has the size needed in its own row and column, so two symbols I in one display may have different sizes.

Practice problem 2.4.P1: Invert a block row-replacement matrix

Let $T = \begin{bmatrix} I & 0 \\ A & I \end{bmatrix}$. Try reversing the addition of A times the first block row to the second:

$$S = \begin{bmatrix} I & 0 \\ -A & I \end{bmatrix}.$$

Block multiplication gives $TS = \begin{bmatrix} I & 0 \\ A - A & I \end{bmatrix} = I$. In the reverse order ST has lower-left block $-A + A = 0$ and all other identity blocks. Thus $S = T^{-1}$.

Practice problem 2.4.P2: Transpose a horizontal partition before multiplying

For $X = [X_1 \ X_2]$, its transpose is vertical:

$$X^T X = \begin{bmatrix} X_1^T \\ X_2^T \end{bmatrix} [X_1 \ X_2] = \begin{bmatrix} X_1^T X_1 & X_1^T X_2 \\ X_2^T X_1 & X_2^T X_2 \end{bmatrix}.$$

The off-diagonal blocks do not disappear; they measure cross-products between the two groups of columns.

Exercise 2.4.1: Add a multiple of one block row

The conformability assumption makes each displayed product valid.

$$\begin{bmatrix} I & 0 \\ E & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ EA + C & EB + D \end{bmatrix}$$

The lower-left block is $EA + IC = EA + C$; the upper row is unchanged.

Exercise 2.4.2: Multiply by a block diagonal matrix

The conformability assumption makes each displayed product valid.

$$\begin{bmatrix} E & 0 \\ 0 & F \end{bmatrix} \begin{bmatrix} P & Q \\ R & S \end{bmatrix} = \begin{bmatrix} EP & EQ \\ FR & FS \end{bmatrix}$$

The first block row is multiplied by E and the second by F; the zero blocks remove the other terms.

Exercise 2.4.3: Interchange two block rows

The conformability assumption makes each displayed product valid.

$$\begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} C & D \\ A & B \end{bmatrix}$$

For example the upper-left product is $0A + IC = C$. This is a block row interchange.

Exercise 2.4.4: Subtract a block multiple

The conformability assumption makes each displayed product valid.

$$\begin{bmatrix} I & 0 \\ -E & I \end{bmatrix} \begin{bmatrix} W & X \\ Y & Z \end{bmatrix} = \begin{bmatrix} W & X \\ Y - EW & Z - EX \end{bmatrix}$$

The minus sign acts on both products in the lower block row. The order EW, rather than WE, comes from the row–column rule.

Exercise 2.4.5: Equate the four blocks before choosing inverses

Multiplication of the two given matrices gives

$$\begin{bmatrix} A + BX & BY \\ C & 0 \end{bmatrix} = \begin{bmatrix} 0 & I \\ Z & 0 \end{bmatrix}.$$

Thus the exact requirements are $BX = -A$, $BY = I$, and $Z = C$. If B is square, these requirements force B invertible, and

$$X = -B^{-1}A, \qquad Y = B^{-1}, \qquad Z = C.$$

Indeed $B(-B^{-1}A) = -A$ and $BB^{-1} = I$, checking both upper blocks. More generally, if B is rectangular and has a right inverse Y, the choice $X = -YA$ still works. Without such a right inverse, $BY = I$ is impossible, so no solution exists.

Exercise 2.4.6: Invert a block lower triangular matrix

The product is

$$\begin{bmatrix} XA & 0 \\ YA + ZB & ZC \end{bmatrix} = I.$$

Assume A and C are square and invertible. Then the diagonal blocks give $X = A^{-1}$ and $Z = C^{-1}$. The lower-left equation is $YA = -C^{-1}B$, so right multiplication by A^{-1} gives

$$Y = -C^{-1}BA^{-1}.$$

Inserting these formulas makes $YA + ZB = -C^{-1}B + C^{-1}B = 0$. For square diagonal blocks, their invertibility is also necessary because $XA = I$ and $ZC = I$ imply it.

Exercise 2.4.7: A zero middle block does not contribute

The block product is

$$\begin{bmatrix} XA & XZ \\ YA+B & YZ+I \end{bmatrix} = I.$$

With A square and invertible, $XA = I$ gives $X = A^{-1}$. Then $XZ = 0$ forces $Z = 0$, and $YA + B = 0$ gives $Y = -BA^{-1}$. The remaining equation is $YZ + I = I$, which is now satisfied.

Conclusion. $X = A^{-1}$, $Y = -BA^{-1}$, and $Z = 0$ under the stated square-invertible assumption.

Exercise 2.4.8: Match three block columns

Multiplication gives

$$\begin{bmatrix} AX & AY & AZ+B \\ 0 & 0 & I \end{bmatrix} = \begin{bmatrix} I & 0 & 0 \\ 0 & 0 & I \end{bmatrix}.$$

If A is square and invertible, the first row requires

$$X = A^{-1}, \qquad Y = 0, \qquad Z = -A^{-1}B.$$

Each formula follows by multiplying its block equation on the left by A^{-1}. If A is rectangular with a right inverse X, one possible solution is instead $Y = 0, Z = -XB$; $AX = I$ checks it. An A without a right inverse cannot satisfy the first equation.

Exercise 2.4.9: Eliminate two blocks below a leading block

The second block row starts with $A_{21}B_{11} + B_{21}$. Setting it to zero and multiplying on the right by B_{11}^{-1} gives $A_{21} = -B_{21}B_{11}^{-1}$. Likewise $A_{31} = -B_{31}B_{11}^{-1}$. The second block in the new second row is consequently

$$C_{22} = A_{21}B_{12} + B_{22} = B_{22} - B_{21}B_{11}^{-1}B_{12}.$$

For completeness $C_{11} = B_{11}$, $C_{12} = B_{12}$, and $C_{32} = B_{32} - B_{31}B_{11}^{-1}B_{12}$. The middle inverse cannot be moved past its neighboring factors.

Exercise 2.4.10: Invert three block levels in order

Multiply the original matrix by the proposed inverse. Its lower blocks must satisfy

$$A + P = 0, \qquad D + R = 0, \qquad B + DP + Q = 0.$$

Thus $P = -A$, $R = -D$, and $Q = DA - B$. In the reverse product the lower-left block is $Q + RA + B = (DA - B) - DA + B = 0$, while the other off-diagonal blocks also vanish. Hence both products are the identity.

Conclusion. $P = -A$, $Q = DA - B$, and $R = -D$.

Exercise 2.4.11: Check conformability rather than only the number of blocks

(a) **True.** Corresponding blocks have equal sizes, so adding entries gives $[A_1 + B_1 \;\; A_2 + B_2]$.

(b) **False.** The drawn partitions do not specify the actual sizes. The number of columns of each A_{ij} must equal the number of rows of the corresponding B_j. For example, take all A_{ij} scalar but each B_j a 2×1 column. The displayed block pattern exists but multiplication is not conformable.

Exercise 2.4.12: Dimensions and transposes of partitions

(a) **False.** A has size $2n \times n$ and B has size $n \times 2n$. Both BA $(n \times n)$ and AB $(2n \times 2n)$ are defined. The blocks of AB are A_iB_j.

(b) **False.** Transposition interchanges block positions as well as transposing each block: $A^T = \begin{bmatrix} P^T & R^T \\ Q^T & S^T \end{bmatrix}$. The stated version puts the off-diagonal blocks in the wrong places.

Exercise 2.4.13: Invertibility of a block diagonal matrix

If B and C are invertible, block multiplication shows that $\operatorname{diag}(B^{-1}, C^{-1})$ is a two-sided inverse. Conversely suppose $\operatorname{diag}(B, C)$ is invertible. If $Bu = 0$, then the block vector $(u, 0)^T$ is in the kernel of the whole matrix; injectivity forces $u = 0$. Since B is square, the Invertible Matrix Theorem makes B invertible. The same argument with $(0, v)^T$ proves C invertible.

Conclusion. The equivalence holds in both directions.

Exercise 2.4.14: Invertibility of a block upper triangular matrix

Write $A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}$ with square diagonal blocks. If they are invertible, the two-sided inverse is

$$\begin{bmatrix} A_{11}^{-1} & -A_{11}^{-1}A_{12}A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{bmatrix}.$$

The upper-right block of the product in the original order is $-A_{12}A_{22}^{-1} + A_{12}A_{22}^{-1} = 0$; in the other order it is $A_{11}^{-1}A_{12} - A_{11}^{-1}A_{12} = 0$. For necessity, invertibility of A first forces A_{11} invertible: a nonzero u with $A_{11}u = 0$ would give kernel vector $(u, 0)^T$. If $A_{22}v = 0$ with $v \neq 0$, put $u = -A_{11}^{-1}A_{12}v$. Then $A(u, v)^T = 0$, another contradiction. Thus A_{22} also has trivial kernel and is invertible.

Exercise 2.4.15: Update an accumulated outer-product sum

The column–row rule gives

$$G_k = X_kX_k^T = \sum_{j=1}^{k} x_jx_j^T.$$

Appending the new column yields

$$G_{k+1} = [X_k \;\; x_{k+1}] \begin{bmatrix} X_k^T \\ x_{k+1}^T \end{bmatrix} = G_k + x_{k+1}x_{k+1}^T.$$

Only one new outer product is needed, followed by entrywise addition into the stored G_k. If each x_j has m entries, the new product uses m^2 scalar products of entries, rather than recomputing $k+1$ outer products. The already processed vectors need not be revisited.

Exercise 2.4.16: Derive the Schur-complement factorization

Multiply the last two factors first:

$$\begin{bmatrix} A_{11} & 0 \\ 0 & S \end{bmatrix} \begin{bmatrix} I & Y \\ 0 & I \end{bmatrix} = \begin{bmatrix} A_{11} & A_{11}Y \\ 0 & S \end{bmatrix}.$$

Multiplying the left factor then gives $\begin{bmatrix} A_{11} & A_{11}Y \\ XA_{11} & XA_{11}Y + S \end{bmatrix}$. Choose $Y = A_{11}^{-1}A_{12}$ and $X = A_{21}A_{11}^{-1}$. The lower-right block becomes $A_{21}A_{11}^{-1}A_{12} + S = A_{22}$ by the definition of S. All four blocks now agree with the required matrix.

Exercise 2.4.17: Invertibility passes to the Schur complement

The outside factors from Exercise 2.4.16 are invertible: reverse X or Y in their off-diagonal block, just as in Practice Problem 1. If $A = L\,\mathrm{diag}(A_{11}, S)U$ is invertible, then $\mathrm{diag}(A_{11}, S) = L^{-1}AU^{-1}$ is invertible. Exercise 2.4.13 implies S is invertible. Inverting the factors in reverse order also gives the explicit formula

$$A^{-1} = \begin{bmatrix} A_{11}^{-1} + A_{11}^{-1}A_{12}S^{-1}A_{21}A_{11}^{-1} & -A_{11}^{-1}A_{12}S^{-1} \\ -S^{-1}A_{21}A_{11}^{-1} & S^{-1} \end{bmatrix}.$$

This expression follows by multiplying U^{-1}, the block diagonal inverse, and L^{-1}, in that order.

Exercise 2.4.18: Append one data column to a cross-product matrix

For $W = [X\ x_0]$, transposition and block multiplication give

$$W^TW = \begin{bmatrix} X^TX & X^Tx_0 \\ x_0^TX & x_0^Tx_0 \end{bmatrix}.$$

Its Schur complement of the invertible upper-left block is the scalar

$$x_0^Tx_0 - x_0^TX(X^TX)^{-1}X^Tx_0 = x_0^T[I - X(X^TX)^{-1}X^T]x_0 = x_0^TMx_0.$$

The asserted reciprocal entry requires this scalar to be nonzero; otherwise W^TW is singular, and an inverse entry is not defined. The question's requested identity itself holds regardless of whether the scalar vanishes.

Exercise 2.4.19: Eliminate the state vector to obtain the transfer function

Put $H = A - sI_n$, assumed invertible. The top block equation is $Hx + Bu = 0$, so $x = -H^{-1}Bu$. Insert this into the bottom equation $Cx + u = y$ to obtain

$$[I_m - CH^{-1}B]u = y.$$

Thus $W(s) = I_m - C(A - sI_n)^{-1}B$. It is exactly the Schur complement of $A - sI_n$ in the original block system matrix. Its size is $m \times m$, so it acts on the input vector $u \in \mathbb{R}^m$.

Exercise 2.4.20: The inverse transfer function is another Schur complement

The Schur complement of $K = A - BC - sI_n$ in the displayed matrix is

$$I_m - (-C)K^{-1}B = I_m + C(A - BC - sI_n)^{-1}B.$$

This is the requested $W(s)^{-1}$ when the stipulated inverses exist. To check the inverse claim, write $K = H - BC$, $H = A - sI_n$. From $K = H - BC$, multiply $KK^{-1}B = B$ on the left by H^{-1} and rearrange to get the identity

$$K^{-1}B = H^{-1}B + H^{-1}BCK^{-1}B.$$

Multiplying it by C shows $CK^{-1}B = CH^{-1}B + CH^{-1}BCK^{-1}B$. Expanding $(I - CH^{-1}B)(I + CK^{-1}B)$ now cancels these three terms and gives I, confirming the formula. The invertibility of H and W also forces K invertible: if $Kv = 0$, then $v = H^{-1}BCv$ and $W(Cv) = 0$; hence $Cv = 0$ and then $v = 0$. The square-matrix theorem completes this check.

Exercise 2.4.21: Verify an involution using blocks

(a) Multiply the 2×2 matrix.

$$\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}^2 = \begin{bmatrix} 1 & 0 \\ 2-2 & 1 \end{bmatrix} = I_2.$$

(b) Recognize the four-dimensional block structure.

For this same A, the given M is $\begin{bmatrix} A & 0 \\ I_2 & -A \end{bmatrix}$. Consequently

$$M^2 = \begin{bmatrix} A^2 & 0 \\ A - A & A^2 \end{bmatrix} = I_4.$$

The lower-left cancellation is essential; squaring only the diagonal blocks would not verify the entire product.

Exercise 2.4.22: Construct a six-dimensional involution

Choose $A = B = I_2$, $D = -I_2$, and $C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \neq 0$. Then

$$M = \begin{bmatrix} I_2 & 0 & 0 \\ 0 & I_2 & 0 \\ C & 0 & -I_2 \end{bmatrix}.$$

It has size 6×6 as required. Its square has identity diagonal blocks and the only possibly nonzero off-diagonal block is $CA + DC = C - C = 0$. Thus $M^2 = I_6$. This construction works with any nonzero 2×2 choice for C.

Exercise 2.4.23: Induct on the dimension of lower triangular matrices

For dimension one the product of two scalars is lower triangular. Assume products of $k \times k$ lower triangular matrices are lower triangular. Partition two matrices of dimension $k+1$ as

$$A_1 = \begin{bmatrix} a & 0^T \\ v & A \end{bmatrix}, \qquad B_1 = \begin{bmatrix} b & 0^T \\ w & B \end{bmatrix}.$$

Their product is

$$A_1 B_1 = \begin{bmatrix} ab & 0^T \\ vb + Aw & AB \end{bmatrix}.$$

The top-right block is zero and AB is lower triangular by the induction hypothesis. These facts account for every entry above the main diagonal. Thus the product in dimension $k+1$ is lower triangular, completing the induction.

Exercise 2.4.24: Invert cumulative sums by successive differences

Let A_n be the lower triangular matrix of ones and B_n the matrix with ones on the diagonal and -1 immediately below it. For $n = 2$, direct multiplication of $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$ in either order gives I_2. Suppose $A_k B_k = B_k A_k = I_k$. With $\mathbf{1}$ the k-entry ones vector and $e_1 = (1, 0, \ldots, 0)^T$, partition

$$A_{k+1} = \begin{bmatrix} 1 & 0^T \\ \mathbf{1} & A_k \end{bmatrix}, \quad B_{k+1} = \begin{bmatrix} 1 & 0^T \\ -e_1 & B_k \end{bmatrix}.$$

The lower-left block of $A_{k+1}B_{k+1}$ is $\mathbf{1} - A_k e_1 = 0$, since $A_k e_1$ is the first column, all ones. The lower-left block of the reverse product is $-e_1 + B_k \mathbf{1} = 0$: successive differences of ones are $(1, 0, \ldots, 0)^T$. The diagonal blocks are identities by induction and the upper-right blocks are zero. Both products are I_{k+1}, which proves the assertion for every $n \geq 2$.

Exercise 2.4.25: Invert the three diagonal blocks separately

The given matrix is block diagonal with blocks $B = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$, $[2]$, and $C = \begin{bmatrix} 7 & 8 \\ 5 & 6 \end{bmatrix}$. The 2×2 formula gives

$$B^{-1} = \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}, \qquad C^{-1} = \begin{bmatrix} 3 & -4 \\ -5/2 & 7/2 \end{bmatrix}.$$

Thus, without row reduction,

$$A^{-1} = \begin{bmatrix} -5 & 2 & 0 & 0 & 0 \\ 3 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 3 & -4 \\ 0 & 0 & 0 & -5/2 & 7/2 \end{bmatrix}.$$

Each product block is an identity, so the whole product is I_5.

Exercise 2.4.26: Concrete commands for extracting and inserting blocks

Using Python/NumPy's zero-based, end-exclusive slices:

(a) `A[4:10, 14:20]` displays rows 5–10 and columns 15–20. The result has size 6×6.

(b) `A[4:9, 9:19] = B` inserts the 5×10 block in rows 5–9 and columns 10–19. The first inserted entry is at row 5, column 10, as required.

(c) `scipy.linalg.block_diag(A, A.T)` constructs the desired 50×50 block diagonal matrix. Alternatively initialize a 50×50 zero array, assign A to `C[:20, :30]` and A^T to `C[20:, 30:]`. The off-diagonal blocks then already have the necessary zero entries.

These are the requested operations, not a request for unspecified data. Their output sizes follow from the slice endpoints and the given sizes.

Exercise 2.4.27: Work entirely with blocks below the size limit

Keep the large matrices on disk or as separately stored blocks; assembling a full 50×50 array would itself violate the limit.

(a) Partition both matrices into four 25×25 blocks. Compute each sum block $C_{ij} = A_{ij} + B_{ij}$ separately and store the four results as the sum's block representation.

(b) With that same partition compute $C_{ij} = A_{i1}B_{1j} + A_{i2}B_{2j}$ for $i, j \in \{1, 2\}$. Every individual multiplication and addition is 25×25, well within the 32×32 bound.

(c) Use the stipulated $20 + 30$ partition, with $b = (b_1, b_2)^T$ and $x = (x_1, x_2)^T$. Since $A_{12} = 0$, first solve the 20×20 system $A_{11}x_1 = b_1$. Compute $r = b_2 - A_{21}x_1$ (A_{21} is 30×20), then solve $A_{22}x_2 = r$, a 30×30 system. Both diagonal matrices are invertible, so both solves have unique answers. Store x_1, x_2 separately as the solution; no explicit inverse or overlarge matrix is needed.

2.5 Matrix factorizations

An LU factorization writes $A = LU$, where L is unit lower triangular and U is in echelon form. "Unit" means the diagonal entries of L are 1. In elimination, if we replace row i by $R_i - mR_j$ to clear an entry below the current pivot, record m in the corresponding lower column of L. The same row replacements change A to U and L to I; reversing them proves $A = LU$. No row interchanges are needed for the matrices factored below.

To solve $Ax = b$, first solve $Ly = b$ from top to bottom, and then $Ux = y$ from bottom to top. Reuse L, U when the right side changes; do not compute an inverse merely to solve one system.

Practice problem 2.5.P1: A rectangular matrix with only three pivot columns

The given matrix has five rows and four columns. Its first-column multipliers are $3, 1, 2, -3$. After that elimination, the second-column multipliers are $-1, 2, -3$. Column 3 then has no new pivot; the next pivot is the entry 5 in column 4 of row 3. Its multipliers are $-1, 2$. The resulting factors are

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & 0 & 0 \\ 2 & 2 & -1 & 1 & 0 \\ -3 & -3 & 2 & 0 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 2 & -4 & -2 & 3 \\ 0 & 3 & 1 & -1 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The unused fourth and fifth columns of L are those of I_5. Multiplying these displayed factors gives the original 5×4 matrix exactly. In particular, L is 5×5, not 5×4; U carries the rectangular dimensions and its last two rows are zero.

Exercise 2.5.1: Forward substitution followed by backward substitution

What we must show. Solve $Ax = b$ using the given factors.

$$A = \begin{bmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{bmatrix}, \qquad b = \begin{bmatrix} -7 \\ 5 \\ 2 \end{bmatrix}.$$

The supplied factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -5 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{bmatrix}.$$

First solve $Ly = b$ by forward substitution.

$$\begin{aligned} y_1 &= -7 \\ y_2 &= 5 - \big[(-1)(-7)\big] = -2 \\ y_3 &= 2 - \big[(2)(-7) + (-5)(-2)\big] = 6 \end{aligned}$$

Then solve $Ux = y$ by backward substitution.

$$\begin{aligned} x_3 &= \frac{6}{-1} = -6 \\ x_2 &= \frac{-2 - \big[(-1)(-6)\big]}{-2} = 4 \\ x_1 &= \frac{-7 - \big[(-7)(4) + (-2)(-6)\big]}{3} = 3 \end{aligned}$$

Also carry out ordinary elimination.

Apply these row replacements to the augmented matrix $[A \mid b]$:

$$\begin{aligned} R_2 &\leftarrow R_2 + R_1 \\ R_3 &\leftarrow R_3 - (2)R_1 \\ R_3 &\leftarrow R_3 + (5)R_2 \end{aligned}$$

They produce $[U \mid y]$, with the same U, y displayed above. The backward substitution therefore gives the same vector as the factorization method. This explicitly performs the requested second method.

Conclusion. $x = \begin{bmatrix} 3 \\ 4 \\ -6 \end{bmatrix}$. Multiplying by the original A returns the displayed b, an independent check of the solution.

Exercise 2.5.2: Forward substitution followed by backward substitution

What we must show. Solve $Ax = b$ using the given factors.

$$A = \begin{bmatrix} 2 & -6 & 4 \\ -4 & 8 & 0 \\ 0 & -4 & 6 \end{bmatrix}, \qquad b = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}.$$

The supplied factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 2 & -6 & 4 \\ 0 & -4 & 8 \\ 0 & 0 & -2 \end{bmatrix}.$$

First solve $Ly = b$ by forward substitution.

$$\begin{aligned} y_1 &= 2 \\ y_2 &= -4 - [(-2)(2)] = 0 \\ y_3 &= 6 - [(1)(0)] = 6 \end{aligned}$$

Then solve $Ux = y$ by backward substitution.

$$\begin{aligned} x_3 &= \frac{6}{-2} = -3 \\ x_2 &= \frac{0 - [(8)(-3)]}{-4} = -6 \\ x_1 &= \frac{2 - [(-6)(-6) + (4)(-3)]}{2} = -11 \end{aligned}$$

Also carry out ordinary elimination.

Apply these row replacements to the augmented matrix $[A \mid b]$:

$$\begin{aligned} R_2 &\leftarrow R_2 + (2)R_1 \\ R_3 &\leftarrow R_3 - R_2 \end{aligned}$$

They produce $[U \mid y]$, with the same U, y displayed above. The backward substitution therefore gives the same vector as the factorization method. This explicitly performs the requested second method.

Conclusion. $x = \begin{bmatrix} -11 \\ -6 \\ -3 \end{bmatrix}$. Multiplying by the original A returns the displayed b, an independent check of the solution.

Exercise 2.5.3: Forward substitution followed by backward substitution

What we must show. Solve $Ax = b$ using the given factors.

$$A = \begin{bmatrix} 2 & -4 & 2 \\ -4 & 5 & 2 \\ 6 & -9 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 6 \\ 0 \\ 6 \end{bmatrix}.$$

The supplied factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 2 & -4 & 2 \\ 0 & -3 & 6 \\ 0 & 0 & 1 \end{bmatrix}.$$

First solve $Ly = b$ by forward substitution.

$$\begin{aligned} y_1 &= 6 \\ y_2 &= 0 - \big[(-2)(6)\big] = 12 \\ y_3 &= 6 - \big[(3)(6) + (-1)(12)\big] = 0 \end{aligned}$$

Then solve $Ux = y$ by backward substitution.

$$\begin{aligned} x_3 &= 0 = 0 \\ x_2 &= \frac{12 - \big[(6)(0)\big]}{-3} = -4 \\ x_1 &= \frac{6 - \big[(-4)(-4) + (2)(0)\big]}{2} = -5 \end{aligned}$$

Conclusion. $x = \begin{bmatrix} -5 \\ -4 \\ 0 \end{bmatrix}$. Multiplying by the original A returns the displayed b, an independent check of the solution.

Exercise 2.5.4: Forward substitution followed by backward substitution

What we must show. Solve $Ax = b$ using the given factors.

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 1 & -3 & 1 \\ 3 & 7 & 5 \end{bmatrix}, \qquad b = \begin{bmatrix} 0 \\ -5 \\ 7 \end{bmatrix}.$$

The supplied factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 3 & -5 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 1 & -1 & 2 \\ 0 & -2 & -1 \\ 0 & 0 & -6 \end{bmatrix}.$$

First solve $Ly = b$ by forward substitution.

$$\begin{aligned} y_1 &= 0 \\ y_2 &= -5 - \big[(1)(0)\big] = -5 \\ y_3 &= 7 - \big[(3)(0) + (-5)(-5)\big] = -18 \end{aligned}$$

Then solve $Ux = y$ by backward substitution.

$$x_3 = \frac{-18}{-6} = 3$$
$$x_2 = \frac{-5 - [(-1)(3)]}{-2} = 1$$
$$x_1 = 0 - [(-1)(1) + (2)(3)] = -5$$

Conclusion. $x = \begin{bmatrix} -5 \\ 1 \\ 3 \end{bmatrix}$. Multiplying by the original A returns the displayed b, an independent check of the solution.

Exercise 2.5.5: Forward substitution followed by backward substitution

What we must show. Solve $Ax = b$ using the given factors.

$$A = \begin{bmatrix} 1 & -2 & -2 & -3 \\ 3 & -9 & 0 & -9 \\ -1 & 2 & 4 & 7 \\ -3 & -6 & 26 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 6 \\ 0 \\ 3 \end{bmatrix}.$$

The supplied factors are

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -3 & 4 & -2 & 1 \end{bmatrix}.$$
$$U = \begin{bmatrix} 1 & -2 & -2 & -3 \\ 0 & -3 & 6 & 0 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

First solve $Ly = b$ by forward substitution.

$$y_1 = 1$$
$$y_2 = 6 - [(3)(1)] = 3$$
$$y_3 = 0 - [(-1)(1)] = 1$$
$$y_4 = 3 - [(-3)(1) + (4)(3) + (-2)(1)] = -4$$

Then solve $Ux = y$ by backward substitution.

$$x_4 = -4 = -4$$
$$x_3 = \frac{1 - [(4)(-4)]}{2} = \frac{17}{2}$$

$$x_2 = \frac{3 - [(6)(\frac{17}{2})]}{-3} = 16$$
$$x_1 = 1 - [(-2)(16) + (-2)(\frac{17}{2}) + (-3)(-4)] = 38$$

Conclusion. $x = \begin{bmatrix} 38 \\ 16 \\ \frac{17}{2} \\ -4 \end{bmatrix}$. Multiplying by the original A returns the displayed b, an independent check of the solution.

Exercise 2.5.6: Forward substitution followed by backward substitution

What we must show. Solve $Ax = b$ using the given factors.

$$A = \begin{bmatrix} 1 & 3 & 2 & 0 \\ -2 & -3 & -4 & 12 \\ 3 & 0 & 4 & -36 \\ -5 & -3 & -8 & 49 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ -2 \\ -1 \\ 2 \end{bmatrix}.$$

The supplied factors are

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 3 & -3 & 1 & 0 \\ -5 & 4 & -1 & 1 \end{bmatrix}.$$
$$U = \begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 3 & 0 & 12 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

First solve $Ly = b$ by forward substitution.

$$y_1 = 1$$
$$y_2 = -2 - [(-2)(1)] = 0$$
$$y_3 = -1 - [(3)(1) + (-3)(0)] = -4$$
$$y_4 = 2 - [(-5)(1) + (4)(0) + (-1)(-4)] = 3$$

Then solve $Ux = y$ by backward substitution.

$$x_4 = 3 = 3$$
$$x_3 = \frac{-4}{-2} = 2$$
$$x_2 = \frac{0 - [(12)(3)]}{3} = -12$$

$$x_1 = 1 - [(3)(-12) + (2)(2)] = 33$$

Conclusion. $x = \begin{bmatrix} 33 \\ -12 \\ 2 \\ 3 \end{bmatrix}$. Multiplying by the original A returns the displayed b, an independent check of the solution.

Exercise 2.5.7: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} 2 & 5 \\ -3 & -4 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$R_2 \leftarrow R_2 + (\frac{3}{2})R_1$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 \\ -\frac{3}{2} & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 2 & 5 \\ 0 & \frac{7}{2} \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly.

Exercise 2.5.8: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} 6 & 4 \\ 12 & 5 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$R_2 \leftarrow R_2 - (2)R_1$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 6 & 4 \\ 0 & -3 \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly.

Exercise 2.5.9: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} 3 & 1 & 2 \\ -9 & 0 & -4 \\ 9 & 9 & 14 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$\begin{aligned} R_2 &\leftarrow R_2 + (3)R_1 \\ R_3 &\leftarrow R_3 - (3)R_1 \\ R_3 &\leftarrow R_3 - (2)R_2 \end{aligned}$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly.

Exercise 2.5.10: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} -5 & 0 & 4 \\ 10 & 2 & -5 \\ 10 & 10 & 16 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$\begin{aligned} R_2 &\leftarrow R_2 + (2)R_1 \\ R_3 &\leftarrow R_3 + (2)R_1 \\ R_3 &\leftarrow R_3 - (5)R_2 \end{aligned}$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -2 & 5 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} -5 & 0 & 4 \\ 0 & 2 & 3 \\ 0 & 0 & 9 \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly.

Exercise 2.5.11: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} 3 & 7 & 2 \\ 6 & 19 & 4 \\ -3 & -2 & 3 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$\begin{aligned} R_2 &\leftarrow R_2 - (2)R_1 \\ R_3 &\leftarrow R_3 + R_1 \\ R_3 &\leftarrow R_3 - R_2 \end{aligned}$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 3 & 7 & 2 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly.

Exercise 2.5.12: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} 2 & 3 & 2 \\ 4 & 13 & 9 \\ -6 & 5 & 4 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$\begin{aligned} R_2 &\leftarrow R_2 - (2)R_1 \\ R_3 &\leftarrow R_3 + (3)R_1 \\ R_3 &\leftarrow R_3 - (2)R_2 \end{aligned}$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & 2 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 2 & 3 & 2 \\ 0 & 7 & 5 \\ 0 & 0 & 0 \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly. A zero final row does not invalidate an LU factorization; the matrix need not be invertible.

Exercise 2.5.13: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} 1 & 3 & -5 & -3 \\ -1 & -5 & 8 & 4 \\ 4 & 2 & -5 & -7 \\ -2 & -4 & 7 & 5 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$\begin{aligned} R_2 &\leftarrow R_2 + R_1 \\ R_3 &\leftarrow R_3 - (4)R_1 \\ R_4 &\leftarrow R_4 + (2)R_1 \\ R_3 &\leftarrow R_3 - (5)R_2 \\ R_4 &\leftarrow R_4 + R_2 \end{aligned}$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 4 & 5 & 1 & 0 \\ -2 & -1 & 0 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 1 & 3 & -5 & -3 \\ 0 & -2 & 3 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly. A zero final row does not invalidate an LU factorization; the matrix need not be invertible.

Exercise 2.5.14: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} 1 & 3 & 1 & 5 \\ 5 & 20 & 6 & 31 \\ -2 & -1 & -1 & -4 \\ -1 & 7 & 1 & 7 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$\begin{aligned} R_2 &\leftarrow R_2 - (5)R_1 \\ R_3 &\leftarrow R_3 + (2)R_1 \\ R_4 &\leftarrow R_4 + R_1 \\ R_3 &\leftarrow R_3 - R_2 \\ R_4 &\leftarrow R_4 - (2)R_2 \end{aligned}$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 5 & 1 & 0 & 0 \\ -2 & 1 & 1 & 0 \\ -1 & 2 & 0 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 1 & 3 & 1 & 5 \\ 0 & 5 & 1 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly. A zero final row does not invalidate an LU factorization; the matrix need not be invertible.

Exercise 2.5.15: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} 2 & 0 & 5 & 2 \\ -6 & 3 & -13 & -3 \\ 4 & 9 & 16 & 17 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$R_2 \leftarrow R_2 + (3)R_1$$
$$R_3 \leftarrow R_3 - (2)R_1$$
$$R_3 \leftarrow R_3 - (3)R_2$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 2 & 3 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 2 & 0 & 5 & 2 \\ 0 & 3 & 2 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly. Because A is rectangular, L is 3×3 and U has the same 3×4 size as A.

Exercise 2.5.16: Record the elimination multipliers to construct L

Start with the given matrix

$$A = \begin{bmatrix} 2 & -3 & 4 \\ -4 & 8 & -7 \\ 6 & -5 & 14 \\ -6 & 9 & -12 \\ 8 & -6 & 19 \end{bmatrix}.$$

Use the following row replacements, in the order listed. The multiplier subtracted in each replacement is recorded below the relevant diagonal entry of L:

$$\begin{aligned} R_2 &\leftarrow R_2 + (2)R_1 \\ R_3 &\leftarrow R_3 - (3)R_1 \\ R_4 &\leftarrow R_4 + (3)R_1 \\ R_5 &\leftarrow R_5 - (4)R_1 \\ R_3 &\leftarrow R_3 - (2)R_2 \\ R_5 &\leftarrow R_5 - (3)R_2 \end{aligned}$$

The resulting factors are

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 & 0 \\ 3 & 2 & 1 & 0 & 0 \\ -3 & 0 & 0 & 1 & 0 \\ 4 & 3 & 0 & 0 & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 2 & -3 & 4 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Here the diagonal of L consists of ones and all its entries above the diagonal are zero. The displayed U is in echelon form. Direct multiplication verifies $LU = A$ exactly. A zero final row does not invalidate an LU factorization; the matrix need not be invertible. Because A is rectangular, L is 5×5 and U has the same 5×3 size as A.

Exercise 2.5.17: Invert the two triangular factors in reverse order

Use the factors from Exercise 2.5.2. Solving $Lz = e_j$ column by column (forward substitution), and $Uw = e_j$ column by column (backward substitution), gives

$$L^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -2 & -1 & 1 \end{bmatrix}, \qquad U^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{3}{4} & -2 \\ 0 & -\frac{1}{4} & -1 \\ 0 & 0 & -\frac{1}{2} \end{bmatrix}.$$

This is the augmented-inverse algorithm: reduce $[L \mid I]$ and $[U \mid I]$ to $[I \mid L^{-1}]$ and $[I \mid U^{-1}]$ using their triangular equations. Each column is checked by multiplying it by its original factor and obtaining the corresponding coordinate vector. Since $(LU)^{-1} = U^{-1}L^{-1}$, the required inverse is

$$A^{-1} = U^{-1}L^{-1} = \begin{bmatrix} 3 & \frac{5}{4} & -2 \\ \frac{3}{2} & \frac{3}{4} & -1 \\ 1 & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}.$$

Both AA^{-1} and $A^{-1}A$ equal the identity. The order of the two inverse factors matters.

Exercise 2.5.18: Invert the two triangular factors in reverse order

Use the factors from Exercise 2.5.3. Solving $Lz = e_j$ column by column (forward substitution), and $Uw = e_j$ column by column (backward substitution), gives

$$L^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}, \qquad U^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{2}{3} & 3 \\ 0 & -\frac{1}{3} & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

This is the augmented-inverse algorithm: reduce $[L \mid I]$ and $[U \mid I]$ to $[I \mid L^{-1}]$ and $[I \mid U^{-1}]$ using their triangular equations. Each column is checked by multiplying it by its original factor and obtaining the corresponding coordinate vector. Since $(LU)^{-1} = U^{-1}L^{-1}$, the required inverse is

$$A^{-1} = U^{-1}L^{-1} = \begin{bmatrix} -\frac{23}{6} & \frac{7}{3} & 3 \\ -\frac{8}{3} & \frac{5}{3} & 2 \\ -1 & 1 & 1 \end{bmatrix}.$$

Both AA^{-1} and $A^{-1}A$ equal the identity. The order of the two inverse factors matters.

Exercise 2.5.19: Invert a triangular matrix without losing triangular form

Every diagonal entry is nonzero, so forward substitution solves $Ax = b$ uniquely for every b. Hence A is invertible by the Invertible Matrix Theorem. To follow the suggested row-operation argument, clear entries below the successive diagonal pivots using replacements by earlier rows, then scale the diagonal pivots to 1. No swaps or additions of later rows into earlier rows are needed. Applying these same operations to I keeps it lower triangular: an earlier lower triangular row has zeros in every column beyond its own index, hence also beyond the index of the later row it is added to. Scaling preserves these zeros. Thus reduction of $[A \mid I]$ to $[I \mid A^{-1}]$ leaves the right block lower triangular. This proves the requested structure as well as invertibility.

Exercise 2.5.20: Reduce the unit lower factor to the identity

Because L is unit lower triangular, its diagonal entries are already 1. Clear below each diagonal using only replacements of a later row by itself minus a multiple of an earlier row. Let E be the product of these elementary row-operation matrices. Then $EL = I$. Applying the same replacements to $A = LU$ gives

$$EA = ELU = IU = U.$$

Thus only row replacements are needed, even if A or U is rectangular or singular.

Exercise 2.5.21: One sequence of row operations acts on every factorization

Let E be the product of all elementary matrices for the stated operations. Since they reduce B to I, $EB = I$; therefore

$$EA = E(BC) = (EB)C = IC = C.$$

This proves the claim for any such sequence, not merely a particular algorithm. The converse can fail: take $A = 0$, any invertible $B \neq I$, and $C = 0$. The empty sequence already leaves $A = C$, but does not change B into I.

Exercise 2.5.22: Discard the zero rows and their unused multiplier columns

For the Practice Problem factors, take the first three columns of L and the first three rows of U:

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & -1 & 1 \\ 2 & 2 & -1 \\ -3 & -3 & 2 \end{bmatrix}, \qquad C = \begin{bmatrix} 2 & -4 & -2 & 3 \\ 0 & 3 & 1 & -1 \\ 0 & 0 & 0 & 5 \end{bmatrix}.$$

They have the required sizes 5×3 and 3×4, and their product is A. To see why, write $L = [L_1 \; L_2]$ and $U = \begin{bmatrix} U_1 \\ 0 \end{bmatrix}$. Block multiplication gives $LU = L_1U_1 + L_2(0) = L_1U_1$. The same argument works for any $m \times n$ matrix whose U has exactly three nonzero rows: use an $m \times 3$ factor and a $3 \times n$ factor. Those rows are first because U is echelon.

Exercise 2.5.23: A compact representation by four outer products

(a) Write $C = [c_1 \; c_2 \; c_3 \; c_4]$ and let the rows of D be $d_1^T, \ldots, d_4^T$. The column–row rule gives $A = c_1d_1^T + c_2d_2^T + c_3d_3^T + c_4d_4^T$. Each term is an $m \times n$ outer product.

(b) A 400×100 array stores 40,000 numbers. The factors require $400(4) + 4(100) = 2{,}000$ numbers, a factor of 20 less storage, ignoring small bookkeeping costs. For a product Ax, first compute Dx (four entries) and then $C(Dx)$; storing the full matrix is unnecessary for that use.

Exercise 2.5.24: Solve a QR-factored system by an orthogonal multiplication

From $Q^TQ = I$ and the fact that Q is square, Q is invertible with inverse Q^T. Thus $QRx = b$ is equivalent to $Rx = Q^Tb$. Compute $c = Q^Tb$, then solve the upper triangular system $Rx = c$ by backward substitution. Invertibility of R guarantees a nonzero diagonal and a unique solution.

Conclusion. For every b, this process yields the unique x. It requires no explicit inverse of R.

Exercise 2.5.25: Invert a singular-value factorization with positive diagonal

The square matrices U, V are invertible because $U^TU = V^TV = I$; their inverses are U^T, V^T. Each $\sigma_j > 0$, so $D^{-1} = \operatorname{diag}(1/\sigma_1, \ldots, 1/\sigma_n)$ exists. Reverse the factor order:

$$A^{-1} = (V^T)^{-1}D^{-1}U^{-1} = VD^{-1}U^T.$$

The product check is $AA^{-1} = UDV^TVD^{-1}U^T = UU^T = I$, and the reverse order similarly gives $VV^T = I$. Strict positivity is stronger than the nonzero condition needed for the inverse, but supplies it here.

Exercise 2.5.26: Cancel neighboring change-of-basis factors in powers

Associativity and $P^{-1}P = I$ give

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD^2P^{-1},$$

$$A^3 = PD^3P^{-1}, \qquad A^k = PD^kP^{-1}.$$

The general formula follows inductively by multiplying the formula for A^k by PDP^{-1}. Since $D = \operatorname{diag}(2, 3, 1)$,

$$A^2 = P\operatorname{diag}(4, 9, 1)P^{-1}, \quad A^3 = P\operatorname{diag}(8, 27, 1)P^{-1},$$
$$A^k = P\operatorname{diag}(2^k, 3^k, 1)P^{-1}.$$

Only scalar powers of three diagonal entries need repeated powering.

Circuit convention for the next four exercises. A series resistor of resistance R has transfer matrix $S(R) = \begin{bmatrix} 1 & -R \\ 0 & 1 \end{bmatrix}$; a shunt resistor has $T(R) = \begin{bmatrix} 1 & 0 \\ -1/R & 1 \end{bmatrix}$. The state is $(v, i)^T$ and is mapped from input to output. If the signal passes through circuit 1 and then circuit 2, the total matrix is A_2A_1, so physical order is reversed in the matrix product.

Exercise 2.5.27: Two different networks with the specified input and output

First design: a series resistor followed by a shunt.

A series resistance $R = 1/2$ changes $(12, 6)^T$ to $(12 - (1/2)6, 6)^T = (9, 6)^T$. A shunt resistance $9/2$ then changes this to $(9, 6 - 9/(9/2))^T = (9, 4)^T$. Its matrix is $T(9/2)S(1/2)$.

Second design: a shunt followed by a series resistor.

A shunt resistance 6 changes the input to $(12, 6 - 12/6)^T = (12, 4)^T$. A series resistance $3/4$ then gives $(12 - (3/4)4, 4)^T = (9, 4)^T$. Its matrix is $S(3/4)T(6)$. All four resistances are positive and both calculations verify the required output, despite the different physical orders and resistance values.

Exercise 2.5.28: Add conductances for three successive shunt circuits

Multiplying two shunt matrices adds their lower-left entries. Thus three of them give

$$T(R_3)T(R_2)T(R_1) = \begin{bmatrix} 1 & 0 \\ -(1/R_1 + 1/R_2 + 1/R_3) & 1 \end{bmatrix}.$$

This is $T(R)$ with

$$R = \frac{1}{1/R_1 + 1/R_2 + 1/R_3}.$$

The positive-resistance assumption makes the denominator positive. One may also write $R = R_1R_2R_3/(R_1R_2 + R_1R_3 + R_2R_3)$.

Exercise 2.5.29: Factor the given series–shunt–series network

(a) Multiply in the physical order's reverse.

The signal passes through R_1 in series, R_2 as a shunt, and R_3 in series. Hence

$$S(R_3)T(R_2)S(R_1) = \begin{bmatrix} 1 + R_3/R_2 & -R_1 - R_3 - R_1R_3/R_2 \\ -1/R_2 & 1 + R_1/R_2 \end{bmatrix}.$$

For example, multiplying the right two factors first gives $\begin{bmatrix} 1 & -R_1 \\ -1/R_2 & 1 + R_1/R_2 \end{bmatrix}$; the final left multiplication gives the displayed first row.

(b) Match entries of the target matrix.

The lower-left entry $-1/3$ forces $R_2 = 3$. The lower-right entry $5/3$ then gives $R_1 = 2$. The upper-left entry 3 gives $R_3 = 6$. The remaining entry checks: $-2 - 6 - 2(6)/3 = -12$.

Conclusion. Use a 2-ohm series resistor, a 3-ohm shunt, then a 6-ohm series resistor. Their product is exactly the specified matrix.

Exercise 2.5.30: A shunt–series–shunt realization of the same matrix

Try a shunt of resistance a, then a series resistance b, then a shunt of resistance c. Multiplying gives

$$T(c)S(b)T(a) = \begin{bmatrix} 1 + b/a & -b \\ -1/a - (1 + b/a)/c & 1 + b/c \end{bmatrix}.$$

Match the target's upper-right entry to obtain $b = 12$. Then $1 + 12/a = 3$ gives $a = 6$, and $1 + 12/c = 5/3$ gives $c = 18$. The lower-left entry checks as $-1/6 - 3/18 = -1/3$.

Conclusion. The distinct physical design is a 6-ohm shunt, a 12-ohm series resistor, and an 18-ohm shunt, with exactly the same overall transfer matrix as in Exercise 2.5.29.

Exercise 2.5.31: Factor and solve the supplied eight-node heat plate

All omitted entries in the printed matrix are zero; the actual coefficient matrix used here is

$$A = \begin{bmatrix} 4 & -1 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 4 & -1 & -1 & 0 & 0 & 0 \\ 0 & -1 & -1 & 4 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 4 & -1 & -1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 4 & 0 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & -1 & 4 \end{bmatrix}.$$

(a) Perform elimination without row swaps.

The successive pivots are the diagonal entries of the following U. At pivot row j, subtract l_{ij} times that row from row i; the multipliers l_{ij} are precisely the displayed entries of L.

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{4} & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{4} & -\frac{1}{15} & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{4}{15} & -\frac{2}{7} & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{15}{56} & -\frac{1}{12} & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{7}{24} & -\frac{26}{89} & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{24}{89} & -\frac{208}{2415} & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{712}{2415} & -\frac{2623}{8948} & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 4 & -1 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{15}{4} & -\frac{1}{4} & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{56}{15} & -\frac{16}{15} & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{24}{7} & -\frac{2}{7} & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{89}{24} & -\frac{13}{12} & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2415}{712} & -\frac{26}{89} & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{8948}{2415} & -\frac{2623}{2415} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{30305}{8948} \end{bmatrix}.$$

Both factors have zeros more than two positions below or above their main diagonals. Exact multiplication gives $LU - A = 0_{8\times 8}$, checking every entry rather than just the diagonals.

(b) Solve using the factors.

For $b = (5, 15, 0, 10, 0, 10, 20, 30)^T$, forward substitution gives

$$y = \begin{bmatrix} 5 \\ \frac{65}{4} \\ \frac{7}{3} \\ 15 \\ \frac{15}{8} \\ \frac{10625}{712} \\ \frac{10525}{483} \\ \frac{364965}{8948} \end{bmatrix}.$$

The rule is $y_i = b_i - \sum_{j<i} l_{ij} y_j$, starting at row 1. Backward substitution uses $x_i = (y_i - \sum_{j>i} u_{ij} x_j)/u_{ii}$, starting at row 8. It gives

$$x = \frac{1}{209} \begin{bmatrix} 827 \\ 1377 \\ 886 \\ 1546 \\ 1171 \\ 1831 \\ 1967 \\ 2517 \end{bmatrix}.$$

Substitution verifies $Ax = b$ exactly. For instance row 1 gives $(4(827) - 1377 - 886)/209 = 1045/209 = 5$.

(c) Compare the inverse with the banded factors.

The exact inverse is

$$A^{-1} = \frac{1}{30305} \begin{bmatrix} 8948 & 2623 & 2864 & 1544 & 964 & 689 & 303 & 248 \\ 2623 & 8948 & 1544 & 2864 & 689 & 964 & 248 & 303 \\ 2864 & 1544 & 9912 & 3312 & 3167 & 1792 & 964 & 689 \\ 1544 & 2864 & 3312 & 9912 & 1792 & 3167 & 689 & 964 \\ 964 & 689 & 3167 & 1792 & 9912 & 3312 & 2864 & 1544 \\ 689 & 964 & 1792 & 3167 & 3312 & 9912 & 1544 & 2864 \\ 303 & 248 & 964 & 689 & 2864 & 1544 & 8948 & 2623 \\ 248 & 303 & 689 & 964 & 1544 & 2864 & 2623 & 8948 \end{bmatrix}.$$

Every entry is nonzero. Unlike L, U, it has no empty outer bands. For an $n \times n$ matrix with fixed band width, storing its factors requires a number of entries proportional to n, whereas a dense inverse

requires n^2. This example verifies the stated storage advantage with actual factors and an actual inverse.

Exercise 2.5.32: Reuse one bidiagonal factorization for four time steps

(a) Set $C = 1$ before factoring.

Then A has diagonal 3 and adjacent off-diagonal entries -1. Its factors are

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -\frac{1}{3} & 1 & 0 & 0 \\ 0 & -\frac{3}{8} & 1 & 0 \\ 0 & 0 & -\frac{8}{21} & 1 \end{bmatrix}.$$

$$U = \begin{bmatrix} 3 & -1 & 0 & 0 \\ 0 & \frac{8}{3} & -1 & 0 \\ 0 & 0 & \frac{21}{8} & -1 \\ 0 & 0 & 0 & \frac{55}{21} \end{bmatrix}.$$

Their off-diagonal nonzero entries lie on just one neighboring diagonal, so both factors are bidiagonal. Multiplication checks $LU = A$.

(b) Solve the same two triangular systems at each step.

Starting with $t_0 = (10, 15, 15, 10)^T$, repeatedly solve $Ly = t_k$ and $Ut_{k+1} = y$. The results are

k	$(t_k)_1$	$(t_k)_2$	$(t_k)_3$	$(t_k)_4$
1	7	11	11	7
2	5	8	8	5
3	18/5	29/5	29/5	18/5
4	13/5	21/5	21/5	13/5

.

A short independent check uses symmetry. For a vector $(a, b, b, a)^T$, $A(a, b, b, a)^T = (3a-b, 2b-a, 2b-a, 3a-b)^T$. Thus if (a_0, b_0) is the preceding pair, the next pair is $a = (2a_0 + b_0)/5$, $b = (a_0 + 3b_0)/5$. These formulas reproduce all four rows and verify $At_{k+1} = t_k$ at each step. The factorization is computed once, not recomputed for each new temperature distribution.

2.6 The Leontief input–output model

The production vector x must supply both intermediate consumption Cx and final demand d. Thus

$$x = Cx + d, \qquad (I - C)x = d.$$

Column j of C lists the amounts each sector supplies as inputs per unit produced in sector j. These columns need not sum to one; the remaining part includes value added. This is a different model from the closed exchange economy of Section 1.6.

The vector d is the external demand, not the total required production. When $I - C$ is invertible, $x = (I - C)^{-1}d$. For a single right side, solving $(I - C)x = d$ by elimination is often more efficient than computing an inverse. Respect the instructions against forming inverses in Exercises 2 and 3 below.

Practice problem 2.6.P1: Write each unit-consumption vector as a column

The Goods column is $(.2, .5)^T$ and the Services column is $(.4, .3)^T$. Hence

$$x = \begin{bmatrix} .2 & .4 \\ .5 & .3 \end{bmatrix} x + \begin{bmatrix} 20 \\ 30 \end{bmatrix}, \quad \text{or} \quad \begin{bmatrix} .8 & -.4 \\ -.5 & .7 \end{bmatrix} x = \begin{bmatrix} 20 \\ 30 \end{bmatrix}.$$

The first row accounts for all Goods required, and the second accounts for all Services required. This sets up the requested model.

Exercise 2.6.1: Construct the three-sector consumption matrix

Use Manufacturing, Agriculture, Services as the order of rows and columns. The input columns described in the exercise give

$$C = \begin{bmatrix} .10 & .60 & .60 \\ .30 & .20 & 0 \\ .30 & .10 & .10 \end{bmatrix}.$$

For example producing one agricultural unit requires .60 unit from Manufacturing, .20 from Agriculture, and .10 from Services. If Agriculture produces 100 units, its intermediate-demand vector is 100 times that column:

$$(60, 20, 10)^T.$$

These are the inputs needed for that production plan, not a final-demand vector or a complete equilibrium production plan for all sectors.

Exercise 2.6.2: Meet agricultural final demand without computing an inverse

The final-demand vector is $(0, 20, 0)^T$. Multiplying the equations $(I - C)x = d$ by 10 gives

$$\left[\begin{array}{rrr|r} 9 & -6 & -6 & 0 \\ -3 & 8 & 0 & 200 \\ -3 & -1 & 9 & 0 \end{array}\right].$$

Swap rows 1 and 2. Then use $R_2 \leftarrow R_2 + 3R_1$, $R_3 \leftarrow R_3 - R_1$, and $R_3 \leftarrow 2R_3 + R_2$. The result is

$$\left[\begin{array}{ccc|c} -3 & 8 & 0 & 200 \\ 0 & 18 & -6 & 600 \\ 0 & 0 & 12 & 200 \end{array}\right].$$

Back substitution gives $x_S = 50/3$, $x_A = (600+6(50/3))/18 = 350/9$, and $x_M = (8(350/9)-200)/3 = 1000/27$.

Conclusion. $x = (1000/27, 350/9, 50/3)^T$. Substitution into the original model gives final demand $(0, 20, 0)^T$ exactly.

Exercise 2.6.3: Meet manufacturing final demand by the same elimination

Now $d = (20, 0, 0)^T$, so the initial augmented right side after multiplying by 10 is $(200, 0, 0)^T$. Use the same row swap and replacements listed in Exercise 2.6.2. They produce

$$\left[\begin{array}{ccc|c} -3 & 8 & 0 & 0 \\ 0 & 18 & -6 & 200 \\ 0 & 0 & 12 & 200 \end{array}\right].$$

Thus $x_S = 50/3$, $x_A = (200 + 100)/18 = 50/3$, and $x_M = 8(50/3)/3 = 400/9$.

Conclusion. $x = (400/9, 50/3, 50/3)^T$. No inverse matrix was used.

Exercise 2.6.4: Add the production responses to two demand vectors

The new demand $(20, 20, 0)^T$ is the sum of the demands in the two previous exercises. If $(I-C)x^{(a)} = d^{(a)}$ and $(I-C)x^{(b)} = d^{(b)}$, additivity gives $(I-C)(x^{(a)}+x^{(b)}) = d^{(a)}+d^{(b)}$. Add the actual production vectors:

$$x = \begin{bmatrix} 1000/27 \\ 350/9 \\ 50/3 \end{bmatrix} + \begin{bmatrix} 400/9 \\ 50/3 \\ 50/3 \end{bmatrix} = \begin{bmatrix} 2200/27 \\ 500/9 \\ 100/3 \end{bmatrix}.$$

The elimination pivots in Exercises 2 and 3 are nonzero, so the coefficient matrix is invertible and this solution is unique.

Exercise 2.6.5: Use the inverse requested in the problem

Here

$$I - C = \begin{bmatrix} 1 & -1/2 \\ -3/5 & 4/5 \end{bmatrix}, \qquad \det(I - C) = \frac{4}{5} - \frac{3}{10} = \frac{1}{2}.$$

The 2×2 inverse formula therefore gives

$$(I - C)^{-1} = \begin{bmatrix} 8/5 & 1 \\ 6/5 & 2 \end{bmatrix}.$$

Multiplication by $(50, 30)^T$ yields

$$x = (80 + 30, 60 + 60)^T = (110, 120)^T.$$

Check intermediate demand: $Cx = (60, 90)^T$, so $Cx + d = (60 + 50, 90 + 30)^T = x$.

Exercise 2.6.6: An inverse with a different determinant

Now

$$I - C = \begin{bmatrix} 4/5 & -1/2 \\ -3/5 & 9/10 \end{bmatrix}, \qquad \det(I - C) = \frac{36}{50} - \frac{3}{10} = \frac{21}{50}.$$

Consequently

$$(I - C)^{-1} = \frac{50}{21} \begin{bmatrix} 9/10 & 1/2 \\ 3/5 & 4/5 \end{bmatrix} = \begin{bmatrix} 15/7 & 25/21 \\ 10/7 & 40/21 \end{bmatrix}.$$

With $d = (16, 12)^T$, the two coordinates are $240/7 + 100/7 = 340/7$ and $160/7 + 160/7 = 320/7$.

Conclusion. $x = (340/7, 320/7)^T$. This vector satisfies $(I - C)x = (16, 12)^T$.

Exercise 2.6.7: The effect of increasing final demand by one unit

Use the inverse from Exercise 2.6.5.

(a) For demand $(1, 0)^T$, the production vector is the first column of the inverse: $(8/5, 6/5)^T$.

(b) For demand $(51, 30)^T$, multiplication gives $x = ((8/5)51 + 30, (6/5)51 + 60)^T = (558/5, 606/5)^T$.

(c) The new demand is $(50, 30)^T + (1, 0)^T$. By distributivity of matrix multiplication, the new production is $(110, 120)^T + (8/5, 6/5)^T = (558/5, 606/5)^T$. Thus the first inverse column records the whole economy's extra production needed for the one-unit increase in sector 1's final demand.

Exercise 2.6.8: Prove superposition and interpret an inverse column

The column-sum hypothesis guarantees $I - C$ invertible for a nonnegative consumption matrix. For instance if $q < 1$ is the largest column sum, $z = Cz$ would give $\|z\|_1 \le q\|z\|_1$, forcing $z = 0$; the square-matrix theorem then gives invertibility.

(a) The given equations are $(I-C)x = d$ and $(I-C)\Delta x = \Delta d$. Adding them gives $(I-C)(x+\Delta x) = d + \Delta d$. Uniqueness makes $x + \Delta x$ the new production vector.

(b) If $\Delta d = e_1 = (1, 0, \ldots, 0)^T$, then $\Delta x = (I - C)^{-1}e_1$. Multiplication by e_1 selects column 1 of any matrix, so it selects exactly the claimed inverse column. The same interpretation applies to column j for an increase in sector j's final demand.

Exercise 2.6.9: Solve a three-sector numerical production model

Multiply $(I - C)x = d$ by 10:

$$8x_1 - 2x_2 = 400, \qquad -3x_1 + 9x_2 - 3x_3 = 600, \qquad -x_1 + 8x_3 = 800.$$

The third equation gives $x_1 = 8x_3 - 800$. Substituting into the first gives $x_2 = 32x_3 - 3400$. Insert both expressions in the middle equation:

$$-3(8x_3 - 800) + 9(32x_3 - 3400) - 3x_3 = 600.$$

Thus $261x_3 = 28800$, giving $x_3 = 3200/29$, $x_1 = 2400/29$, and $x_2 = 3800/29$.

Conclusion. $x = (2400/29, 3800/29, 3200/29)^T$; direct substitution checks all three equations.

Exercise 2.6.10: A demand increase affects every sector when every inverse entry is positive

An increase $h > 0$ in final demand for sector j is $\Delta d = he_j$. The production change is $\Delta x = h(I - C)^{-1}e_j$, namely h times inverse column j. Every entry of that column is strictly positive by hypothesis.

Conclusion. Production must rise in every sector, including sectors whose final demand did not change directly. Intermediate purchases transmit the increase throughout this model.

Exercise 2.6.11: Verify the two GDP expressions algebraically

The production equation gives $d = (I-C)x$. The dual price equation gives $v = p-C^Tp$, so transposing yields $v^T = p^T - p^TC = p^T(I - C)$. Therefore

$$p^Td = p^T(I - C)x = v^Tx.$$

This uses associativity and the transpose identity $(C^Tp)^T = p^TC$; all three expressions are scalar values. It proves the equality without assuming an inverse for either equation.

Exercise 2.6.12: An iterative inverse from finite geometric sums

By definition $D_m = I + C + \cdots + C^m$. Multiplying by C shifts every power up by one, so

$$D_{m+1} = I + CD_m, \qquad D_0 = I.$$

Equivalently $D_{m+1} - D_m = C^{m+1}$. Thus repeatedly multiply the current matrix by C and add I to obtain the next approximation. To verify the limit, the finite telescoping identity is $(I - C)D_m = I - C^{m+1}$. The hypothesis $C^m \to 0$ implies $I - C$ invertible: if $z = Cz$, then $z = C^mz \to 0$, so $z = 0$. Consequently

$$D_m = (I - C)^{-1}(I - C^{m+1}) \longrightarrow (I - C)^{-1}.$$

The recurrence therefore approximates the requested inverse, not just an unexplained matrix sequence.

Exercise 2.6.13: Compute production for the supplied seven-sector economy

Enter the supplied consumption data exactly.

To specify the data without ambiguous rounding, write

$$C = \frac{1}{10000}\begin{bmatrix} 1588 & 64 & 25 & 304 & 14 & 83 & 1594 \\ 57 & 2645 & 436 & 99 & 83 & 201 & 3413 \\ 264 & 1506 & 3557 & 139 & 142 & 70 & 236 \\ 3299 & 565 & 495 & 3636 & 204 & 483 & 649 \\ 89 & 81 & 333 & 295 & 3412 & 237 & 20 \\ 1190 & 901 & 996 & 1260 & 1722 & 2368 & 3369 \\ 63 & 126 & 196 & 98 & 64 & 132 & 12 \end{bmatrix}.$$

Use the demand vector $d = (74000, 56000, 10500, 25000, 17500, 196000, 5000)^T$ in the same sector order as the problem. Solve $Mx = d$ with $M = I - C$; this is a seven-equation linear system, not a demand for external data.

Solve and check the residual before rounding.

Exact rational elimination gives the following values, rounded to six decimal places for display (units: millions of dollars):

$$x \approx \begin{bmatrix} 99575.653398 \\ 97703.022863 \\ 51230.523166 \\ 131569.921929 \\ 49488.491372 \\ 329554.452570 \\ 13835.335715 \end{bmatrix}.$$

The exact rational solution satisfies $Mx - d = 0$ in every coordinate. The six-decimal display is only an approximation and need not have an identically zero numerical residual. To reproduce that exact calculation, define C using the integer matrix above divided by 10000, and run:

```
from sympy import Matrix, eye
# C is the displayed integer matrix divided by 10000.
d = Matrix([74000,56000,10500,25000,17500,196000,5000])
M = eye(7) - C
x = M.LUsolve(d)      # exact rational elimination
assert M*x == d
print(x.evalf(12))
```

This solves the system without constructing an inverse. The displayed vector is the actual production plan, not merely an algorithm for finding it.

Exercise 2.6.14: Use the same coefficient matrix for a new demand

Keep the consumption matrix from Exercise 2.6.13 and change only the demand to

$$d = (99640, 75548, 14444, 33501, 23527, 263985, 6526)^T.$$

Solving $(I - C)x = d$ by exact arithmetic, then displaying six decimals, gives (again in millions of dollars)

$$x \approx \begin{bmatrix} 134033.615328 \\ 131686.642968 \\ 69471.699079 \\ 176912.002023 \\ 66595.556706 \\ 443772.902077 \\ 18431.177683 \end{bmatrix}.$$

The same reproduction code works with this new d. Checking the exact solution gives $(I-C)x-d = 0$ coordinate by coordinate. A previously computed LU factorization of $I - C$ can be reused; only its two triangular solves need be repeated.

Exercise 2.6.15: Count the actual number of iteration steps

Start with the specified $x^{(0)} = d$ from Exercise 2.6.13 and iterate $x^{(k+1)} = d + Cx^{(k)}$ without rounding intermediate values. We interpret "to four significant figures" as agreement, in all seven coordinates,

with rounding the exact solution to four significant figures. The resulting target vector is

$$\begin{bmatrix} 9.958 \times 10^4 \\ 9.770 \times 10^4 \\ 5.123 \times 10^4 \\ 1.316 \times 10^5 \\ 4.949 \times 10^4 \\ 3.296 \times 10^5 \\ 1.384 \times 10^4 \end{bmatrix}.$$

Actual iteration gives these checks:

k	$\max_i \lvert x_i^{(k)} - x_i \rvert / \lvert x_i \rvert$	all rounded entries agree?
0	8.09987×10^{-1}	no
5	4.04043×10^{-2}	no
10	1.62676×10^{-3}	no
15	6.49063×10^{-5}	no
16	3.40686×10^{-5}	yes

.

At step 16, before rounding,

$$x^{(16)} \approx \begin{bmatrix} 99575.105594 \\ 97701.855867 \\ 51228.777814 \\ 131566.362702 \\ 49487.009311 \\ 329550.480997 \\ 13835.050858 \end{bmatrix}.$$

No preceding iterate has all seven required rounded values. At step 15, for instance, the first coordinate is 99574.609463, which rounds to 9.957×10^4, not 9.958×10^4. Here is a reproducible counting procedure using the exact C, d:

```
x_exact = (eye(7)-C).LUsolve(d)
four = lambda v: [format(float(z), '.4g') for z in v]
target = four(x_exact)
xk = d
k = 0
while four(xk) != target:
    xk = d + C*xk
    k += 1
assert k == 16
```

The rational arithmetic prevents iteration-roundoff accumulation; the final formatting is used only for the specified comparison.

Conclusion. With $x^{(0)} = d$ counted as step zero, 16 updates are needed. The usual relative-error check $\max_i \lvert x_i^{(k)} - x_i \rvert / \lvert x_i \rvert < 5 \times 10^{-5}$ also first holds at step 16 for these data.

2.7 Applications to computer graphics

For two-dimensional graphics, write a point as $(x, y, 1)^T$. An affine transformation is then represented by

$$H = \begin{bmatrix} a & b & h \\ c & d & k \\ 0 & 0 & 1 \end{bmatrix}, \qquad H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} ax + by + h \\ cx + dy + k \\ 1 \end{bmatrix}.$$

The upper-left block controls the linear part and the last column controls translation. A positive angle denotes counterclockwise rotation in the usual oriented plane. To perform transformation F first and G second, multiply their matrices as GF: the first action is the rightmost factor. We use $R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ and write $T_p = \begin{bmatrix} I_2 & p \\ 0^T & 1 \end{bmatrix}$ for translation.

Practice problem 2.7.P1: Rotate about a point other than the origin

Let $p = (-2, 6)^T$ and $\theta = -30°$. Subtract p to move the center to zero, rotate, then add p back. Algebraically this is $x \mapsto R_\theta(x - p) + p = R_\theta x + (p - R_\theta p)$. With $c = \sqrt{3}/2$, $s = -1/2$, the matrix is

$$T_p \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix} T_{-p} = \boxed{\begin{bmatrix} \sqrt{3}/2 & 1/2 & \sqrt{3} - 5 \\ -1/2 & \sqrt{3}/2 & 5 - 3\sqrt{3} \\ 0 & 0 & 1 \end{bmatrix}}.$$

For example $R_\theta p = (-\sqrt{3} + 3, 1 + 3\sqrt{3})^T$, which explains the translation column. Substituting $x = p$ gives p again, as a rotation center must be fixed.

Exercise 2.7.1: Embed the shear into homogeneous coordinates

Example 2 uses $(x, y) \mapsto (x + y/4, y)$. Preserve the final coordinate 1:

$$\boxed{\begin{bmatrix} 1 & 1/4 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}.$$

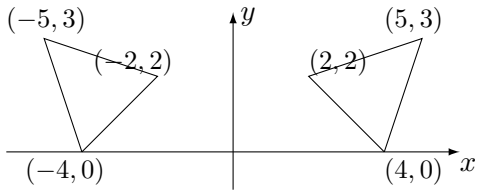
Multiplication by $(x, y, 1)^T$ gives $(x + y/4, y, 1)^T$, so the extra coordinate introduces no translation and does not change the original shear.

Exercise 2.7.2: Reflect and sketch a triangle

Reflection through the y-axis negates x and preserves y. Thus

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 2 & 5 \\ 0 & 2 & 3 \end{bmatrix} = \begin{bmatrix} -4 & -2 & -5 \\ 0 & 2 & 3 \end{bmatrix}.$$

The original vertices $(4, 0), (2, 2), (5, 3)$ therefore become $(-4, 0), (-2, 2), (-5, 3)$. Connect corresponding triples of vertices; reflection preserves straight segments and lengths.



Exercise 2.7.3: Translate first, rotate second

Translation by $(2,1)$ takes (x,y) to $(x+2,y+1)$. A 90° rotation then gives $(-(y+1),x+2)$. Reading off coefficients yields

$$R_{90^\circ}T_{(2,1)} = \boxed{\begin{bmatrix} 0 & -1 & -1 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}},$$

where the rotation matrix is understood in its homogeneous 3×3 form. The translated origin goes to $(-1,2)$, checking the last column.

Exercise 2.7.4: Translate before unequal scaling

The first action gives $(x-1,y+4)$, and the second gives $((x-1)/2, 3(y+4)/2)$. Therefore the matrix is

$$\boxed{\begin{bmatrix} 1/2 & 0 & -1/2 \\ 0 & 3/2 & 6 \\ 0 & 0 & 1 \end{bmatrix}}.$$

The translation column is scaled too, because the scaling is performed *after* translation.

Exercise 2.7.5: Reflect, then rotate

Reflection in the x-axis sends (x,y) to $(x,-y)$. Put $a=\sqrt{2}/2$. Rotation by 45° gives $(ax+ay, ax-ay)$, so

$$\boxed{\begin{bmatrix} a & a & 0 \\ a & -a & 0 \\ 0 & 0 & 1 \end{bmatrix}}.$$

Equivalently its upper-left block is $R_{45^\circ}\,\mathrm{diag}(1,-1)$.

Exercise 2.7.6: Reverse the preceding order

First rotation gives $(ax-ay, ax+ay)$. Reflecting through the x-axis now gives $(ax-ay, -ax-ay)$. Thus the matrix is

$$\boxed{\begin{bmatrix} a & -a & 0 \\ -a & -a & 0 \\ 0 & 0 & 1 \end{bmatrix}}, \qquad a=\sqrt{2}/2.$$

It differs from Exercise 5, illustrating why the order cannot be ignored.

Exercise 2.7.7: Rotate sixty degrees around a fixed point

Use the general formula $x \mapsto R_\theta x + p - R_\theta p$ with $p = (6,8)^T$, $\theta = 60°$. Since $R_\theta p = (3-4\sqrt{3}, 3\sqrt{3}+4)^T$, the result is

$$\boxed{\begin{bmatrix} 1/2 & -\sqrt{3}/2 & 3+4\sqrt{3} \\ \sqrt{3}/2 & 1/2 & 4-3\sqrt{3} \\ 0 & 0 & 1 \end{bmatrix}}.$$

Its action fixes $(6,8)$, which independently checks the translation terms.

Exercise 2.7.8: Rotate forty-five degrees around another point

Now $p = (3,7)^T$, and $R_{45°}p = (-2\sqrt{2}, 5\sqrt{2})^T$. Therefore

$$\boxed{\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 & 3+2\sqrt{2} \\ \sqrt{2}/2 & \sqrt{2}/2 & 7-5\sqrt{2} \\ 0 & 0 & 1 \end{bmatrix}}.$$

This is $T_p R_{45°} T_{-p}$, not $R_{45°} T_p$.

Exercise 2.7.9: Precompute a composite transformation

For $A(BD)$, the 2×100 product BD has 200 entries, each requiring two multiplications: 400 multiplications. Multiplication by A costs another 400, for 800 total. For $(AB)D$, forming the four entries of AB costs $4 \cdot 2 = 8$, and multiplying its result by D costs 400, for 408 total. Thus precomputing the 2×2 composite saves 392 multiplications. The advantage grows when the same transformations are applied to more vertices: combine the small transformation matrices once, then transform all data points with their product.

Exercise 2.7.10: Which geometric transformations commute?

Let $D(x) = \lambda x$, $R(x) = Rx$, and $T(x) = x + p$. Then $D(Rx) = \lambda Rx = R(\lambda x)$, so a uniform dilation and an origin-centered rotation always commute. By contrast,

$$D(Tx) = \lambda x + \lambda p, \qquad T(Dx) = \lambda x + p.$$

They commute exactly when $(\lambda - 1)p = 0$, and generally do not. Likewise $R(Tx) = Rx + Rp$ and $T(Rx) = Rx + p$, so rotation and translation commute exactly when $Rp = p$. A nontrivial planar rotation fixes only the zero vector, making a nonzero translation fail this test. For concrete failures take $p = (1,0)$, $\lambda = 2$, and a $90°$ rotation, and evaluate the two orders at $x = 0$.

Exercise 2.7.11: Two shear-and-scale factors produce a rotation

The first action is A_1, the second is A_2, so calculate $A_2 A_1$. Its upper-left block is

$$\begin{bmatrix} \sec\varphi & -\tan\varphi \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \sin\varphi & \cos\varphi \end{bmatrix} = \begin{bmatrix} \sec\varphi - \tan\varphi \sin\varphi & -\tan\varphi\cos\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix} = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}.$$

The first simplification uses $(1-\sin^2\varphi)/\cos\varphi = \cos\varphi$. The last homogeneous row and column remain $(0,0,1)$, so this is rotation by φ. The stated factors require $\cos\varphi \neq 0$; the rotation itself still exists at other angles, but these particular factors do not.

Exercise 2.7.12: Three shears can also implement rotation

Write $t = \tan(\varphi/2)$, $s = \sin\varphi$. Ignoring the unchanged homogeneous coordinate for a moment, multiply the three 2×2 blocks:

$$\begin{bmatrix} 1 & -t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ s & 1 \end{bmatrix} \begin{bmatrix} 1 & -t \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-ts & -2t+t^2 s \\ s & 1-st \end{bmatrix}.$$

Half-angle identities give $st = 1-\cos\varphi$ and $-2t+t^2 s = -\sin\varphi$, so the result is R_φ. For instance $\sin\varphi = 2t/(1+t^2)$ makes the second identity $-2t+2t^3/(1+t^2) = -2t/(1+t^2)$. Appending the homogeneous row and column yields precisely the claimed 3×3 rotation matrix. Each shear needs one nontrivial multiplication per point, whereas the usual rotation needs four. The factorization requires $\tan(\varphi/2)$ to be defined, excluding odd multiples of $180°$.

Exercise 2.7.13: Factor a general affine transformation

Block multiplication gives

$$\begin{bmatrix} A & p \\ 0^T & 1 \end{bmatrix} = \begin{bmatrix} I_2 & p \\ 0^T & 1 \end{bmatrix} \begin{bmatrix} A & 0 \\ 0^T & 1 \end{bmatrix}.$$

The right factor acts first and sends $(x,1)$ to $(Ax,1)$. The left then adds p, giving $(Ax+p,1)$. Thus the transformation is a linear transformation of the plane followed by a translation, as requested. No inverse or invertibility of A is needed.

Exercise 2.7.14: Separate the linear rotation from its final translation

The transformation from Exercise 7 is $x \mapsto R_{60°}x + (3+4\sqrt{3}, 4-3\sqrt{3})^T$. Hence it is rotation about zero followed by translation by

$$\boxed{p = (3+4\sqrt{3},\ 4-3\sqrt{3})^T}.$$

The translation is the original center minus its rotated image; it is not simply the center $(6,8)$ itself.

Exercise 2.7.15: Normalize a homogeneous four-vector

A homogeneous point (X,Y,Z,w) with $w \neq 0$ represents $(X/w, Y/w, Z/w)$. Dividing by $1/24$ multiplies the first three entries by 24, so the represented vector is $\boxed{(12,-6,-3)^T}$ in $\mathbb{R}^3$.

Exercise 2.7.16: Different representatives of one point

Yes: $(10,-20,-30,40) = 10(1,-2,-3,4)$. Dividing either representative by its final coordinate gives $(1/4,-1/2,-3/4)$. A common nonzero scale changes the homogeneous representative but not the point it represents.

Exercise 2.7.17: A three-dimensional rotation about the x-axis

The x-coordinate is unchanged, while the (y, z) coordinates undergo the usual 60° planar rotation in the orientation shown. The homogeneous matrix is

$$\boxed{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2 & -\sqrt{3}/2 & 0 \\ 0 & \sqrt{3}/2 & 1/2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}.$$

It sends e_2 toward e_3, confirming the positive orientation about e_1 by the right-hand convention.

Exercise 2.7.18: Rotate about z, then translate

The rotation leaves z fixed and has planar angle -30°. Translation after rotation appends $(5, -2, 1)^T$ as the last column:

$$\boxed{\begin{bmatrix} \sqrt{3}/2 & 1/2 & 0 & 5 \\ -1/2 & \sqrt{3}/2 & 0 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}}.$$

For example the origin is sent to $(5, -2, 1)$, checking the order of actions.

Exercise 2.7.19: Project a triangle from a finite viewing point

A line from $c = (0, 0, 10)$ through $v = (x, y, z)$ is $c + t(v - c) = (tx, ty, 10 + t(z - 10))$. Its intersection with the image plane $z = 0$ has $t = 10/(10 - z)$. Consequently the projection formula is

$$(x, y, z) \longmapsto \left(\frac{10x}{10 - z}, \frac{10y}{10 - z}, 0 \right).$$

For the three given vertices the scale factors are $5/3, 5/4, 5/2$, so their images are

$$\boxed{(7, 2, 0), \quad (15/2, 5, 0), \quad (5, 5, 0)}.$$

All original points have $z < 10$, so each line meets the image plane, and the triangular region projects to the triangle with these vertices.

Exercise 2.7.20: A second perspective triangle

Apply the same formula. At $(7, 3, -5)$ the scale factor is $2/3$; at $(12, 8, 2)$ it is $5/4$; at $(1, 2, 1)$ it is $10/9$. Thus the image triangle has vertices

$$\boxed{(14/3, 2, 0), \quad (15, 10, 0), \quad (10/9, 20/9, 0)}.$$

Perspective projection is not a single ordinary 3×3 linear map on the original three coordinates: the required division depends on z. Homogeneous coordinates encode that division by a final normalization.

Exercise 2.7.21: Convert standard color data into screen data

Let M denote the stated CIE conversion matrix, so $M(R,G,B)^T = (X,Y,Z)^T$. To recover the gun data solve this system:

$$\boxed{\begin{bmatrix} R \\ G \\ B \end{bmatrix} = M^{-1} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}}, \qquad M^{-1} = \frac{1}{20224} \begin{bmatrix} 45677 & -21023 & -7023 \\ -27293 & 47407 & 1407 \\ 1840 & -6160 & 25840 \end{bmatrix}.$$

This exact rational inverse treats the printed decimals as exact input. Its decimal approximation is

$$\begin{bmatrix} 2.258554 & -1.039508 & -.347261 \\ -1.349535 & 2.344096 & .069571 \\ .090981 & -.304589 & 1.277690 \end{bmatrix}.$$

Exact matrix multiplication checks $MM^{-1} = I$; in particular the conversion is uniquely reversible. A matrix program can reproduce it with SymPy rational decimal entries and `M.inv()`; do not reverse the conversion by multiplying by M a second time.

Exercise 2.7.22: Recover RGB from a television signal

Write the supplied matrix as N, so the transmitted vector is $(Y,I,Q)^T = N(R,G,B)^T$. Solving for RGB gives

$$\boxed{\begin{bmatrix} R \\ G \\ B \end{bmatrix} = N^{-1} \begin{bmatrix} Y \\ I \\ Q \end{bmatrix}},$$

$$N^{-1} = \frac{1}{50845523} \begin{bmatrix} 51002600 & 48549800 & 31415400 \\ 50681600 & -13764200 & -32784600 \\ 51277600 & -56463200 & 86415400 \end{bmatrix} \approx \begin{bmatrix} 1.003089 & .954849 & .617860 \\ .996776 & -.270706 & -.644788 \\ 1.008498 & -1.110485 & 1.699568 \end{bmatrix}.$$

The exact form was calculated and verified by rational matrix multiplication. It uses the coefficients as printed, including $.212, -.528, .311$ in the last row. Different engineering conventions or more accurate calibration coefficients would produce a different numerical inverse; no such change has been silently substituted here.

2.8 Subspaces of Rn

A subspace contains zero and is closed under *every* vector sum and *every* scalar multiple of its members. A single failure disproves subspace status. A span is always a subspace. For an $m \times n$ matrix,

$$\operatorname{Col} A = \{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m, \qquad \operatorname{Nul} A = \{x \in \mathbb{R}^n : Ax = 0\} \subseteq \mathbb{R}^n.$$

Column space concerns attainable *outputs*; null space concerns *inputs* with zero output. A basis is an independent spanning set. Row reduction preserves null space and dependence relations between columns, but generally changes the column space itself. To construct a column-space basis, take the *original* columns in the pivot positions. To construct a null-space basis, solve $Ax = 0$, choose the free variables as independent parameters, and read off their coefficient vectors.

Practice problem 2.8.P1: Membership in two very different spaces

For the supplied A and $u = (-7, 3, 2)^T$,

$$Au = \begin{bmatrix} -7 - 3 + 10 \\ -14 + 14 \\ 21 - 15 - 6 \end{bmatrix} = 0.$$

Thus $u \in \operatorname{Nul} A$. It is not in $\operatorname{Col} A$. Indeed row 3 of A equals five times row 1 minus four times row 2. Every possible output $b = Ax$ must consequently obey $b_3 = 5b_1 - 4b_2$. For u, that would require $2 = 5(-7) - 4(3) = -47$, which is false. Thus the same vector can belong to the null space but not the column space, even when these spaces happen to share the same ambient dimension.

Practice problem 2.8.P2: Read a shift matrix

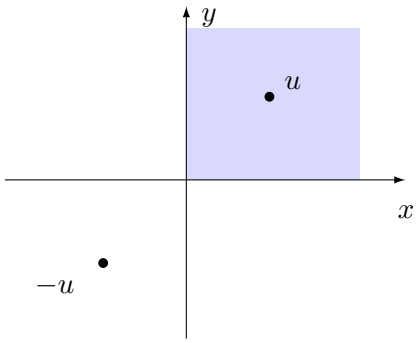
Here $A(x_1, x_2, x_3)^T = (x_2, x_3, 0)^T$. Thus $e_1 = (1, 0, 0)^T$ is a nonzero null vector. The second column of A is e_1, so e_1 is also a nonzero vector in $\operatorname{Col} A$. Alternatively e_2 belongs to the column space as column 3. It is legitimate for a vector to lie in both spaces in this example.

Practice problem 2.8.P3: The spaces of an invertible matrix

If A is invertible, every $b \in \mathbb{R}^n$ equals $A(A^{-1}b)$, so $\operatorname{Col} A = \mathbb{R}^n$. Also $Ax = 0$ implies $x = A^{-1}0 = 0$, so $\operatorname{Nul} A = \{0\}$. The zero subspace contains one vector; it is not an empty set.

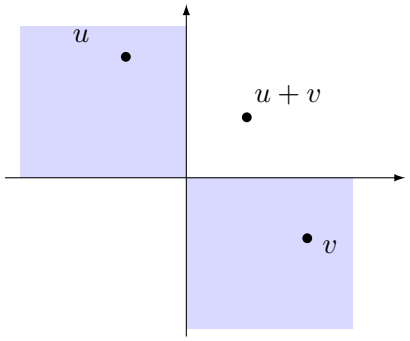
Exercise 2.8.1: A quadrant fails negative scaling

The shaded set is the first quadrant, including its axes. The vector $u = (1, 1)$ belongs to it, but $-u = (-1, -1)$ does not. Since -1 is an allowed scalar, closure under scalar multiplication fails. The following is a finite-window sketch; the quadrant continues outward.



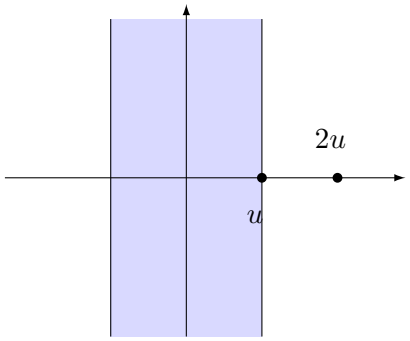
Exercise 2.8.2: Two allowed quadrants fail addition

The set consists of quadrants II and IV, with boundary axes. Take $u = (-1, 2)$ in quadrant II and $v = (2, -1)$ in quadrant IV. Their sum is $(1, 1)$ in quadrant I, outside the set. Thus addition is not closed.



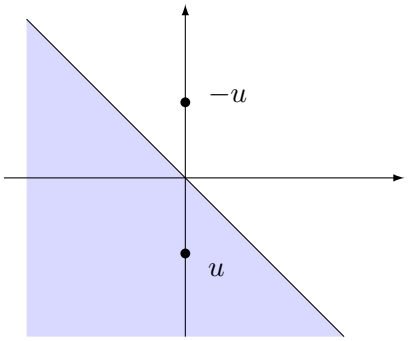
Exercise 2.8.3: A strip fails large scaling

Let the positive vertical boundary have coordinate $x = a > 0$. The point $u = (a, 0)$ lies on that included boundary, but $2u = (2a, 0)$ lies to its right, outside the strip. Hence closure under scalar multiplication fails. The finite upper and lower ends in the sketch are viewing limits, not extra boundaries.



Exercise 2.8.4: One side of a line is not a vector space

The shaded half-plane includes the region below a line through the origin. Take a point on the negative vertical axis, such as $u = (0, -1)$. It belongs to the shaded region, but $-u = (0, 1)$ lies on the unshaded side. Thus multiplying by -1 leaves the set, so it is not a subspace.



Exercise 2.8.5: Exhibit the required linear combination

Compute

$$v_1 + 2v_2 = (1, 3, -4)^T + 2(-2, -3, 7)^T = (-3, -3, 10)^T = w.$$

Thus $w \in \text{Span}\{v_1, v_2\}$. Giving coefficients that reproduce every coordinate proves membership directly; merely noticing that all vectors have three entries would not prove it.

Exercise 2.8.6: A consistency obstruction in four dimensions

Form $A = [v_1\ v_2\ v_3]$. Its four rows obey the relation $-11\,\text{row}_1 A + 5\,\text{row}_2 A + 10\,\text{row}_3 A + 2\,\text{row}_4 A = 0$. This identity is checked by its three entries: $-11 - 15 + 20 + 6 = 0$, $-44 - 20 + 50 + 14 = 0$, and $-55 - 15 + 60 + 10 = 0$. If $Ac = u$ had a solution, the same combination of its right-hand sides would be zero. Instead it equals $-11(-1) + 5(-7) + 10(-1) + 2(2) = -30$. This contradiction proves $u \notin \text{Span}\{v_1, v_2, v_3\}$. Equivalently, elimination of $[A \mid u]$ creates an inconsistent row.

Exercise 2.8.7: A finite generating set versus its infinite span

(a) The set $\{v_1, v_2, v_3\}$ contains exactly three distinct vectors.

(b) Its column space contains infinitely many vectors: all multiples tv_1 are in it, and $v_1 \neq 0$ makes these distinct for distinct real t.

(c) Yes, $p \in \text{Col}\, A$. Solving $Ac = p$ gives the reduced equations $c_1 + (7/4)c_3 = -9/4$, $c_2 + (5/2)c_3 = -7/2$. Choosing the free variable $c_3 = 0$ yields $p = (-9/4)v_1 - (7/2)v_2$. Its coordinates are $(6, -10, 11)$, as required.

Exercise 2.8.8: Another column-space membership test

A convenient combination with integer coefficients is

$$v_1 + 2v_2 + v_3 = (-2, 0, 6)^T + 2(-2, 3, 3)^T + (0, -5, 5)^T = (-6, 1, 17)^T = p.$$

Thus $p \in \operatorname{Col} A$. One could instead reduce $[A \mid p]$; the displayed coefficients give a direct, independently checkable witness.

Exercise 2.8.9: Column membership is not null membership

With A, p from Exercise 7, compute

$$Ap = \begin{bmatrix} 2(6) - 3(-10) - 4(11) \\ -8(6) + 8(-10) + 6(11) \\ 6(6) - 7(-10) - 7(11) \end{bmatrix} = \begin{bmatrix} -2 \\ -62 \\ 29 \end{bmatrix} \neq 0.$$

Therefore $p \notin \operatorname{Nul} A$, even though Exercise 7 placed p in $\operatorname{Col} A$.

Exercise 2.8.10: A nonzero null vector

For $u = (-5, 5, 3)^T$ and the matrix in Exercise 8,

$$Au = \begin{bmatrix} 10 - 10 + 0 \\ 0 + 15 - 15 \\ -30 + 15 + 15 \end{bmatrix} = 0.$$

Hence $u \in \operatorname{Nul} A$. Every coordinate has been checked; the nonzero input being sent to zero is exactly the definition of null-space membership.

Exercise 2.8.11: Ambient dimensions from the matrix shape

The given matrix has three rows and four columns. An input to A has four entries and an output has three, so $\operatorname{Nul} A \subseteq \mathbb{R}^4$ and $\operatorname{Col} A \subseteq \mathbb{R}^3$. Thus $\boxed{p = 4, q = 3}$. These are ambient dimensions, not necessarily the dimensions of the two subspaces themselves.

Exercise 2.8.12: Five outputs and three inputs

Here A is 5×3. Consequently $\operatorname{Nul} A \subseteq \mathbb{R}^3$ and $\operatorname{Col} A \subseteq \mathbb{R}^5$, giving $\boxed{p = 3, q = 5}$. Count columns for inputs and rows for outputs; these roles should not be reversed.

Exercise 2.8.13: One null vector and one column vector

For the matrix in Exercise 11, use

$$z = (1, -2, 1, 0)^T, \qquad a_1 = (3, -9, 9)^T.$$

Multiplication gives $Az = (3 - 4 + 1, -9 + 8 + 1, 9 - 4 - 5)^T = 0$, so the nonzero z is in $\operatorname{Nul} A$. The nonzero a_1 belongs to $\operatorname{Col} A$ because it is an actual column, namely Ae_1. The two vectors have different numbers of entries, as their respective ambient spaces require.

Exercise 2.8.14: Choose a null vector with integer entries

For the 5×3 matrix in Exercise 12, take $z = (1, -5, 3)^T$. Its row products are $1 - 10 + 9$, $4 - 25 + 21$, $-5 + 5 + 0$, $2 - 35 + 33$, and $3 - 15 + 12$, all zero. Thus $z \neq 0$ belongs to $\operatorname{Nul} A$. A nonzero column-space vector is $a_1 = (1, 4, -5, 2, 3)^T$. It belongs to $\operatorname{Col} A$ by its definition as column 1.

Exercise 2.8.15: A basis of the plane

The two vectors $(4, -2)^T$ and $(16, -3)^T$ have determinant $4(-3) - 16(-2) = 20 \neq 0$. Their 2×2 column matrix is invertible, so they are independent and span $\mathbb{R}^2$. Therefore they form a basis for $\mathbb{R}^2$.

Exercise 2.8.16: A dependent pair cannot be a basis

The second vector $(4, -10)^T$ is -2 times the first $(-2, 5)^T$. Thus the set is dependent and spans only one line. It is not a basis for $\mathbb{R}^2$, which requires two independent directions.

Exercise 2.8.17: Three pivots after a cyclic row reordering

Place the given vectors into columns: $A = \begin{bmatrix} 0 & 5 & 6 \\ 0 & 0 & 3 \\ -2 & 4 & 2 \end{bmatrix}$. Moving row 3 to the top gives the echelon matrix $\begin{bmatrix} -2 & 4 & 2 \\ 0 & 5 & 6 \\ 0 & 0 & 3 \end{bmatrix}$. Its three pivots show independence and spanning, so the given vectors form a basis for $\mathbb{R}^3$.

Exercise 2.8.18: An explicit dependence in three dimensions

The third vector equals twice the first plus the second: $2(1, 1, -3)^T + (3, -1, 2)^T = (5, 1, -4)^T$. Thus $2v_1 + v_2 - v_3 = 0$ is a nontrivial dependence relation. The set is not a basis for $\mathbb{R}^3$.

Exercise 2.8.19: Too few vectors to span three-dimensional space

Only two vectors are supplied. Their 3×2 column matrix has at most two pivots, so it cannot have a pivot in every one of its three rows. They do not span $\mathbb{R}^3$ and therefore do not form a basis for it. Their own span can have a basis of two vectors; that is a different question.

Exercise 2.8.20: Too many vectors for independence in three dimensions

Four vectors in $\mathbb{R}^3$ are necessarily dependent: the associated 3×4 homogeneous system has at least one free variable. Hence these four vectors do not form a basis for $\mathbb{R}^3$, regardless of whether they happen to span the whole space.

Exercise 2.8.21: Subspace definitions and preserved dependence

(a) **False as written.** The closure requirements must say *for every* $u, v \in H$, $u + v \in H$ and *for every* $u \in H$ *and every scalar* c, $cu \in H$. Simply listing that some $u, v, u + v, cu$ are in H is insufficient. For example $H = \{0, e_1, -e_1\}$ has zero and satisfies those listed facts with $u = v = 0$, but $2e_1 \notin H$. With the universal conditional quantifiers restored, the statement is exactly the correct subspace definition.

(b) **True.** $[v_1 \ \cdots \ v_p]x$ is precisely a linear combination of the listed vectors, so its set of possible values is their span.

(c) **False.** There are n unknowns, so the solution vectors have n entries; the subspace is in $\mathbb{R}^n$, not generally $\mathbb{R}^m$.

(d) **True.** The IMT supplies both independence and spanning for the columns of an invertible $n \times n$ matrix.

(e) **True.** If row operations give $B = EA$ with invertible E, then $Ac = 0$ if and only if $Bc = 0$. Thus precisely the same coefficient vectors express dependencies among the corresponding columns.

Exercise 2.8.22: Distinguish spaces from their descriptions

(a) **False.** $H = \{0, e_1\}$ contains zero but not $2e_1$, so it fails scalar closure.

(b) **False.** Pivot columns must be selected from A itself. For example $A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ reduces to $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$. Its pivot column e_1 is not in $\operatorname{Col} A = \operatorname{Span}\{e_2\}$.

(c) **True.** The set contains zero (all coefficients zero), and sums or scalar multiples of linear combinations are linear combinations. These are exactly the three subspace tests.

(d) **False.** Closure requires both added vectors in H. Take $H = \operatorname{Span}\{e_1\} \subset \mathbb{R}^2$, $x = 0$, and $y = e_2$; then $x + y = e_2 \notin H$.

(e) **False.** $\operatorname{Col} A$ is the set of right sides b for which $Ax = b$ has a solution. It is not the set of solution inputs x to one fixed equation; input and output spaces can even have different dimensions.

Exercise 2.8.23: Two pivot columns and two free variables

The given echelon form has pivots in columns 1 and 2. Therefore a column-space basis is

$$\mathcal{C} = \{(4, 6, 3)^T, (5, 5, 4)^T\},$$

taken from the *original* matrix. For $Ax = 0$, the echelon equations are $x_1 + 2x_2 + 6x_3 - 5x_4 = 0$ and $x_2 + 5x_3 - 6x_4 = 0$. Set $x_3 = s$, $x_4 = t$; then $x_2 = -5s + 6t$ and $x_1 = 4s - 7t$. Thus

$$x = s(4, -5, 1, 0)^T + t(-7, 6, 0, 1)^T,$$

and the two displayed coefficient vectors form a basis for $\operatorname{Nul} A$. They span because this is the general solution, and they are independent because their free coordinates x_3, x_4 force $s = t = 0$ in a zero combination.

Exercise 2.8.24: A nonadjacent pair of pivot columns

The pivot columns are 1 and 3, so a basis for $\operatorname{Col} A$ is $\{(3,2,3)^T, (9,7,6)^T\}$. The echelon equations reduce to $x_3 = -2x_4$ and $x_1 - 2x_2 - 6x_4 = 0$. Let $x_2 = s$, $x_4 = t$. Then

$$x = s(2,1,0,0)^T + t(6,0,-2,1)^T.$$

The coefficient vectors form a null-space basis: they describe all solutions and their free-coordinate entries make them independent.

Exercise 2.8.25: Three pivot columns among five inputs

The echelon form has pivot columns 1,2,4. A column-space basis is

$$\{(1,-1,-2,3)^T, (4,2,2,6)^T, (-3,3,5,-5)^T\}.$$

The three equations are $x_1 + 4x_2 + 8x_3 + 5x_5 = 0$, $2x_2 + 5x_3 - x_5 = 0$, and $x_4 + 4x_5 = 0$. Taking $x_3 = s$, $x_5 = t$ gives $x_2 = -5s/2 + t/2$, $x_4 = -4t$, $x_1 = 2s - 7t$. Hence a basis for $\operatorname{Nul} A$ is

$$\{(2,-5/2,1,0,0)^T, (-7,1/2,0,-4,1)^T\}.$$

Substituting either vector into the three echelon equations yields zero; independence follows from their third and fifth coordinates.

Exercise 2.8.26: Another three-pivot example

The pivot positions are in columns 1,2,4 of the original matrix. A column-space basis is

$$\{(3,3,0,6)^T, (-1,1,3,3)^T, (-1,0,-1,-2)^T\}.$$

The echelon equations give $x_4 = 2x_5$, $x_2 = -3x_3 + 2x_5$, and $3x_1 - x_2 - 3x_3 + 6x_5 = 0$, so $x_1 = -4x_5/3$. Thus a null-space basis is

$$\{(0,-3,1,0,0)^T, (-4/3,2,0,2,1)^T\}.$$

The free variables are x_3, x_5; this explains both why there are two basis vectors and why they are independent.

Exercise 2.8.27: A column combination different from every column

Choose $A = I_3$ and $b = (1,1,0)^T$. Then $b = e_1 + e_2 = A(1,1,0)^T \in \operatorname{Col} A$, and $b \neq 0$. It differs from each of e_1, e_2, e_3, the columns of A. Column space contains all combinations, not just the listed columns.

Exercise 2.8.28: Construct an unattainable output

Take $A = \operatorname{diag}(1,1,0)$ and $b = e_3$. Every output Ax has third coordinate zero, whereas b has third coordinate one. Hence $b \notin \operatorname{Col} A$.

Exercise 2.8.29: A nonzero input annihilated by a nonzero matrix

Take $A = \operatorname{diag}(1, 0, 0)$ and $b = e_2$. Both A and b are nonzero, but $Ab = 0$, so $b \in \operatorname{Nul} A$. This does not violate any rule about scalars: nonzero matrices can annihilate nonzero vectors.

Exercise 2.8.30: Independent columns are already a basis for their own span

By definition $\operatorname{Col} A = \operatorname{Span}\{a_1, \ldots, a_p\}$, so the columns span this space. The given assumption supplies independence. These are exactly the two conditions for a basis, hence the columns are a basis for $\operatorname{Col} A$. No claim that the column space equals the whole codomain is needed.

Exercise 2.8.31: A proper column space of a square matrix

Since $\operatorname{Col} F \neq \mathbb{R}^5$, the 5×5 matrix is not onto. The IMT makes it singular, so $\operatorname{Nul} F$ contains a nonzero vector. It is a subspace of $\mathbb{R}^5$ and contains infinitely many vectors, since all real multiples of any nonzero null vector are distinct members. The exact size of a basis is not determined by the premise.

Exercise 2.8.32: Full column space in seven dimensions

$\operatorname{Col} B = \mathbb{R}^7$ says every b is attained. Because B is square, the IMT makes it invertible. Thus for every $b \in \mathbb{R}^7$, $Bx = b$ has exactly one solution, $B^{-1}b$. In particular $\operatorname{Nul} B = \{0\}$; there are no free variables.

Exercise 2.8.33: A trivial null space in six dimensions

The only homogeneous solution is zero, so the square matrix C is invertible by the IMT. Therefore for every $b \in \mathbb{R}^6$ the equation $Cx = b$ has a unique solution $C^{-1}b$, and $\operatorname{Col} C = \mathbb{R}^6$. The conclusion would require qualification for a nonsquare matrix.

Exercise 2.8.34: A column basis of the entire codomain fixes the shape

Spanning $\mathbb{R}^m$ requires a pivot in every row, so there are m pivots. Independence requires every column to be a pivot column, so there are n pivots. Hence $m = n$: A is square. It is then invertible by the IMT. Both basis requirements, rather than spanning alone, forced equality.

Exercise 2.8.35: A nontrivial null space implies a proper column space

The given nonzero null vector makes the 5×5 matrix singular. Therefore its column space is a proper subspace of $\mathbb{R}^5$ by the IMT. Some right sides are unattainable. Every attainable right side has infinitely many preimages: if $Bx = b$ and $Bz = 0$ with $z \neq 0$, then $B(x + tz) = b$ for every real t.

Exercise 2.8.36: Independent columns of a tall matrix

The four columns are independent, so $Cx = 0$ has only $x = 0$. Hence $\boxed{\operatorname{Nul} C = \{0\} \subseteq \mathbb{R}^4}$. Although there are six output coordinates, inputs and null vectors have four coordinates. The independent columns do not span $\mathbb{R}^6$, so not every right side is attained, despite uniqueness whenever one is attained.

Exercise 2.8.37: An exact computer-assisted pair of bases

Exact row reduction of the supplied matrix gives

$$R = \begin{bmatrix} 1 & 0 & 5/2 & -9/2 & 7/2 \\ 0 & 1 & 3/2 & -5/2 & 3/2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1 and 2, so $\{(3,-7,-5,3)^T, (-5,9,7,-7)^T\}$ is a column-space basis. For the null space, solve $x_1 = -5x_3/2 + 9x_4/2 - 7x_5/2$ and $x_2 = -3x_3/2 + 5x_4/2 - 3x_5/2$. Taking each free variable in turn equal to 2 gives the integer basis

$$\{(-5,-3,2,0,0)^T, (9,5,0,2,0)^T, (-7,-3,0,0,2)^T\}.$$

Scaling basis vectors by 2 does not change their span or independence. These were checked by exact multiplication by the original A. A reproducible computation is `R,pivots=Matrix(A).rref()` in SymPy, followed by `Matrix(A).nullspace()`; the latter may return scalar multiples of the chosen vectors.

Exercise 2.8.38: Four pivot columns and one free coordinate

Exact row reduction gives

$$R = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Thus original columns 1,2,4,5 form a column-space basis. Explicitly they are the columns of

$$C = \begin{bmatrix} 5 & 3 & -6 & -8 \\ 4 & 1 & -8 & -7 \\ 5 & 1 & 5 & 19 \\ -7 & -5 & 8 & 5 \end{bmatrix}.$$

For $Rx = 0$, $x_1 = -x_3$, $x_2 = x_3$, and $x_4 = x_5 = 0$. Hence $\{(-1,1,1,0,0)^T\}$ is a basis for $\operatorname{Nul} A$. Multiplication by the original matrix checks the null vector, and the four pivot positions establish independence of the selected original columns.

2.9 Dimension and rank

For an ordered basis $\mathcal{B} = (b_1, \ldots, b_p)$, the coordinate vector $[x]_{\mathcal{B}} = (c_1, \ldots, c_p)^T$ means $x = c_1 b_1 + \cdots + c_p b_p$. Coordinates depend on the basis and its order. The dimension of a subspace is the number of vectors in any basis for it. For an $m \times n$ matrix, rank counts pivots, while nullity counts free variables; therefore

$$\dim \operatorname{Col} A = \operatorname{rank} A, \qquad \dim \operatorname{Nul} A = n - \operatorname{rank} A.$$

A subspace's dimension is different from the number of coordinates in its vectors. A two-dimensional subspace of $\mathbb{R}^6$ still consists of six-entry vectors, not two-entry vectors.

Practice problem 2.9.P1: Count independent directions rather than generators

Put the three vectors into a matrix. Eliminating below column 1 gives

$$\begin{bmatrix} 2 & 3 & -1 \\ -8 & -7 & 6 \\ 6 & -1 & -7 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 3 & -1 \\ 0 & 5 & 2 \\ 0 & -10 & -4 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 3 & -1 \\ 0 & 5 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 1 and 2 are pivot columns. Thus a basis for H is $\{(2, -8, 6)^T, (3, -7, -1)^T\}$, and $\dim H = 2$. The third vector is redundant: $v_3 = -11v_1/10 + 2v_2/5$.

Practice problem 2.9.P2: Turn coordinates into a vector

The coordinate entries are weights, so

$$x = 3(1, .2)^T + 2(.2, 1)^T = (3 + .4, .6 + 2)^T = (3.4, 2.6)^T.$$

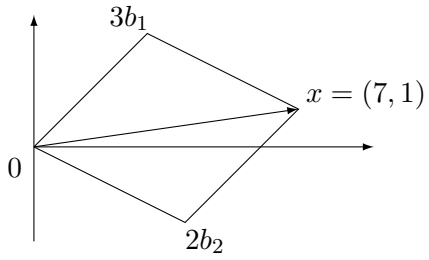
This ordinary coordinate vector differs from $(3, 2)^T$, because the basis is not the standard basis.

Practice problem 2.9.P3: A subspace cannot have more independent directions than its ambient space

No subspace of $\mathbb{R}^3$ can have dimension four. A four-dimensional subspace would have a basis of four independent vectors in $\mathbb{R}^3$, but any 3×4 homogeneous system has a free variable, making four such columns dependent. In fact possible dimensions are only $0, 1, 2, 3$; a three-dimensional subspace of $\mathbb{R}^3$ must be $\mathbb{R}^3$ itself.

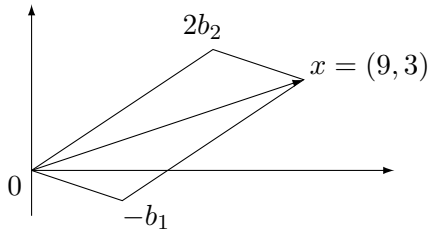
Exercise 2.9.1: Combine two basis vectors and sketch the result

The supplied coordinate vector gives $x = 3(1, 1)^T + 2(2, -1)^T = (7, 1)^T$. The sketch shows the parallelogram with sides $3b_1 = (3, 3)$ and $2b_2 = (4, -2)$; its diagonal is x.



Exercise 2.9.2: A negative basis coordinate reverses a direction

Now $x = -1(-3, 1)^T + 2(3, 2)^T = (3, -1)^T + (6, 4)^T = (9, 3)^T$. A negative coefficient means travel opposite to b_1, not negate the whole answer. The parallelogram below has sides $-b_1$ and $2b_2$.



Exercise 2.9.3: Recover coordinates from two equations

Write $x = c_1b_1 + c_2b_2$. The two coordinates require $2c_1 - c_2 = 0$ and $-3c_1 + 5c_2 = 7$. The first gives $c_2 = 2c_1$; substituting into the second gives $7c_1 = 7$, hence $c_1 = 1, c_2 = 2$. Thus $\boxed{[x]_{\mathcal{B}} = (1, 2)^T}$, and $(2, -3)^T + 2(-1, 5)^T = (0, 7)^T$ verifies the answer.

Exercise 2.9.4: Coordinates may both be negative

The equations are $c_1 - 2c_2 = 1$ and $-5c_1 + 3c_2 = 9$. Substitute $c_1 = 1 + 2c_2$ into the second: $-5 - 10c_2 + 3c_2 = 9$, so $c_2 = -2$ and $c_1 = -3$. Therefore $\boxed{[x]_{\mathcal{B}} = (-3, -2)^T}$. Indeed $-3(1, -5)^T - 2(-2, 3)^T = (1, 9)^T$.

Exercise 2.9.5: Two coordinates describe a three-entry vector

From the first two entries of $c_1b_1 + c_2b_2 = x$, $c_1 - 2c_2 = 2$ and $4c_1 - 7c_2 = 9$. Subtract four times the first equation from the second to get $c_2 = 1$; then $c_1 = 4$. The third entry also matches: $-3(4) + 5(1) = -7$. Thus $\boxed{[x]_{\mathcal{B}} = (4, 1)^T}$. Checking the third coordinate is important: a solution to only two equations would not by itself prove the vector equality in $\mathbb{R}^3$.

Exercise 2.9.6: Verify the remaining coordinate after solving

The first two equations are $-3c_1 + 7c_2 = 5$ and $2c_1 - 3c_2 = 0$. The second gives $c_1 = 3c_2/2$; the first becomes $(5/2)c_2 = 5$. Thus $c_2 = 2, c_1 = 3$. The last coordinate is $-4(3) + 5(2) = -2$, as required.

Therefore $\boxed{[x]_{\mathcal{B}} = (3,2)^T}$.

Exercise 2.9.7: Read the skew grid and confirm algebraically

The diagram suggests two units in the b_1 direction and minus one unit in the b_2 direction for w, and one and a half b_1 units plus half a b_2 unit for x. Algebra confirms

$$[w]_{\mathcal{B}} = (2,-1)^T, \qquad [x]_{\mathcal{B}} = (3/2, 1/2)^T.$$

Indeed $2(3,0)^T - (-1,2)^T = (7,-2)^T = w$, and $(3/2)(3,0)^T + (1/2)(-1,2)^T = (4,1)^T = x$. These computations turn approximate visual readings into exact answers.

Exercise 2.9.8: A second skew coordinate grid

Here $c_1 b_1 + c_2 b_2 = (2c_2, 2c_1 + c_2)^T$. Thus for a vector $(u,v)^T$, $c_2 = u/2$ and $c_1 = (v - u/2)/2$. Applying this formula gives

$$[x]_{\mathcal{B}} = (2,-1)^T, \quad [y]_{\mathcal{B}} = (3/2, 1)^T, \quad [z]_{\mathcal{B}} = (-1, -1/2)^T.$$

The first matches the picture's movement two b_1 steps and minus one b_2 step. The requested confirmations are $(3/2)(0,2)^T + (2,1)^T = (2,4)^T = y$ and $-(0,2)^T - (1/2)(2,1)^T = (-1,-5/2)^T = z$.

Exercise 2.9.9: Rank three, nullity one

The supplied echelon form has pivots in columns 1,3,4, so an original-column basis is

$$\mathcal{C} = \{(1,3,2,5)^T, (2,1,-1,0)^T, (-6,5,9,14)^T\}.$$

The bottom nonzero equation gives $x_4 = 0$; the next gives $x_3 = 0$; the first then gives $x_1 = -3x_2$. Thus $\mathcal{N} = \{(-3,1,0,0)^T\}$ is a null-space basis. Consequently $\dim \operatorname{Col} A = 3$ and $\dim \operatorname{Nul} A = 1$. Their sum is four, the number of columns, not necessarily the number of rows in other examples.

Exercise 2.9.10: Four pivots among five columns

The pivot positions are columns 1,2,4,5. A column-space basis consists of the columns of

$$C = \begin{bmatrix} 1 & -2 & 5 & 4 \\ 2 & -1 & 5 & 6 \\ -2 & 0 & 1 & -6 \\ 3 & 1 & 1 & 5 \end{bmatrix}.$$

Reading the supplied echelon equations from the bottom upward gives $x_5 = 0$, $x_4 = 0$, $x_2 = -x_3$, and $x_1 = -x_3$. Thus $\{(-1,-1,1,0,0)^T\}$ is a null-space basis. The dimensions are $\boxed{4 \text{ and } 1}$ for column and null spaces, respectively. The basis vectors for the column space are from the original A, while the equations for the null space come from its echelon form.

Exercise 2.9.11: Two pivot directions and three degrees of freedom

The pivots are in columns 1 and 3. Thus a column-space basis is

$$\{(2,3,0,-3)^T,(-5,-8,9,-7)^T\}.$$

The second echelon equation gives $x_3=-x_5$. Substitution in the first gives $x_1=-2x_2-x_4-x_5$. Taking x_2,x_4,x_5 as independent parameters gives a null-space basis

$$\{(-2,1,0,0,0)^T,(-1,0,0,1,0)^T,(-1,0,-1,0,1)^T\}.$$

These vectors have independent free-coordinate entries, so they are independent and span all solutions. Therefore $\dim\operatorname{Col}A=2$, $\dim\operatorname{Nul}A=3$.

Exercise 2.9.12: A third free-variable calculation

The echelon form has pivots in columns 1,2,3. A column-space basis is

$$\{(1,5,4,3)^T,(2,1,6,4)^T,(-4,-9,-9,-5)^T\}.$$

The bottom nonzero equation gives $x_3=x_5$. The preceding equation then gives $x_2=-2x_4-x_5$, and the first gives $x_1=0$. Hence a null-space basis is

$$\{(0,-2,0,1,0)^T,(0,-1,1,0,1)^T\}.$$

The dimensions are $\boxed{3 \text{ and } 2}$. Direct substitution into the original A checks both null vectors, and the free coordinates prove their independence.

Exercise 2.9.13: Discard one redundant spanning vector

Place the four supplied vectors into $A=[v_1\ v_2\ v_3\ v_4]$. Exact row reduction gives

$$R=\begin{bmatrix}1&-3&0&0\\0&0&1&0\\0&0&0&1\\0&0&0&0\end{bmatrix}.$$

The pivot columns are 1,3,4, so

$$\{(1,-3,2,-4)^T,(2,-1,4,2)^T,(-4,5,-3,7)^T\}$$

is a basis for the supplied span, whose dimension is 3. The omitted vector is $v_2=-3v_1$. The other three are independent by the pivot test; discarding a multiple does not by itself establish that last fact without the reduction.

Exercise 2.9.14: Select three of five generators

The column matrix of the five vectors reduces exactly to

$$R=\begin{bmatrix}1&0&-2&0&15/2\\0&1&1&0&-7/2\\0&0&0&1&-5/2\\0&0&0&0&0\end{bmatrix}.$$

Thus v_1, v_2, v_4 are pivot columns, giving the basis

$$\{(1,-1,-2,3)^T, (2,-3,-1,4)^T, (-1,4,-7,7)^T\}$$

and dimension 3. For a check on spanning, $v_3 = -2v_1 + v_2$ and $v_5 = (15/2)v_1 - (7/2)v_2 - (5/2)v_4$. Thus the two omitted vectors are indeed combinations of the selected ones.

Exercise 2.9.15: Equal dimensions do not mean equal sets

Four pivot columns in a 4×6 matrix give four pivot rows, so $\operatorname{Col} A = \mathbb{R}^4$. There are $6 - 4 = 2$ free variables, so $\dim \operatorname{Nul} A = 2$. However $\operatorname{Nul} A \neq \mathbb{R}^2$: its vectors have six entries and it is a subspace of $\mathbb{R}^6$. A basis identifies it with $\mathbb{R}^2$ through coordinates; an identification by coordinates is different from literal equality of the sets.

Exercise 2.9.16: A three-dimensional column space inside four dimensions

The column space has dimension 3 but consists of four-entry vectors. It is a proper subspace of $\mathbb{R}^4$, not literally $\mathbb{R}^3$. Rank–nullity gives $\dim \operatorname{Nul} A = 7 - 3 = \boxed{4}$. The five words "three-dimensional subspace of $\mathbb{R}^4$" specify both its internal dimension and its ambient space.

Exercise 2.9.17: True–false questions about dimension

(a) **True.** For an ordered basis, the unique weights in $x = \sum c_i v_i$ are precisely its basis coordinates.

(b) **False.** A line must pass through zero to be a subspace. The line $\{(1, t) : t \in \mathbb{R}\}$ does not contain zero, for instance. A line through zero does form a one-dimensional subspace.

(c) **True.** Pivot columns form a basis for $\operatorname{Col} A$, so their number equals its dimension.

(d) **True.** Pivots plus free variables count all n columns: $\dim \operatorname{Col} A + \dim \operatorname{Nul} A = n$.

(e) **True.** A spanning set of p vectors in a p-dimensional space is a basis. If it were dependent, one vector could be deleted without changing its span, leaving at most $p - 1$ generators for a space of dimension p, which is impossible.

Exercise 2.9.18: Uniqueness, nullity, and coordinate maps

(a) **True.** Two coefficient representations would subtract to a dependence relation among basis vectors; independence forces the corresponding coefficients to be identical.

(b) **False.** The null-space dimension is the number of *free* variables, not all variables. For $A = I_n$ it is zero despite there being n variables.

(c) **True.** Rank is the number of pivots, which is the column-space dimension.

(d) **True.** The coordinate map is a linear bijection to $\mathbb{R}^p$, with inverse $(c_1, \ldots, c_p)^T \mapsto \sum c_i v_i$. Thus it preserves addition and scalar multiplication and assigns one coordinate vector to every vector of H.

(e) **True.** If p independent vectors in a p-dimensional space did not span it, a vector outside their span could be appended to produce $p + 1$ independent vectors, contradicting the dimension.

Exercise 2.9.19: Recover rank from nullity

The null space has a basis of three vectors, so its dimension is 3. There are seven columns. Thus $\operatorname{rank} A = 7 - 3 = \boxed{4}$ by rank–nullity. The five rows impose the separate upper bound $\operatorname{rank} A \leq 5$, which this answer respects.

Exercise 2.9.20: A six-by-eight matrix

There are eight columns and nullity three, so $\operatorname{rank} A = 8 - 3 = \boxed{5}$. This is possible because five pivots fit into six rows.

Exercise 2.9.21: The dimension of a homogeneous solution space

The solution space is $\operatorname{Nul} A$. Rank–nullity with eight columns and rank seven gives $\dim \operatorname{Nul} A = 8 - 7 = \boxed{1}$. Its vectors have eight entries even though only one parameter is needed to describe them.

Exercise 2.9.22: Five generators of a four-dimensional span are dependent

If the five vectors were independent, they would be a basis for their own span, making its dimension five by the definition of dimension. The given dimension is four, a contradiction. Hence the vectors are dependent. The ambient value of n does not change this argument.

Exercise 2.9.23: Construct prescribed rank and nullity

Take

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Its column-space basis is $\{e_1, e_2\} \subset \mathbb{R}^3$, so its rank is 2. Its homogeneous solutions are $(0, 0, s, t, u)^T$, so nullity is 3. Thus the requested 3×5 shape and both dimensions are achieved.

Exercise 2.9.24: Construct rank one

Choose

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Every column is a multiple of the nonzero first column, so the column space is exactly one line. Hence A has rank 1; it has the required 3×4 shape.

Exercise 2.9.25: Full column-space dimension forces independent columns

The p columns span $\operatorname{Col} A$ by definition, and this space has dimension p by hypothesis. A spanning set with exactly its dimension many vectors is a basis. Therefore the columns are independent. Equivalently rank p in a matrix with p columns leaves no free variables.

Exercise 2.9.26: A basis need not be the leftmost pivot selection

The five named columns lie in $\operatorname{Col} A$, which has dimension five because $\operatorname{rank} A = 5$. They are independent by hypothesis. An independent set of five vectors in a five-dimensional space is a basis, so these columns span $\operatorname{Col} A$. The pivot-column algorithm is one way to find a basis, not a claim that all bases must use the same indices.

Exercise 2.9.27: More vectors than a spanning set must be dependent

(a) Since $a_j \in W = \operatorname{Span}\{b_1, \ldots, b_p\}$, there are coefficients $c_{1j}, \ldots, c_{pj}$ with $a_j = \sum_i c_{ij} b_i$. Collect them in $c_j \in \mathbb{R}^p$ to obtain $a_j = Bc_j$.

(b) The matrix $C = [c_1 \ \cdots \ c_q]$ has p rows and $q > p$ columns. Its homogeneous system has more variables than possible pivots, so at least one variable is free. Thus some $u \neq 0$ satisfies $Cu = 0$.

(c) The column identities in part (a) give $A = BC$. Therefore $Au = B(Cu) = B0 = 0$. Since $u \neq 0$, its entries furnish a nontrivial linear dependence among $a_1, \ldots, a_q$.

This proves the claim without assuming the two supplied lists were bases.

Exercise 2.9.28: Every basis has the same number of vectors

Let $\mathcal{A}, \mathcal{B}$ be bases of W, with q, p vectors, respectively. If $q > p$, then Exercise 27 applied to the spanning set $\mathcal{B}$ would make $\mathcal{A}$ dependent, contrary to its being a basis. Hence $q \leq p$. Interchanging the roles gives $p \leq q$. Thus $p = q$. This justifies defining dimension using *any* basis.

Exercise 2.9.29: Compute coordinates in a two-dimensional subspace of R4

The vectors v_1, v_2 are independent: their first two coordinates have 2×2 determinant $15(-10) - 14(-5) = -80 \neq 0$. Thus they are a basis for their span H. A direct calculation gives

$$2v_1 - v_2 = (30 - 14, -10 + 10, 24 - 13, 14 - 17)^T = (16, 0, 11, -3)^T = x.$$

Therefore $x \in H$ and $\boxed{[x]_{\mathcal{B}} = (2, -1)^T}$. These exact integer products also verify the computer-assisted answer.

Exercise 2.9.30: Three coordinates in a four-dimensional ambient space

Put $V = [v_1 \ v_2 \ v_3]$ and row reduce the augmented matrix. Exact arithmetic gives

$$[V \mid x] \ \sim \ \left[\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

The three coefficient pivots show that v_1, v_2, v_3 are independent, so $\mathcal{B}$ is a basis for H. The consistent right block gives $[x]_{\mathcal{B}} = \boxed{(-2, 1, 1)^T}$. For an independent check,

$$-2v_1 + v_2 + v_3 = (12 + 8 - 9, -6 + 0 + 4, 18 + 7 - 8, -8 - 3 + 3)^T = (11, -2, 17, -8)^T = x.$$

Thus the coordinate vector has three entries while x itself has four.

2.S Chapter 2 supplementary exercises

These exercises combine matrix shapes, multiplication order, inverse identities, and the geometry of column and null spaces. Unless the statement explicitly says a matrix is square, do not add that hypothesis silently. A rectangular matrix can have a left inverse or a right inverse without having both. For a true–false question, a false statement is settled by one valid counterexample, while a true statement needs a general argument.

Exercise 2.S.1: Sixteen structural true–false statements

(a) **True.** If A, B are $m \times n$, then AB^T has shape $(m \times n)(n \times m)$ and $A^T B$ has shape $(n \times m)(m \times n)$. Both inner dimensions agree.

(b) **False.** The product has as many columns as B, not A. For example a 1×3 matrix times a 3×2 matrix has two columns, even though the first factor has three columns.

(c) **True.** For $A = \operatorname{diag}(a_1, \ldots, a_m)$, entry (i, j) of AB is $a_i b_{ij}$. Thus row i is scaled by a_i.

(d) **False.** Let $B = 0$, $C = I$, $D = 0$ of the same square size. Then $BC = BD = 0$ but $C \neq D$. Cancellation needs an injective left factor.

(e) **False.** $A = \operatorname{diag}(1, 0)$ and $C = \operatorname{diag}(0, 1)$ are nonzero but $AC = 0$.

(f) **False.** Expanding gives $A^2 - B^2 + BA - AB$. For $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $B = \operatorname{diag}(1, 2)$, one has $AB \neq BA$, so the extra term does not vanish.

(g) **True.** A row swap or nonzero row scaling of I_n leaves exactly n nonzero entries. Adding a nonzero multiple of a different row introduces one off-diagonal nonzero entry, giving $n + 1$. A zero-multiple replacement leaves the identity with n entries.

(h) **True.** A swap matrix and a scaling matrix are symmetric. The transpose of $I + re_i e_j^T$ is $I + re_j e_i^T$, which is the matrix for the corresponding row replacement with the indices exchanged.

(i) **True.** An elementary matrix is obtained by one row operation on a square identity matrix; its shape remains square.

(j) **False.** Every product of elementary matrices is invertible, whereas a square zero matrix is not. Only invertible square matrices admit such a factorization.

(k) **True.** Three pivots in a 3×3 matrix imply reduced form I_3. List the elementary row operations that carry out that reduction; their matrices $E_1, \ldots, E_p$ give $E_p \cdots E_1 A = I$.

(l) **False without a square hypothesis.** Take $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$. Then $AB = I_2$ but A is rectangular and has a nonzero null vector e_3, so it has no two-sided inverse. If A, B were assumed square, the assertion would instead be true by the IMT.

(m) **False.** The inverse is $B^{-1}A^{-1}$, with reversed order. For the noncommuting invertible pair in part (f), the same-order product is not an inverse.

(n) **True.** From $AB = BA$, left-multiply by A^{-1} and then right-multiply by A^{-1}: $BA^{-1} = A^{-1}B$.

(o) **False.** The correct scalar factor is $1/r$: $(rA)((1/r)A^{-1}) = I$. Taking $A = I$, $r = 2$ disproves the claimed factor r.

(p) **True.** Let x_0 be the unique solution to the stated right side. If $Az = 0$, then $A(x_0 + z) = Ax_0$, so uniqueness forces $z = 0$. Thus the square matrix has trivial null space and is invertible.

Exercise 2.S.2: Invert a supplied inverse

Since $C^{-1} = \begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}$ has determinant $28 - 30 = -2$, invert this 2×2 matrix:

$$\boxed{C = \frac{1}{-2}\begin{bmatrix} 7 & -5 \\ -6 & 4 \end{bmatrix} = \begin{bmatrix} -7/2 & 5/2 \\ 3 & -2 \end{bmatrix}}.$$

The inverse of an inverse is the original matrix, which explains why this operation recovers C rather than another new matrix.

Exercise 2.S.3: A nilpotent shift and a finite geometric sum

The given A sends $(x_1, x_2, x_3)^T$ to $(0, x_1, x_2)^T$. Therefore $A^2x = (0, 0, x_1)^T$ and $A^3x = 0$ for every x, proving $A^3 = 0$. Distribute the requested product in its existing order:

$$(I - A)(I + A + A^2) = I + A + A^2 - A - A^2 - A^3 = I.$$

The cancellation works because all terms are powers of the same matrix.

Exercise 2.S.4: Invert identity minus a nilpotent matrix

Let $S = I + A + \cdots + A^{n-1}$. Then telescoping gives

$$(I - A)S = I - A^n = I.$$

Multiplying in the other order gives the same telescoping sum, $S(I - A) = I - A^n = I$, since a matrix commutes with its own powers. Thus $\boxed{(I - A)^{-1} = I + A + \cdots + A^{n-1}}$. No inverse of A was used; in fact a nilpotent matrix of positive size cannot itself be invertible.

Exercise 2.S.5: Reduce higher powers using a quadratic identity

The hypothesis says $A^2 = 2A - I$. Multiply by A and substitute again:

$$A^3 = 2A^2 - A = 2(2A - I) - A = 3A - 2I.$$

Next multiply this result by A:

$$A^4 = 3A^2 - 2A = 3(2A - I) - 2A = 4A - 3I.$$

Every step uses multiplication and addition of already equal matrices; we did not divide by A or assume it was diagonalizable.

Exercise 2.S.6: A pair of anticommuting matrices

Direct multiplication gives

$$A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad B^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

$$AB = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \qquad BA = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = -AB.$$

Thus both are their own inverses, but they anticommute rather than commute. This is a concrete illustration of behavior unavailable for nonzero real scalars, since scalar multiplication commutes.

Exercise 2.S.7: Solve a two-column matrix equation efficiently

Compute $A^{-1}B$ by solving $AX = B$. Begin with $[A \mid B]$. Use $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - R_1$, then $R_3 \leftarrow R_3 - (1/2)R_2$:

$$\left[\begin{array}{ccc|cc} 1 & 3 & 8 & -3 & 5 \\ 2 & 4 & 11 & 1 & 5 \\ 1 & 2 & 5 & 3 & 4 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|cc} 1 & 3 & 8 & -3 & 5 \\ 0 & -2 & -5 & 7 & -5 \\ 0 & 0 & -1/2 & 5/2 & 3/2 \end{array}\right].$$

Back-substitution gives the last row of X as $(-5, -3)$, then the second as $(9, 10)$, and the first as $(10, -1)$. Therefore

$$\boxed{A^{-1}B = \begin{bmatrix} 10 & -1 \\ 9 & 10 \\ -5 & -3 \end{bmatrix}}.$$

The three nonzero pivots verify invertibility of A, and exact multiplication checks $AX = B$. No entire inverse was computed.

Exercise 2.S.8: Determine a map from two prescribed images

Place the input vectors into $X = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$ and outputs into $Y = \begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix}$. The two prescribed equations are exactly $AX = Y$. Since $\det X = 7 - 6 = 1$, $X^{-1} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$. Thus

$$\boxed{A = YX^{-1} = \begin{bmatrix} -2 & 1 \\ 4 & -1 \end{bmatrix}}.$$

Indeed $A(1, 3)^T = (1, 1)^T$ and $A(2, 7)^T = (3, 1)^T$. The independent inputs ensure that this matrix is uniquely determined.

Exercise 2.S.9: Recover the left factor of a known product

The given B has determinant $7-6=1$, so $B^{-1}=\begin{bmatrix}1 & -3\\ -2 & 7\end{bmatrix}$. Right multiplication by this inverse yields

$$A=(AB)B^{-1}=\begin{bmatrix}5 & 4\\ -2 & 3\end{bmatrix}\begin{bmatrix}1 & -3\\ -2 & 7\end{bmatrix}=\boxed{\begin{bmatrix}-3 & 13\\ -8 & 27\end{bmatrix}}.$$

For example the first output row checks as $[-3,13]B=[-21+26,-9+13]=[5,4]$.

Exercise 2.S.10: Another formula for an inverse

If A is invertible, so is A^T. Therefore A^TA is a product of invertible square matrices and is invertible. Set $X=(A^TA)^{-1}A^T$. Then $XA=(A^TA)^{-1}(A^TA)=I$. Since A^{-1} already exists, right multiplication by it gives $X=XAA^{-1}=A^{-1}$, establishing

$$\boxed{A^{-1}=(A^TA)^{-1}A^T}.$$

This is an algebraic identity, not a recommendation to compute the inverse by forming A^TA in floating-point arithmetic.

Exercise 2.S.11: Vandermonde matrices and interpolation

(a) Row i of Vc is $c_0+c_1x_i+\cdots+c_{n-1}x_i^{n-1}=p(x_i)$. Thus $Vc=y$ states precisely $p(x_i)=y_i$ for every i.

(b) Suppose $Vc=0$. Then p vanishes at the n distinct numbers $x_1,\ldots,x_n$. A nonzero polynomial of degree at most $n-1$ can have at most $n-1$ distinct roots, so p must be the zero polynomial. All its coefficients c_j are therefore zero. Thus $Vc=0$ has only the trivial solution, proving that the columns of V are independent.

(c) The square matrix V is invertible by the IMT, using part (b). For every arbitrary data vector y, choose $c=V^{-1}y$ and form the polynomial with those coefficients. Part (a) shows it interpolates every data point, and its degree is at most $n-1$. The polynomial is also unique: any second coefficient vector would be another solution of $Vc=y$.

Exercise 2.S.12: Read the first two columns of an LU product

The first column of upper triangular U is $(u_{11},0,\ldots,0)^T$, so $a_1=Lu_1=u_{11}l_1$. The second column of U is $(u_{12},u_{22},0,\ldots,0)^T$, so $a_2=u_{12}l_1+u_{22}l_2$. Thus the first column of A is a multiple of column 1 of L, and the second is a combination of columns 1 and 2 of L. Triangularity, not a mysterious cancellation, makes the later coefficients zero.

Exercise 2.S.13: Projection and reflection identities

Since $u^Tu=1$ is a scalar, associativity gives

$$\text{(a)}\quad P^2=(uu^T)(uu^T)=u(u^Tu)u^T=uu^T=P.$$

For the transpose,

$$\text{(b)} \quad P^T = (uu^T)^T = (u^T)^T u^T = uu^T = P.$$

Finally distribute the product using $P^2 = P$:

$$\text{(c)} \quad Q^2 = (I - 2P)^2 = I - 4P + 4P^2 = I.$$

Thus Q undoes itself: applying the same reflection twice returns every vector to its original position.

Exercise 2.S.14: A coordinate-plane reflection

With $u = e_3$,

$$P = uu^T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad Q = I - 2P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}.$$

For $x = (1, 5, 3)^T$ this gives $Px = (0, 0, 3)^T$ and $Qx = (1, 5, -3)^T$. The vector $x - Px = (1, 5, 0)^T$ is the component in the coordinate plane; Q preserves it and reverses the perpendicular component Px.

Exercise 2.S.15: A product of elementary operations gives row equivalence

Multiplication by E_1 performs one elementary row operation on B, then E_2 performs another and E_3 a third. Consequently $C = E_3E_2E_1B$ is obtained from B by a finite sequence of row operations, which is the definition of row equivalence. The reverse sequence uses $E_3^{-1}, E_2^{-1}, E_1^{-1}$ in that temporal order, so the relationship works both ways.

Exercise 2.S.16: Construct a nonzero matrix with zero product

Because square A is singular, $Av = 0$ has a nonzero solution v. Choose one by row reduction, then set $B = [v \ 0 \ \cdots \ 0]$. The first column makes $B \neq 0$, whereas $AB = [Av \ A0 \ \cdots \ A0] = 0$. This construction works even if the null space is only one-dimensional.

Exercise 2.S.17: A product that factors through four dimensions

Every column of AB is a combination of the four columns of the 6×4 matrix A. Therefore $\text{Col}(AB) \subseteq \text{Col}\, A$, whose dimension is at most four. It cannot equal $\mathbb{R}^6$, whose dimension is six. The 6×6 matrix AB is thus not onto and cannot be invertible by the IMT.

Exercise 2.S.18: A left inverse gives uniqueness for attainable outputs

If $Ax = b$, then $x = CAx = Cb$. Thus every solution must equal Cb. The premise already says a solution exists, so it is unique. Equivalently, if $Ax_1 = Ax_2$, multiplication by C yields $x_1 = x_2$. The 5×3 matrix need not attain every $b \in \mathbb{R}^5$; the proof establishes uniqueness when existence has separately been supplied.

Exercise 2.S.19: Observe and explain limits of stochastic powers

Both matrices have nonnegative entries and each column sums to one. Such a matrix is called column-stochastic: it preserves the sum of a vector's coordinates. Exact rational powers give

$$A^2 = \begin{bmatrix} .31 & .26 & .30 \\ .39 & .48 & .39 \\ .30 & .26 & .31 \end{bmatrix}, \qquad B^2 = \begin{bmatrix} .29 & .18 & .18 \\ .33 & .44 & .33 \\ .38 & .38 & .49 \end{bmatrix}.$$

For $k = 8$, the powers are approximately

$$A^8 \approx \begin{bmatrix} .285728 & .285696 & .285728 \\ .428543 & .428609 & .428543 \\ .285728 & .285696 & .285728 \end{bmatrix}, \quad B^8 \approx \begin{bmatrix} .202364 & .202218 & .202218 \\ .370732 & .370879 & .370732 \\ .426904 & .426904 & .427050 \end{bmatrix}.$$

By $k = 16$ their columns are close to the respective vectors

$$p_A = (2/7, 3/7, 2/7)^T, \qquad p_B = (18/89, 33/89, 38/89)^T.$$

These are verified exactly by $Ap_A = p_A$, $Bp_B = p_B$, and coordinate sums one. The powers tend toward matrices having every column equal to the corresponding p. An optional explanation from later eigenvalue theory is that A's other eigenvalues are $.3, .1$, and B's are $\pm\sqrt{11}/10$; all have magnitude less than one, so their powers die out.

Here is a further reproducible investigation with positive column-stochastic matrices:

```
import numpy as np
rng = np.random.default_rng(22019)
C = rng.integers(1,10,(3,3)).astype(float)
C = C/C.sum(axis=0)
print(np.linalg.matrix_power(C,16))
```

The executed result has nearly identical columns $(.20882917, .35930902, .43186180)^T$. A precise conjecture supported by these examples is that positive stochastic matrices, and more generally stochastic matrices with some positive power, have powers converging to a matrix with identical columns. It would be false to remove that extra condition: the permutation matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is stochastic but its powers alternate instead of converging. Thus the experiment suggests a qualified theorem, not a claim about every stochastic matrix.

Exercise 2.S.20: A closed formula for inverses of zero-diagonal ones matrices

Let J_n have every entry equal to one. Then $A_n = J_n - I_n$ and $J_n^2 = nJ_n$, since each entry of the square is a sum of n ones. For $n \geq 2$ propose

$$B_n = -I_n + \frac{1}{n-1} J_n.$$

Its diagonal entries are $-(n-2)/(n-1)$ and off-diagonal entries are $1/(n-1)$. In particular the computed inverses for the three requested sizes are

$$B_4 = \frac{1}{3} \begin{bmatrix} -2 & 1 & 1 & 1 \\ 1 & -2 & 1 & 1 \\ 1 & 1 & -2 & 1 \\ 1 & 1 & 1 & -2 \end{bmatrix},$$

$$B_5 = \frac{1}{4}\begin{bmatrix} -3 & 1 & 1 & 1 & 1 \\ 1 & -3 & 1 & 1 & 1 \\ 1 & 1 & -3 & 1 & 1 \\ 1 & 1 & 1 & -3 & 1 \\ 1 & 1 & 1 & 1 & -3 \end{bmatrix},$$

$$B_6 = \frac{1}{5}\begin{bmatrix} -4 & 1 & 1 & 1 & 1 & 1 \\ 1 & -4 & 1 & 1 & 1 & 1 \\ 1 & 1 & -4 & 1 & 1 & 1 \\ 1 & 1 & 1 & -4 & 1 & 1 \\ 1 & 1 & 1 & 1 & -4 & 1 \\ 1 & 1 & 1 & 1 & 1 & -4 \end{bmatrix}.$$

To turn the conjecture into a proof, multiply symbolically:

$$(J_n - I_n)B_n = I_n + \left(-1 + \frac{n}{n-1} - \frac{1}{n-1}\right) J_n = I_n.$$

The reverse product is identical because both factors are combinations of I_n and J_n. Hence $B_n = A_n^{-1}$ for all $n \geq 2$. For $n = 1$, $A_1 = [0]$ is singular and the denominator $n-1$ correctly signals the excluded case.